

3.4.3 Nested for Loops ¹

Control structures in Java are statements that contain statements. In particular, control structures can contain control structures. You've already seen several examples of `if` statements inside loops, and one example of a `while` loop inside another `while`, but any combination of one control structure inside another is possible. We say that one structure is *nested* inside another. You can even have multiple levels of nesting, such as a `while` loop inside an `if` statement inside another `while` loop. The syntax of Java does not set a limit on the number of levels of nesting. As a practical matter, though, it's difficult to understand a program that has more than a few levels of nesting.

Nested `for` loops arise naturally in many algorithms, and it is important to understand how they work. Let's look at a couple of examples. First, consider the problem of printing out a multiplication table like this one:

1	2	3	4	5	6	7	8	9	10	11	12
2	4	6	8	10	12	14	16	18	20	22	24
3	6	9	12	15	18	21	24	27	30	33	36
4	8	12	16	20	24	28	32	36	40	44	48
5	10	15	20	25	30	35	40	45	50	55	60
6	12	18	24	30	36	42	48	54	60	66	72
7	14	21	28	35	42	49	56	63	70	77	84
8	16	24	32	40	48	56	64	72	80	88	96
9	18	27	36	45	54	63	72	81	90	99	108
10	20	30	40	50	60	70	80	90	100	110	120
11	22	33	44	55	66	77	88	99	110	121	132
12	24	36	48	60	72	84	96	108	120	132	144

The data in the table are arranged into 12 rows and 12 columns. The process of printing them out can be expressed in a pseudocode algorithm as

```
for each rowNumber = 1, 2, 3, ..., 12:
    Print the first twelve multiples of rowNumber on one line
    Output a carriage return
```

The first step in the `for` loop can itself be expressed as a `for` loop. We can expand "Print the first twelve multiples of `rowNumber` on one line" as:

```
for N = 1, 2, 3, ..., 12:
    Print N * rowNumber
```

so a refined algorithm for printing the table has one `for` loop nested inside another:

```
for each rowNumber = 1, 2, 3, ..., 12:
    for N = 1, 2, 3, ..., 12:
        Print N * rowNumber
    Output a carriage return
```

We want to print the output in neat columns, with each output number taking up four spaces. This can be done using formatted output with format specifier `%4d`. Assuming that `rowNumber` and `N` have been declared to be variables of type `int`, the algorithm can be expressed in Java as

```
for ( rowNumber = 1; rowNumber <= 12; rowNumber++ ) {
    for ( N = 1; N <= 12; N++ ) {
        // print in 4-character columns
        System.out.printf( "%4d;", N * rowNumber ); // No carriage return !
    }
}
```

```

    System.out.println(); // Add a carriage return at end of the line.
}

```

This section has been weighed down with lots of examples of numerical processing. For our next example, let's do some text processing. Consider the problem of finding which of the 26 letters of the alphabet occur in a given string. For example, the letters that occur in "Hello World" are D, E, H, L, O, R, and W. More specifically, we will write a program that will list all the letters contained in a string and will also count the number of different letters. The string will be input by the user. Let's start with a pseudocode algorithm for the program.

```

Ask the user to input a string
Read the response into a variable, str
Let count = 0 (for counting the number of different letters)
for each letter of the alphabet:
    if the letter occurs in str:
        Print the letter
        Add 1 to count
Output the count

```

Since we want to process the entire line of text that is entered by the user, we'll use `TextIO.getln()` to read it. The line of the algorithm that reads "for each letter of the alphabet" can be expressed as "`for (letter='A'; letter<='Z'; letter++)`". But the body of this `for` loop needs more thought. How do we check whether the given letter, `letter`, occurs in `str`? One idea is to look at each character in the string in turn, and check whether that character is equal to `letter`. We can get the `i`-th character of `str` with the function call `str.charAt(i)`, where `i` ranges from 0 to `str.length() - 1`. One more difficulty: A letter such as 'A' can occur in `str` in either upper or lower case, 'A' or 'a'. We have to check for both of these. But we can avoid this difficulty by converting `str` to upper case before processing it. Then, we only have to check for the upper case letter. We can now flesh out the algorithm fully. Note the use of `break` in the nested `for` loop. It is required to avoid printing or counting a given letter more than once (in the case where it occurs more than once in the string). The `break` statement breaks out of the inner `for` loop, but not the outer `for` loop. Upon executing the `break`, the computer continues the outer loop with the next value of `letter`.

```

Ask the user to input a string
Read the response into a variable, str
Convert str to upper case
Let count = 0
for letter = 'A', 'B', ..., 'Z':
    for i = 0, 1, ..., str.length()-1:
        if letter == str.charAt(i):
            Print letter
            Add 1 to count
            break // jump out of the loop
Output the count

```

Here is the complete program:

```

/**
 * This program reads a line of text entered by the user.
 * It prints a list of the letters that occur in the text,
 * and it reports how many different letters were found.
 */

```

```

public class ListLetters {
    public static void main(String[] args) {
        String str; // Line of text entered by the user.
        int count; // Number of different letters found in str.
        char letter; // A letter of the alphabet.

        TextIO.putln("Please type in a line of text.");
        str = TextIO.getln();

        str = str.toUpperCase();

        count = 0;
        TextIO.putln("Your input contains the following letters:");
        TextIO.putln();
        TextIO.put(" ");
        for ( letter = 'A'; letter <= 'Z'; letter++ ) {
            int i; // Position of a character in str.
            for ( i = 0; i < str.length(); i++ ) {
                if ( letter == str.charAt(i) ) {
                    TextIO.put(letter);
                    TextIO.put(' ');
                    count++;
                    break;
                }
            }
        }

        TextIO.putln();
        TextIO.putln();
        TextIO.putln("There were " + count + " different letters.");

    } // end main()
} // end class ListLetters

```

In fact, there is actually an easier way to determine whether a given letter occurs in a string, `str`. The built-in function `str.indexOf(letter)` will return `-1` if `letter` does **not** occur in the string. It returns a number greater than or equal to zero if it does occur. So, we could check whether `letter` occurs in `str` simply by checking “if (`str.indexOf(letter)` `>= 0`)”. If we used this technique in the above program, we wouldn’t need a nested `for` loop. This gives you a preview of how subroutines can be used to deal with complexity.

3.4.4 Enums and for-each Loops

Java 5.0 introduces a new “enhanced” form of the `for` loop that is designed to be convenient for processing data structures. A **data structure** is a collection of data items, considered as a unit. For example, a **list** is a data structure that consists simply of a sequence of items. The enhanced `for` loop makes it easy to apply the same processing to every element of a list or other data structure. Data structures are a major topic in computer science, but we won’t encounter them in any serious way until Chapter 7. However, one of the applications of the enhanced `for` loop is to **enum** types, and so we consider it briefly here. (Enums were introduced in Subsection 2.3.3.)

The enhanced for loop can be used to perform the same processing on each of the enum constants that are the possible values of an enumerated type. The syntax for doing this is: 1

```
for ( <enum-type-name> <variable-name> : <enum-type-name>.values() ) 2
    <statement>
```

or

```
for ( <enum-type-name> <variable-name> : <enum-type-name>.values() ) {
    <statements>
}
```

If *MyEnum* is the name of any enumerated type, then *MyEnum.values()* is a function call that returns a list containing all of the values of the enum. (*values()* is a static member function in *MyEnum* and of any other enum.) For this enumerated type, the for loop would have the form: 3

```
for ( MyEnum <variable-name> : MyEnum.values() ) 4
    <statement>
```

The intent of this is to execute the *<statement>* once for each of the possible values of the *MyEnum* type. The *<variable-name>* is the loop control variable. In the *<statement>*, it represents the enumerated type value that is currently being processed. This variable should **not** be declared before the for loop; it is essentially being declared in the loop itself. 5

To give a concrete example, suppose that the following enumerated type has been defined to represent the days of the week: 6

```
enum Day { MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY } 7
```

Then we could write: 8

```
for ( Day d : Day.values() ) { 9
    System.out.print( d );
    System.out.print(" is day number ");
    System.out.println( d.ordinal() );
}
```

Day.values() represents the list containing the seven constants that make up the enumerated type. The first time through this loop, the value of *d* would be the first enumerated type value *Day.MONDAY*, which has ordinal number 0, so the output would be “MONDAY is day number 0”. The second time through the loop, the value of *d* would be *Day.TUESDAY*, and so on through *Day.SUNDAY*. The body of the loop is executed once for each item in the list *Day.values()*, with *d* taking on each of those values in turn. The full output from this loop would be: 10

```
MONDAY is day number 0 11
TUESDAY is day number 1
WEDNESDAY is day number 2
THURSDAY is day number 3
FRIDAY is day number 4
SATURDAY is day number 5
SUNDAY is day number 6
```

Since the intent of the enhanced for loop is to do something “for each” item in a data structure, it is often called a **for-each loop**. The syntax for this type of loop is unfortunate. It would be better if it were written something like “foreach Day d in Day.values()”, which conveys the meaning much better and is similar to the syntax used in other programming languages for similar types of loops. It’s helpful to think of the colon (:) in the loop as meaning “in.” 12

3.5 The if Statement¹

THE FIRST OF THE TWO BRANCHING STATEMENTS in Java is the `if` statement, which you² have already seen in Section 3.1. It takes the form

```
if (⟨boolean-expression⟩)3
    ⟨statement-1⟩
else
    ⟨statement-2⟩
```

As usual, the statements inside an `if` statements can be blocks. The `if` statement represents⁴ a two-way branch. The `else` part of an `if` statement—consisting of the word “else” and the statement that follows it—can be omitted.

3.5.1 The Dangling else Problem⁵

Now, an `if` statement is, in particular, a statement. This means that either `⟨statement-1⟩`⁶ or `⟨statement-2⟩` in the above `if` statement can itself be an `if` statement. A problem arises, however, if `⟨statement-1⟩` is an `if` statement that has no `else` part. This special case is effectively forbidden by the syntax of Java. Suppose, for example, that you type

```
if ( x > 0 )7
    if ( y > 0 )
        System.out.println("First case");
else
    System.out.println("Second case");
```

Now, remember that the way you’ve indented this doesn’t mean anything at all to the computer.⁸ You might think that the `else` part is the second half of your “`if ( x > 0 )`” statement, but the rule that the computer follows attaches the `else` to “`if ( y > 0 )`”, which is closer. That is, the computer reads your statement as if it were formatted:

```
if ( x > 0 )9
    if ( y > 0 )
        System.out.println("First case");
else
    System.out.println("Second case");
```

You can force the computer to use the other interpretation by enclosing the nested `if` in a¹⁰ block:

```
if ( x > 0 ) {11
    if ( y > 0 )
        System.out.println("First case");
}
else
    System.out.println("Second case");
```

These two `if` statements have different meanings: In the case when `x <= 0`, the first statement¹² doesn’t print anything, but the second statement prints “Second case.”.

3.5.2 The if...else if Construction¹³

Much more interesting than this technicality is the case where `⟨statement-2⟩`, the `else` part¹⁴ of the `if` statement, is itself an `if` statement. The statement would look like this (perhaps without the final `else` part):

```

if (<boolean-expression-1>) 1
    <statement-1>
else
    if (<boolean-expression-2>)
        <statement-2>
    else
        <statement-3>

```

However, since the computer doesn't care how a program is laid out on the page, this is almost always written in the format: 2

```

if (<boolean-expression-1>) 3
    <statement-1>
else if (<boolean-expression-2>)
    <statement-2>
else
    <statement-3>

```

You should think of this as a single statement representing a three-way branch. When the computer executes this, one and only one of the three statements—*<statement-1>*, *<statement-2>*, or *<statement-3>*—will be executed. The computer starts by evaluating *<boolean-expression-1>*. If it is **true**, the computer executes *<statement-1>* and then jumps all the way to the end of the outer if statement, skipping the other two *<statement>*s. If *<boolean-expression-1>* is **false**, the computer skips *<statement-1>* and executes the second, nested if statement. To do this, it tests the value of *<boolean-expression-2>* and uses it to decide between *<statement-2>* and *<statement-3>*. 4

Here is an example that will print out one of three different messages, depending on the value of a variable named `temperature`: 5

```

if (temperature < 50) 6
    System.out.println("It's cold.");
else if (temperature < 80)
    System.out.println("It's nice.");
else
    System.out.println("It's hot.");

```

If `temperature` is, say, 42, the first test is **true**. The computer prints out the message “It’s cold”, and skips the rest—without even evaluating the second condition. For a temperature of 75, the first test is **false**, so the computer goes on to the second test. This test is **true**, so the computer prints “It’s nice” and skips the rest. If the temperature is 173, both of the tests evaluate to **false**, so the computer says “It’s hot” (unless its circuits have been fried by the heat, that is). 7

You can go on stringing together “else-if’s” to make multi-way branches with any number of cases: 8

```

if (<boolean-expression-1>) 9
    <statement-1>
else if (<boolean-expression-2>)
    <statement-2>
else if (<boolean-expression-3>)
    <statement-3>
.
. // (more cases)
.

```

```

else if (⟨boolean-expression-N⟩) 1
    ⟨statement-N⟩
else
    ⟨statement-(N+1)⟩

```

The computer evaluates boolean expressions one after the other until it comes to one that is **true**. It executes the associated statement and skips the rest. If none of the boolean expressions evaluate to **true**, then the statement in the **else** part is executed. This statement is called a multi-way branch because only one of the statements will be executed. The final **else** part can be omitted. In that case, if all the boolean expressions are false, none of the statements is executed. Of course, each of the statements can be a block, consisting of a number of statements enclosed between { and }. (Admittedly, there is lot of syntax here; as you study and practice, you'll become comfortable with it.) ²

3.5.3 If Statement Examples ³

As an example of using if statements, let's suppose that **x**, **y**, and **z** are variables of type **int**, ⁴ and that each variable has already been assigned a value. Consider the problem of printing out the values of the three variables in increasing order. For examples, if the values are 42, 17, and 20, then the output should be in the order 17, 20, 42.

One way to approach this is to ask, where does **x** belong in the list? It comes first if it's ⁵ less than both **y** and **z**. It comes last if it's greater than both **y** and **z**. Otherwise, it comes in the middle. We can express this with a 3-way if statement, but we still have to worry about the order in which **y** and **z** should be printed. In pseudocode,

```

if (x < y && x < z) { 6
    output x, followed by y and z in their correct order
}
else if (x > y && x > z) {
    output y and z in their correct order, followed by x
}
else {
    output x in between y and z in their correct order
}

```

Determining the relative order of **y** and **z** requires another if statement, so this becomes ⁷

```

if (x < y && x < z) {           // x comes first 8
    if (y < z)
        System.out.println( x + " " + y + " " + z );
    else
        System.out.println( x + " " + z + " " + y );
}
else if (x > y && x > z) {      // x comes last
    if (y < z)
        System.out.println( y + " " + z + " " + x );
    else
        System.out.println( z + " " + y + " " + x );
}
else {                         // x in the middle
    if (y < z)
        System.out.println( y + " " + x + " " + z );
    else

```

```

        System.out.println( z + " " + x + " " + y); 1
    }

```

You might check that this code will work correctly even if some of the values are the same. If the values of two variables are the same, it doesn't matter which order you print them in. 2

Note, by the way, that even though you can say in English “if x is less than y and z,”, 3 you can't say in Java “if (x < y && z)”. The && operator can only be used between boolean values, so you have to make separate tests, x<y and x<z, and then combine the two tests with &&.

There is an alternative approach to this problem that begins by asking, “which order should x and y be printed in?” Once that's known, you only have to decide where to stick in z. This line of thought leads to different Java code: 4

```

if ( x < y ) { // x comes before y
    if ( z < x ) // z comes first
        System.out.println( z + " " + x + " " + y);
    else if ( z > y ) // z comes last
        System.out.println( x + " " + y + " " + z);
    else // z is in the middle
        System.out.println( x + " " + z + " " + y);
}
else { // y comes before x
    if ( z < y ) // z comes first
        System.out.println( z + " " + y + " " + x);
    else if ( z > x ) // z comes last
        System.out.println( y + " " + x + " " + z);
    else // z is in the middle
        System.out.println( y + " " + z + " " + x);
}

```

Once again, we see how the same problem can be solved in many different ways. The two approaches to this problem have not exhausted all the possibilities. For example, you might start by testing whether x is greater than y. If so, you could swap their values. Once you've done that, you know that x should be printed before y. 6

* * *

Finally, let's write a complete program that uses an if statement in an interesting way. I 7 want a program that will convert measurements of length from one unit of measurement to another, such as miles to yards or inches to feet. So far, the problem is extremely under-specified. Let's say that the program will only deal with measurements in inches, feet, yards, and miles. It would be easy to extend it later to deal with other units. The user will type in a measurement in one of these units, such as “17 feet” or “2.73 miles”. The output will show the length in terms of **each** of the four units of measure. (This is easier than asking the user which units to use in the output.) An outline of the process is

```

Read the user's input measurement and units of measure 8
Express the measurement in inches, feet, yards, and miles
Display the four results

```

The program can read both parts of the user's input from the same line by using 9 `TextIO.getDouble()` to read the numerical measurement and `TextIO.getlnWord()` to read the unit of measure. The conversion into different units of measure can be simplified by first

converting the user's input into inches. From there, the number of inches can easily be converted into feet, yards, and miles. Before converting into inches, we have to test the input to determine which unit of measure the user has specified:

```

Let measurement = TextIO.getDouble()
Let units = TextIO.getlnWord()
if the units are inches
    Let inches = measurement
else if the units are feet
    Let inches = measurement * 12           // 12 inches per foot
else if the units are yards
    Let inches = measurement * 36          // 36 inches per yard
else if the units are miles
    Let inches = measurement * 12 * 5280   // 5280 feet per mile
else
    The units are illegal!
    Print an error message and stop processing
Let feet = inches / 12.0
Let yards = inches / 36.0
Let miles = inches / (12.0 * 5280.0)
Display the results

```

Since `units` is a *String*, we can use `units.equals("inches")` to check whether the specified unit of measure is “inches”. However, it would be nice to allow the units to be specified as “inch” or abbreviated to “in”. To allow these three possibilities, we can check if `(units.equals("inches") || units.equals("inch") || units.equals("in"))`. It would also be nice to allow upper case letters, as in “Inches” or “IN”. We can do this by converting `units` to lower case before testing it or by substituting the function `units.equalsIgnoreCase` for `units.equals`.

In my final program, I decided to make things more interesting by allowing the user to enter a whole sequence of measurements. The program will end only when the user inputs 0. To do this, I just have to wrap the above algorithm inside a **while** loop, and make sure that the loop ends when the user inputs a 0. Here's the complete program:

```
/*  
 * This program will convert measurements expressed in inches,  
 * feet, yards, or miles into each of the possible units of  
 * measure. The measurement is input by the user, followed by  
 * the unit of measure. For example: "17 feet", "1 inch",  
 * "2.73 mi". Abbreviations in, ft, yd, and mi are accepted.  
 * The program will continue to read and convert measurements  
 * until the user enters an input of 0.  
 */  
  
public class LengthConverter {  
  
    public static void main(String[] args) {  
  
        double measurement; // Numerical measurement, input by user.  
        String units;       // The unit of measure for the input, also  
                           // specified by the user.  
  
        double inches, feet, yards, miles; // Measurement expressed in  
                                           // each possible unit of  
                                           // measure.
```

```

TextIO.putln("Enter measurements in inches, feet, yards, or miles.");1
TextIO.putln("For example: 1 inch    17 feet    2.73 miles");
TextIO.putln("You can use abbreviations:  in  ft  yd  mi");
TextIO.putln("I will convert your input into the other units");
TextIO.putln("of measure.");
TextIO.putln();

while (true) {

    /* Get the user's input, and convert units to lower case. */

    TextIO.put("Enter your measurement, or 0 to end: ");
    measurement = TextIO.getDouble();
    if (measurement == 0)
        break; // Terminate the while loop.
    units = TextIO.getlnWord();
    units = units.toLowerCase();

    /* Convert the input measurement to inches. */

    if (units.equals("inch") || units.equals("inches")
        || units.equals("in")) {
        inches = measurement;
    }
    else if (units.equals("foot") || units.equals("feet")
        || units.equals("ft")) {
        inches = measurement * 12;
    }
    else if (units.equals("yard") || units.equals("yards")
        || units.equals("yd")) {
        inches = measurement * 36;
    }
    else if (units.equals("mile") || units.equals("miles")
        || units.equals("mi")) {
        inches = measurement * 12 * 5280;
    }
    else {
        TextIO.putln("Sorry, but I don't understand \""
            + units + "\".");
        continue; // back to start of while loop
    }

    /* Convert measurement in inches to feet, yards, and miles. */

    feet = inches / 12;
    yards = inches / 36;
    miles = inches / (12*5280);

    /* Output measurement in terms of each unit of measure. */

    TextIO.putln();
    TextIO.putln("That's equivalent to:");
    TextIO.printf("%12.5g", inches);
    TextIO.putln(" inches");
    TextIO.printf("%12.5g", feet);
    TextIO.putln(" feet");
    TextIO.printf("%12.5g", yards);

```

```

        TextIO.putln(" yards");
        TextIO.printf("%12.5g", miles);
        TextIO.putln(" miles");
        TextIO.putln();

    } // end while

    TextIO.putln();
    TextIO.putln("OK! Bye for now.");

} // end main()
} // end class LengthConverter

```

1

(Note that this program uses formatted output with the “g” format specifier. In this program, we have no control over how large or how small the numbers might be. It could easily make sense for the user to enter very large or very small measurements. The “g” format will print a real number in exponential form if it is very large or very small, and in the usual decimal form otherwise. Remember that in the format specification `%12.5g`, the 5 is the total number of significant digits that are to be printed, so we will always get the same number of significant digits in the output, no matter what the size of the number. If we had used an “f” format specifier such as `%12.5f`, the output would be in decimal form with 5 digits after the decimal point. This would print the number 0.0000000007454 as 0.00000, with no **significant** digits at all! With the “g” format specifier, the output would be 7.454e-10.)

2

3.5.4 The Empty Statement 3

As a final note in this section, I will mention one more type of statement in Java: the **empty statement**. This is a statement that consists simply of a semicolon and which tells the computer to do nothing. The existence of the empty statement makes the following legal, even though you would not ordinarily see a semicolon after a `}` :

```

if (x < 0) {
    x = -x;
};

```

5

The semicolon is legal after the `}`, but the computer considers it to be an empty statement, not part of the `if` statement. Occasionally, you might find yourself using the empty statement when what you mean is, in fact, “do nothing”. For example, the rather contrived `if` statement

6

```

if ( done )
    ; // Empty statement
else
    System.out.println( "Not done yet. ");

```

7

does nothing when the **boolean** variable `done` is true, and prints out “Not done yet” when it is false. You can’t just leave out the semicolon in this example, since Java syntax requires an actual statement between the `if` and the `else`. I prefer, though, to use an empty block, consisting of `{` and `}` with nothing between, for such cases.

8

Occasionally, stray empty statements can cause annoying, hard-to-find errors in a program. For example, the following program segment prints out “Hello” just **once**, not ten times:

9

```

for (int i = 0; i < 10; i++);
    System.out.println("Hello");

```

10

Why? Because the “;” at the end of the first line is a statement, and it is this statement that is executed ten times. The `System.out.println` statement is not really inside the `for` statement at all, so it is executed just once, after the `for` loop has completed.

3.6 The switch Statement

THE SECOND BRANCHING STATEMENT in Java is the `switch` statement, which is introduced in this section. The `switch` statement is used far less often than the `if` statement, but it is sometimes useful for expressing a certain type of multi-way branch.

3.6.1 The Basic switch Statement

A switch statement allows you to test the value of an expression and, depending on that value, to jump directly to some location within the switch statement. Only expressions of certain types can be used. The value of the expression can be one of the primitive integer types `int`, `short`, or `byte`. It can be the primitive `char` type. Or, as we will see later in this section, it can be an enumerated type. In particular, the expression **cannot** be a *String* or a real number. The positions that you can jump to are marked with *case labels* that take the form: “case *<constant>*:". This marks the position the computer jumps to when the expression evaluates to the given *<constant>*. As the final case in a switch statement you can, optionally, use the label “default:”, which provides a default jump point that is used when the value of the expression is not listed in any case label.

A `switch` statement, as it is most often used, has the form:

```
switch ((expression)) {
    case <constant-1>:
        <statements-1>
        break;
    case <constant-2>:
        <statements-2>
        break;
    .
    .    // (more cases)
    .
    case <constant-N>:
        <statements-N>
        break;
    default: // optional default case
        <statements-(N+1)>
} // end of switch statement
```

The `break` statements are technically optional. The effect of a `break` is to make the computer jump to the end of the switch statement. If you leave out the `break` statement, the computer will just forge ahead after completing one case and will execute the statements associated with the next case label. This is rarely what you want, but it is legal. (I will note here—although you won’t understand it until you get to the next chapter—that inside a subroutine, the `break` statement is sometimes replaced by a `return` statement.)

Note that you can leave out one of the groups of statements entirely (including the `break`). You then have two case labels in a row, containing two different constants. This just means

that the computer will jump to the same place and perform the same action for each of the two constants. 1

Here is an example of a switch statement. This is not a useful example, but it should be easy for you to follow. Note, by the way, that the constants in the case labels don't have to be in any particular order, as long as they are all different: 2

```
switch ( N ) {    // (Assume N is an integer variable.) 3
    case 1:
        System.out.println("The number is 1.");
        break;
    case 2:
    case 4:
    case 8:
        System.out.println("The number is 2, 4, or 8.");
        System.out.println("(That's a power of 2!)");
        break;
    case 3:
    case 6:
    case 9:
        System.out.println("The number is 3, 6, or 9.");
        System.out.println("(That's a multiple of 3!)");
        break;
    case 5:
        System.out.println("The number is 5.");
        break;
    default:
        System.out.println("The number is 7 or is outside the range 1 to 9.");
}
```

The switch statement is pretty primitive as control structures go, and it's easy to make mistakes when you use it. Java takes all its control structures directly from the older programming languages C and C++. The switch statement is certainly one place where the designers of Java should have introduced some improvements. 4

3.6.2 Menus and switch Statements 5

One application of switch statements is in processing menus. A menu is a list of options. The user selects one of the options. The computer has to respond to each possible choice in a different way. If the options are numbered 1, 2, ..., then the number of the chosen option can be used in a switch statement to select the proper response. 6

In a *TextIO*-based program, the menu can be presented as a numbered list of options, and the user can choose an option by typing in its number. Here is an example that could be used in a variation of the `LengthConverter` example from the previous section: 7

```
int optionNumber;    // Option number from menu, selected by user. 8
double measurement; // A numerical measurement, input by the user.
                    // The unit of measurement depends on which
                    // option the user has selected.
double inches;       // The same measurement, converted into inches.

/* Display menu and get user's selected option number. */

TextIO.putln("What unit of measurement does your input use?");
TextIO.putln();
```

```

TextIO.putln("          1.  inches");
TextIO.putln("          2.  feet");
TextIO.putln("          3.  yards");
TextIO.putln("          4.  miles");
TextIO.putln();
TextIO.putln("Enter the number of your choice: ");
optionNumber = TextIO.getlnInt();

/* Read user's measurement and convert to inches. */

switch ( optionNumber ) {
    case 1:
        TextIO.putln("Enter the number of inches: ");
        measurement = TextIO.getlnDouble();
        inches = measurement;
        break;
    case 2:
        TextIO.putln("Enter the number of feet: ");
        measurement = TextIO.getlnDouble();
        inches = measurement * 12;
        break;
    case 3:
        TextIO.putln("Enter the number of yards: ");
        measurement = TextIO.getlnDouble();
        inches = measurement * 36;
        break;
    case 4:
        TextIO.putln("Enter the number of miles: ");
        measurement = TextIO.getlnDouble();
        inches = measurement * 12 * 5280;
        break;
    default:
        TextIO.putln("Error!  Illegal option number!  I quit!");
        System.exit(1);
} // end switch

/* Now go on to convert inches to feet, yards, and miles... */

```

3.6.3 Enums in switch Statements ²

The type of the expression in a `switch` can be an enumerated type. In that case, the constants ³ in the `case` labels must be values from the enumerated type. For example, if the type of the expression is the enumerated type *Season* defined by

```
enum Season { SPRING, SUMMER, FALL, WINTER } 4
```

then the constants in the case label must be chosen from among the values *Season.SPRING*, *Season.SUMMER*, *Season.FALL*, or *Season.WINTER*. However, there is another quirk in the syntax: when an enum constant is used in a case label, only the simple name, such as “*SPRING*” can be used, not the full name “*Season.SPRING*”. Of course, the computer already knows that the value in the case label must belong to the enumerated type, since it can tell that from the type of expression used, so there is really no need to specify the type name in the constant. As an example, suppose that *currentSeason* is a variable of type *Season*. Then we could have the switch statement: ⁵

```

switch ( currentSeason ) {
    case WINTER:    // ( NOT Season.WINTER ! )
        System.out.println("December, January, February");
        break;
    case SPRING:
        System.out.println("March, April, May");
        break;
    case SUMMER:
        System.out.println("June, July, August");
        break;
    case FALL:
        System.out.println("September, October, November");
        break;
}

```

2

3.6.4 Definite Assignment 3

As a somewhat more realistic example, the following `switch` statement makes a random choice among three possible alternatives. Recall that the value of the expression `(int)(3*Math.random())` is one of the integers 0, 1, or 2, selected at random with equal probability, so the `switch` statement below will assign one of the values "Rock", "Scissors", "Paper" to `computerMove`, with probability 1/3 for each case. Although the switch statement in this example is correct, this code segment as a whole illustrates a subtle syntax error that sometimes comes up:

```

String computerMove;
switch ( (int)(3*Math.random()) ) {
    case 0:
        computerMove = "Rock";
        break;
    case 1:
        computerMove = "Scissors";
        break;
    case 2:
        computerMove = "Paper";
        break;
}
System.out.println("Computer's move is " + computerMove); // ERROR!

```

5

You probably haven't spotted the error, since it's not an error from a human point of view. The computer reports the last line to be an error, because the variable `computerMove` might not have been assigned a value. In Java, it is only legal to use the value of a variable if a value has already been *definitely assigned* to that variable. This means that the computer must be able to prove, just from looking at the code when the program is compiled, that the variable must have been assigned a value. Unfortunately, the computer only has a few simple rules that it can apply to make the determination. In this case, it sees a `switch` statement in which the type of expression is `int` and in which the cases that are covered are 0, 1, and 2. For other values of the expression, `computerMove` is never assigned a value. So, the computer thinks `computerMove` might still be undefined after the `switch` statement. Now, in fact, this isn't true: 0, 1, and 2 are actually the only possible values of the expression `(int)(3*Math.random())`, but the computer isn't smart enough to figure that out. The easiest way to fix the problem is

6

to replace the case label `case 2` with `default`. The computer can see that a value is assigned to `computerMove` in all cases. 1

More generally, we say that a value has been definitely assigned to a variable at a given point in a program if every execution path leading from the declaration of the variable to that point in the code includes an assignment to the variable. This rule takes into account loops and `if` statements as well as `switch` statements. For example, the following two `if` statements both do the same thing as the `switch` statement given above, but only the one on the right definitely assigns a value to `computerMove`: 2

<pre>String computerMove; int rand; rand = (int)(3*Math.random()); if (rand == 0) computerMove = "Rock"; else if (rand == 1) computerMove = "Scissors"; else if (rand == 2) computerMove = "Paper";</pre>	<pre>String computerMove; int rand; rand = (int)(3*Math.random()); if (rand == 0) computerMove = "Rock"; else if (rand == 1) computerMove = "Scissors"; else computerMove = "Paper";</pre>	3
---	--	---

In the code on the left, the test “`if ( rand == 2 )`” in the final `else` clause is unnecessary because if `rand` is not 0 or 1, the only remaining possibility is that `rand == 2`. The computer, however, can’t figure that out. 4

3.7 Introduction to Exceptions and `try..catch` 5

IN ADDITION TO THE CONTROL structures that determine the normal flow of control in a program, Java has a way to deal with “exceptional” cases that throw the flow of control off its normal track. When an error occurs during the execution of a program, the default behavior is to terminate the program and to print an error message. However, Java makes it possible to “catch” such errors and program a response different from simply letting the program crash. This is done with the ***try..catch*** statement. In this section, we will take a preliminary, incomplete look at using `try..catch` to handle errors. Error handling is a complex topic, which we will return to in Chapter 8. 6

3.7.1 Exceptions 7

The term ***exception*** is used to refer to the type of error that one might want to handle with a `try..catch`. An exception is an exception to the normal flow of control in the program. The term is used in preference to “error” because in some cases, an exception might not be considered to be an error at all. You can sometimes think of an exception as just another way to organize a program. 8

Exceptions in Java are represented as objects of type *Exception*. Actual exceptions are defined by subclasses of *Exception*. Different subclasses represent different types of exceptions. We will look at only two types of exception in this section: *NumberFormatException* and *IllegalArgumentException*. 9

A *NumberFormatException* can occur when an attempt is made to convert a string into a number. Such conversions are done by the functions `Integer.parseInt` and `Integer.parseDouble`. (See Subsection 2.5.7.) Consider the function call `Integer.parseInt(str)` where `str` is a variable of type *String*. If the value of `str` is the string “42”, then the function call will correctly convert the string into the `int` 42. However, 10

if the value of `str` is, say, `"fred"`, the function call will fail because `"fred"` is not a legal string representation of an `int` value. In this case, an exception of type *NumberFormatException* occurs. If nothing is done to handle the exception, the program will crash.

An *IllegalArgumentException* can occur when an illegal value is passed as a parameter to a subroutine. For example, if a subroutine requires that a parameter be greater than or equal to zero, an *IllegalArgumentException* might occur when a negative value is passed to the subroutine. How to respond to the illegal value is up to the person who wrote the subroutine, so we can't simply say that every illegal parameter value will result in an *IllegalArgumentException*. However, it is a common response.

One case where an *IllegalArgumentException* can occur is in the `valueOf` function of an enumerated type. Recall from Subsection 2.3.3 that this function tries to convert a string into one of the values of the enumerated type. If the string that is passed as a parameter to `valueOf` is not the name of one of the enumerated type's values, then an *IllegalArgumentException* occurs. For example, given the enumerated type

```
enum Toss { HEADS, TAILS }
```

`Toss.valueOf("HEADS")` correctly returns the value `Toss.HEADS`, while `Toss.valueOf("FEET")` results in an *IllegalArgumentException*.

3.7.2 try..catch

When an exception occurs, we say that the exception is “thrown”. For example, we say that `Integer.parseInt(str)` *throws* an exception of type *NumberFormatException* when the value of `str` is illegal. When an exception is thrown, it is possible to “catch” the exception and prevent it from crashing the program. This is done with a *try..catch* statement. In somewhat simplified form, the syntax for a *try..catch* is:

```
try {
    <statements-1>
}
catch ( <exception-class-name> <variable-name> ) {
    <statements-2>
}
```

The `<exception-class-name>` could be *NumberFormatException*, *IllegalArgumentException*, or some other exception class. When the computer executes this statement, it executes the statements in the `try` part. If no error occurs during the execution of `<statements-1>`, then the computer just skips over the `catch` part and proceeds with the rest of the program. However, if an exception of type `<exception-class-name>` occurs during the execution of `<statements-1>`, the computer immediately jumps to the `catch` part and executes `<statements-2>`, skipping any remaining statements in `<statements-1>`. During the execution of `<statements-2>`, the `<variable-name>` represents the exception object, so that you can, for example, print it out. At the end of the `catch` part, the computer proceeds with the rest of the program; the exception has been caught and handled and does not crash the program. Note that only one type of exception is caught; if some other type of exception occurs during the execution of `<statements-1>`, it will crash the program as usual.

(By the way, note that the braces, `{` and `}`, are part of the syntax of the *try..catch* statement. They are required even if there is only one statement between the braces. This is different from the other statements we have seen, where the braces around a single statement are optional.)

As an example, suppose that `str` is a variable of type *String* whose value might or might not represent a legal real number. Then we could say:

```
try {
    double x;
    x = Double.parseDouble(str);
    System.out.println( "The number is " + x );
}
catch ( NumberFormatException e ) {
    System.out.println( "Not a legal number." );
}
```

If an error is thrown by the call to `Double.parseDouble(str)`, then the output statement in the `try` part is skipped, and the statement in the `catch` part is executed.

It's not always a good idea to catch exceptions and continue with the program. Often that can just lead to an even bigger mess later on, and it might be better just to let the exception crash the program at the point where it occurs. However, sometimes it's possible to recover from an error. For example, suppose that we have the enumerated type

```
enum Day { MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY }
```

and we want the user to input a value belonging to this type. `TextIO` does not know about this type, so we can only read the user's response as a string. The function `Day.valueOf` can be used to convert the user's response to a value of type *Day*. This will throw an exception of type *IllegalArgumentException* if the user's response is not the name of one of the values of type *Day*, but we can respond to the error easily enough by asking the user to enter another response. Here is a code segment that does this. (Converting the user's response to upper case will allow responses such as "Monday" or "monday" in addition to "MONDAY".)

```
Day weekday; // User's response as a value of type Day.
while ( true ) {
    String response; // User's response as a String.
    TextIO.put("Please enter a day of the week: ");
    response = TextIO.getln();
    response = response.toUpperCase();
    try {
        weekday = Day.valueOf(response);
        break;
    }
    catch ( IllegalArgumentException e ) {
        TextIO.putln( response + " is not the name of a day of the week." );
    }
}
```

The `break` statement will be reached only if the user's response is acceptable, and so the loop will end only when a legal value has been assigned to `weekday`.

3.7.3 Exceptions in `TextIO`

When `TextIO` reads a numeric value from the user, it makes sure that the user's response is legal, using a technique similar to the `while` loop and `try..catch` in the previous example. However, `TextIO` can read data from other sources besides the user. (See Subsection 2.4.5.) When it is reading from a file, there is no reasonable way for `TextIO` to recover from an illegal value in the input, so it responds by throwing an exception. To keep things simple, `TextIO` only

throws exceptions of type *IllegalArgumentException*, no matter what type of error it encounters. ¹ For example, an exception will occur if an attempt is made to read from a file after all the data in the file has already been read. In `TextIO`, the exception is of type *IllegalArgumentException*. If you have a better response to file errors than to let the program crash, you can use a `try..catch` to catch exceptions of type *IllegalArgumentException*.

For example, suppose that a file contains nothing but real numbers, and we want a program ² that will read the numbers and find their sum and their average. Since it is unknown how many numbers are in the file, there is the question of when to stop reading. One approach is simply to try to keep reading indefinitely. When the end of the file is reached, an exception occurs. This exception is not really an error—it's just a way of detecting the end of the data, so we can catch the exception and finish up the program. We can read the data in a `while (true)` loop and break out of the loop when an exception occurs. This is an example of the somewhat unusual technique of using an exception as part of the expected flow of control in a program.

To read from the file, we need to know the file's name. To make the program more general, ³ we can let the user enter the file name, instead of hard-coding a fixed file name in the program. However, it is possible that the user will enter the name of a file that does not exist. When we use `TextIO.readFile` to open a file that does not exist, an exception of type *IllegalArgumentException* occurs. We can catch this exception and ask the user to enter a different file name. Here is a complete program that uses all these ideas:

```
4
/**
 * This program reads numbers from a file. It computes the sum and
 * the average of the numbers that it reads. The file should contain
 * nothing but numbers of type double; if this is not the case, the
 * output will be the sum and average of however many numbers were
 * successfully read from the file. The name of the file will be
 * input by the user.
 */
public class ReadNumbersFromFile {

    public static void main(String[] args) {

        while (true) {
            String fileName; // The name of the file, to be input by the user.
            TextIO.put("Enter the name of the file: ");
            fileName = TextIO.getln();
            try {
                TextIO.readFile( fileName ); // Try to open the file for input.
                break; // If that succeeds, break out of the loop.
            }
            catch ( IllegalArgumentException e ) {
                TextIO.putln("Can't read from the file \"" + fileName + "\".");
                TextIO.putln("Please try again.\n");
            }
        }

        // At this point, TextIO is reading from the file.

        double number; // A number read from the data file.
        double sum;     // The sum of all the numbers read so far.
        int count;      // The number of numbers that were read.

        sum = 0;
```

```

count = 0;
try {
    while (true) { // Loop ends when an exception occurs.
        number = TextIO.getDouble();
        count++; // This is skipped when the exception occurs
        sum += number;
    }
}
catch ( IllegalArgumentException e ) {
    // We expect this to occur when the end-of-file is encountered.
    // We don't consider this to be an error, so there is nothing to do
    // in this catch clause. Just proceed with the rest of the program.
}

// At this point, we've read the entire file.

TextIO.putln();
TextIO.putln("Number of data values read: " + count);
TextIO.putln("The sum of the data values: " + sum);
if ( count == 0 )
    TextIO.putln("Can't compute an average of 0 values.");
else
    TextIO.putln("The average of the values: " + (sum/count));
}
}

```

3.8 Introduction to GUI Programming

FOR THE PAST TWO CHAPTERS, you've been learning the sort of programming that is done inside a single subroutine. In the rest of the text, we'll be more concerned with the larger scale structure of programs, but the material that you've already learned will be an important foundation for everything to come.

In this section, before moving on to programming-in-the-large, we'll take a look at how programming-in-the-small can be used in other contexts besides text-based, command-line-style programs. We'll do this by taking a short, introductory look at applets and graphical programming.

An *applet* is a Java program that runs on a Web page. An applet is not a stand-alone application, and it does not have a `main()` routine. In fact, an applet is an **object** rather than a **class**. When Java first appeared on the scene, applets were one of its major appeals. Since then, they have become less important, although they can still be very useful. When we study GUI programming in Chapter 6, we will concentrate on stand-alone GUI programs rather than on applets, but applets are a good place to start for our first look at the subject.

When an applet is placed on a Web page, it is assigned a rectangular area on the page. It is the job of the applet to draw the contents of that rectangle. When the region needs to be drawn, the Web page calls a subroutine in the applet to do so. This is not so different from what happens with stand-alone programs. When such a program needs to be run, the system calls the `main()` routine of the program. Similarly, when an applet needs to be drawn, the Web page calls the `paint()` routine of the applet. The programmer specifies what happens when these routines are called by filling in the bodies of the routines. Programming in the

small! Applets can do other things besides draw themselves, such as responding when the user clicks the mouse on the applet. Each of the applet's behaviors is defined by a subroutine. The programmer specifies how the applet behaves by filling in the bodies of the appropriate subroutines.

A very simple applet, which does nothing but draw itself, can be defined by a class that contains nothing but a `paint()` routine. The source code for the class would then have the form:

```
import java.awt.*;
import java.applet.*;

public class <name-of-applet> extends Applet {
    public void paint(Graphics g) {
        <statements>
    }
}
```

where `<name-of-applet>` is an identifier that names the class, and the `<statements>` are the code that actually draws the applet. This looks similar to the definition of a stand-alone program, but there are a few things here that need to be explained, starting with the first two lines.

When you write a program, there are certain built-in classes that are available for you to use. These built-in classes include *System* and *Math*. If you want to use one of these classes, you don't have to do anything special. You just go ahead and use it. But Java also has a large number of standard classes that are there if you want them but that are not automatically available to your program. (There are just too many of them.) If you want to use these classes in your program, you have to ask for them first. The standard classes are grouped into so-called "packages." Two of these packages are called "java.awt" and "java.applet". The directive "import java.awt.*;" makes all the classes from the package java.awt available for use in your program. The java.awt package contains classes related to graphical user interface programming, including a class called *Graphics*. The *Graphics* class is referred to in the `paint()` routine above. The java.applet package contains classes specifically related to applets, including the class named *Applet*.

The first line of the class definition above says that the class "extends *Applet*." *Applet* is a standard class that is defined in the java.applet package. It defines all the basic properties and behaviors of applet objects. By extending the *Applet* class, the new class we are defining inherits all those properties and behaviors. We only have to define the ways in which our class differs from the basic *Applet* class. In our case, the only difference is that our applet will draw itself differently, so we only have to define the `paint()` routine that does the drawing. This is one of the main advantages of object-oriented programming.

(Actually, in the future, our applets will be defined to extend *JApplet* rather than *Applet*. The *JApplet* class is itself an extension of *Applet*. The *Applet* class has existed since the original version of Java, while *JApplet* is part of the newer "Swing" set of graphical user interface components. For the moment, the distinction is not important.)

One more thing needs to be mentioned—and this is a point where Java's syntax gets unfortunately confusing. Applets are objects, not classes. Instead of being static members of a class, the subroutines that define the applet's behavior are part of the applet object. We say that they are "non-static" subroutines. Of course, objects are related to classes because every object is described by a class. Now here is the part that can get confusing: Even though a non-static subroutine is not actually part of a class (in the sense of being part of the behavior

of the class), it is nevertheless defined in a class (in the sense that the Java code that defines the subroutine is part of the Java code that defines the class). Many objects can be described by the same class. Each object has its own non-static subroutine. But the common definition of those subroutines—the actual Java source code—is physically part of the class that describes all the objects. To put it briefly: static subroutines in a class definition say what the class does; non-static subroutines say what all the objects described by the class do. An applet’s `paint()` routine is an example of a non-static subroutine. A stand-alone program’s `main()` routine is an example of a static subroutine. The distinction doesn’t really matter too much at this point: When working with stand-alone programs, mark everything with the reserved word, “`static`”; leave it out when working with applets. However, the distinction between static and non-static will become more important later in the course.

* * * 2

Let’s write an applet that draws something. In order to write an applet that draws something, you need to know what subroutines are available for drawing, just as in writing text-oriented programs you need to know what subroutines are available for reading and writing text. In Java, the built-in drawing subroutines are found in objects of the class `Graphics`, one of the classes in the `java.awt` package. In an applet’s `paint()` routine, you can use the `Graphics` object `g` for drawing. (This object is provided as a parameter to the `paint()` routine when that routine is called.) `Graphics` objects contain many subroutines. I’ll mention just three of them here. You’ll encounter more of them in Chapter 6.

- `g.setColor(c)`, is called to set the color that is used for drawing. The parameter, `c` is an object belonging to a class named `Color`, another one of the classes in the `java.awt` package. About a dozen standard colors are available as static member variables in the `Color` class. These standard colors include `Color.BLACK`, `Color.WHITE`, `Color.RED`, `Color.GREEN`, and `Color.BLUE`. For example, if you want to draw in red, you would say “`g.setColor(Color.RED);`”. The specified color is used for all subsequent drawing operations up until the next time `setColor` is called.
- `g.drawRect(x,y,w,h)` draws the outline of a rectangle. The parameters `x`, `y`, `w`, and `h` must be integer-valued expressions. This subroutine draws the outline of the rectangle whose top-left corner is `x` pixels from the left edge of the applet and `y` pixels down from the top of the applet. The width of the rectangle is `w` pixels, and the height is `h` pixels.
- `g.fillRect(x,y,w,h)` is similar to `drawRect` except that it fills in the inside of the rectangle instead of just drawing an outline.

This is enough information to write an applet that will draw the following image on a Web page:



The applet first fills its entire rectangular area with red. Then it changes the drawing color to black and draws a sequence of rectangles, where each rectangle is nested inside the previous one. The rectangles can be drawn with a `while` loop. Each time through the loop, the rectangle gets smaller and it moves down and over a bit. We'll need variables to hold the width and height of the rectangle and a variable to record how far the top-left corner of the rectangle is inset from the edges of the applet. The while loop ends when the rectangle shrinks to nothing. In general outline, the algorithm for drawing the applet is

```
Set the drawing color to red (using the g.setColor subroutine)
Fill in the entire applet (using the g.fillRect subroutine)
Set the drawing color to black
Set the top-left corner inset to be 0
Set the rectangle width and height to be as big as the applet
while the width and height are greater than zero:
    draw a rectangle (using the g.drawRect subroutine)
    increase the inset
    decrease the width and the height
```

In my applet, each rectangle is 15 pixels away from the rectangle that surrounds it, so the `inset` is increased by 15 each time through the `while` loop. The rectangle shrinks by 15 pixels on the left **and** by 15 pixels on the right, so the width of the rectangle shrinks by 30 each time through the loop. The height also shrinks by 30 pixels each time through the loop.

It is not hard to code this algorithm into Java and use it to define the `paint()` method of an applet. I've assumed that the applet has a height of 160 pixels and a width of 300 pixels. The size is actually set in the source code of the Web page where the applet appears. In order for an applet to appear on a page, the source code for the page must include a command that specifies which applet to run and how big it should be. (We'll see how to do that later.) It's not a great idea to assume that we know how big the applet is going to be. On the other hand, it's also not a great idea to write an applet that does nothing but draw a static picture. I'll address both these issues before the end of this section. But for now, here is the source code for the applet:

```
import java.awt.*;
import java.applet.Applet;

public class StaticRects extends Applet {
    public void paint(Graphics g) {
        // Draw a set of nested black rectangles on a red background.
        // Each nested rectangle is separated by 15 pixels on
        // all sides from the rectangle that encloses it.

        int inset;    // Gap between borders of applet
                     // and one of the rectangles.

        int rectWidth, rectHeight; // The size of one of the rectangles.

        g.setColor(Color.red);
        g.fillRect(0,0,300,160); // Fill the entire applet with red.

        g.setColor(Color.black); // Draw the rectangles in black.

        inset = 0;

        rectWidth = 299;    // Set size of first rect to size of applet.
```



```

    rectHeight = 159;
    while (rectWidth >= 0 && rectHeight >= 0) {
        g.drawRect(inset, inset, rectWidth, rectHeight);
        inset += 15;      // Rects are 15 pixels apart.
        rectWidth -= 30;   // Width decreases by 15 pixels
                           //           on left and 15 on right.
        rectHeight -= 30;  // Height decreases by 15 pixels
                           //           on top and 15 on bottom.
    }
} // end paint()
} // end class StaticRects

```

(You might wonder why the initial `rectWidth` is set to 299, instead of to 300, since the width of the applet is 300 pixels. It's because rectangles are drawn as if with a pen whose nib hangs below and to the right of the point where the pen is placed. If you run the pen exactly along the right edge of the applet, the line it draws is actually outside the applet and therefore is not seen. So instead, we run the pen along a line one pixel to the left of the edge of the applet. The same reasoning applies to `rectHeight`. Careful graphics programming demands attention to details like these.)

* * * 3

When you write an applet, you get to build on the work of the people who wrote the `Applet` class. The `Applet` class provides a framework on which you can hang your own work. Any programmer can create additional frameworks that can be used by other programmers as a basis for writing specific types of applets or stand-alone programs. I've written a small framework that makes it possible to write applets that display simple animations. One example that we will consider is an animated version of the nested rectangles applet from earlier in this section. You can see the applet in action at the bottom of the on-line version of this page.

A **computer animation** is really just a sequence of still images. The computer displays the images one after the other. Each image differs a bit from the preceding image in the sequence. If the differences are not too big and if the sequence is displayed quickly enough, the eye is tricked into perceiving continuous motion.

In the example, rectangles shrink continually towards the center of the applet, while new rectangles appear at the edge. The perpetual motion is, of course, an illusion. If you think about it, you'll see that the applet loops through the same set of images over and over. In each image, there is a gap between the borders of the applet and the outermost rectangle. This gap gets wider and wider until a new rectangle appears at the border. Only it's not a new rectangle. What has really happened is that the applet has started over again with the first image in the sequence.

The problem of creating an animation is really just the problem of drawing each of the still images that make up the animation. Each still image is called a **frame**. In my framework for animation, which is based on a non-standard class called `SimpleAnimationApplet2`, all you have to do is fill in the code that says how to draw one frame. The basic format is as follows:

```

import java.awt.*; 8

public class <name-of-class> extends SimpleAnimationApplet2 { 9

    public void drawFrame(Graphics g) {
        <statements> // to draw one frame of the animation
    }
}

```



```
    }
}
```

The “`import java.awt.*;`” is required to get access to graphics-related classes such as `Graphics` and `Color`. You get to fill in any name you want for the class, and you get to fill in the statements inside the subroutine. The `drawFrame()` subroutine will be called by the system each time a frame needs to be drawn. All you have to do is say what happens when this subroutine is called. Of course, you have to draw a different picture for each frame, and to do that you need to know which frame you are drawing. The class `SimpleAnimationApplet2` provides a function named `getFrameNumber()` that you can call to find out which frame to draw. This function returns an integer value that represents the frame number. If the value returned is 0, you are supposed to draw the first frame; if the value is 1, you are supposed to draw the second frame, and so on. 1

In the sample applet, the thing that differs from one frame to another is the distance between the edges of the applet and the outermost rectangle. Since the rectangles are 15 pixels apart, this distance increases from 0 to 14 and then jumps back to 0 when a “new” rectangle appears. The appropriate value can be computed very simply from the frame number, with the statement “`inset = getFrameNumber() % 15;`”. The value of the expression `getFrameNumber() % 15` is between 0 and 14. When the frame number reaches 15 or any multiple of 15, the value of `getFrameNumber() % 15` jumps back to 0. 2

Drawing one frame in the sample animated applet is very similar to drawing the single image of the `StaticRects` applet, as given above. The `paint()` method in the `StaticRects` applet becomes, with only minor modification, the `drawFrame()` method of my `MovingRects` animation applet. I’ve chosen to make one improvement: The `StaticRects` applet assumes that the applet is 300 by 160 pixels. The `MovingRects` applet will work for any applet size. To implement this, the `drawFrame()` routine has to know how big the applet is. There are two functions that can be called to get this information. The function `getWidth()` returns an integer value representing the width of the applet, and the function `getHeight()` returns the height. The width and height, together with the frame number, are used to compute the size of the first rectangle that is drawn. Here is the complete source code: 3

```
import java.awt.*; 4
```

```
public class MovingRects extends SimpleAnimationApplet2 { 5
```

```
    public void drawFrame(Graphics g) { 6
```

```
        // Draw one frame in the animation by filling in the background 7
        // with a solid red and then drawing a set of nested black
        // rectangles. The frame number tells how much the first
        // rectangle is to be inset from the borders of the applet.
```

```
        int width;    // Width of the applet, in pixels.
        int height;   // Height of the applet, in pixels.
```

```
        int inset;    // Gap between borders of applet and a rectangle.
                     // The inset for the outermost rectangle goes
                     // from 0 to 14 then back to 0, and so on,
                     // as the frameNumber varies.
```

```
        int rectWidth, rectHeight; // The size of one of the rectangles.
```

```
        width = getWidth();        // Find out the size of the drawing area.
```

```

height = getHeight();

g.setColor(Color.red);          // Fill the frame with red.
g.fillRect(0,0,width,height);

g.setColor(Color.black);        // Switch color to black.

inset = getFrameNumber() % 15;   // Get the inset for the
                                //          outermost rect.

rectWidth = width - 2*inset - 1; // Set size of outermost rect.
rectHeight = height - 2*inset - 1;

while (rectWidth >= 0 && rectHeight >= 0) {
    g.drawRect(inset,inset,rectWidth,rectHeight);
    inset += 15;                // Rects are 15 pixels apart.
    rectWidth -= 30;             // Width decreases by 15 pixels
                                //          on left and 15 on right.
    rectHeight -= 30;           // Height decreases by 15 pixels
                                //          on top and 15 on bottom.
}
} // end drawFrame()
} // end class MovingRects

```

The main point here is that by building on an existing framework, you can do interesting things using the type of local, inside-a-subroutine programming that was covered in Chapter 2 and Chapter 3. As you learn more about programming and more about Java, you'll be able to do more on your own—but no matter how much you learn, you'll always be dependent on other people's work to some extent.

Exercises for Chapter 3¹

1. How many times do you have to roll a pair of dice before they come up snake eyes? You² could do the experiment by rolling the dice by hand. Write a computer program that simulates the experiment. The program should report the number of rolls that it makes before the dice come up snake eyes. (Note: “Snake eyes” means that both dice show a value of 1.) Exercise 2.2 explained how to simulate rolling a pair of dice.

2. Which integer between 1 and 10000 has the largest number of divisors, and how many³ divisors does it have? Write a program to find the answers and print out the results. It is possible that several integers in this range have the same, maximum number of divisors. Your program only has to print out one of them. Subsection 3.4.2 discussed divisors. The source code for that example is *CountDivisors.java*.

You might need some hints about how to find a maximum value. The basic idea is⁴ to go through all the integers, keeping track of the largest number of divisors that you’ve seen *so far*. Also, keep track of the integer that had that number of divisors.

3. Write a program that will evaluate simple expressions such as $17 + 3$ and $3.14159 * 4.7$.⁵ The expressions are to be typed in by the user. The input always consist of a number, followed by an operator, followed by another number. The operators that are allowed are $+$, $-$, $*$, and $/$. You can read the numbers with `TextIO.getDouble()` and the operator with `TextIO.getChar()`. Your program should read an expression, print its value, read another expression, print its value, and so on. The program should end when the user enters 0 as the first number on the line.

4. Write a program that reads one line of input text and breaks it up into words. The⁶ words should be output one per line. A word is defined to be a sequence of letters. Any characters in the input that are not letters should be discarded. For example, if the user inputs the line

He said, "That's not a good idea."⁷

then the output of the program should be⁸

```
He
said
that
s
not
a
good
idea9
```

An improved version of the program would list “that’s” as a single word. An apostrophe¹⁰ can be considered to be part of a word if there is a letter on each side of the apostrophe.

To test whether a character is a letter, you might use `(ch >= 'a' && ch <= 'z') ||11` `(ch >= 'A' && ch <= 'Z')`. However, this only works in English and similar languages. A better choice is to call the standard function `Character.isLetter(ch)`, which returns a boolean value of `true` if `ch` is a letter and `false` if it is not. This works for any Unicode character.

5. Suppose that a file contains information about sales figures for a company in various cities. Each line of the file contains a city name, followed by a colon (:) followed by the data for that city. The data is a number of type **double**. However, for some cities, no data was available. In these lines, the data is replaced by a comment explaining why the data is missing. For example, several lines from the file might look like:

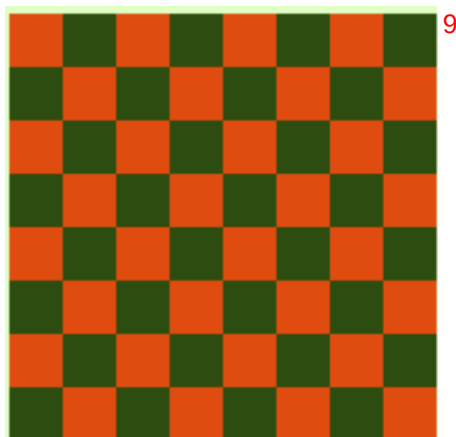
```
San Francisco: 19887.32
Chicago: no report received
New York: 298734.12
```

Write a program that will compute and print the total sales from all the cities together. The program should also report the number of cities for which data was not available. The name of the file is “sales.dat”.

To complete this program, you’ll need one fact about file input with *TextIO* that was not covered in Subsection 2.4.5. Since you don’t know in advance how many lines there are in the file, you need a way to tell when you have gotten to the end of the file. When *TextIO* is reading from a file, the function `TextIO.eof()` can be used to test for **end of file**. This **boolean**-valued function returns **true** if the file has been entirely read and returns **false** if there is more data to read in the file. This means that you can read the lines of the file in a loop `while (TextIO.eof() == false)...`. The loop will end when all the lines of the file have been read.

Suggestion: For each line, read and ignore characters up to the colon. Then read the rest of the line into a variable of type *String*. Try to convert the string into a number, and use `try..catch` to test whether the conversion succeeds.

6. Write an applet that draws a checkerboard. Assume that the size of the applet is 160 by 160 pixels. Each square in the checkerboard is 20 by 20 pixels. The checkerboard contains 8 rows of squares and 8 columns. The squares are red and black. Here is a tricky way to determine whether a given square is red or black: If the row number and the column number are either both even or both odd, then the square is red. Otherwise, it is black. Note that a square is just a rectangle in which the height is equal to the width, so you can use the subroutine `g.fillRect()` to draw the squares. Here is an image of the checkerboard:



(To run an applet, you need a Web page to display it. A very simple page will do. Assume that your applet class is called `Checkerboard`, so that when you compile it you get a class file named `Checkerboard.class`. Make a file that contains only the lines:

```
<applet code="Checkerboard.class" width=160 height=160> 1
</applet>
```

Call this file `Checkerboard.html`. This is the source code for a simple Web page that 2 shows nothing but your applet. You can open the file in a Web browser or with Sun's appletviewer program. The compiled class file, `Checkerboard.class`, must be in the same directory with the Web-page file, `Checkerboard.html`.

(If you are using the Eclipse Integrated Development Environment, you can simply 3 right-click the name of the source code file in the Package Explorer. In the pop-up menu, go to "Run As" then to "Java Applet". This will open the window in which the applet appears. The default size for the window is bigger than 160-by-160, so the drawing of the checkerboard will not fill the entire window.)

7. Write an animation applet that shows a checkerboard pattern in which the even numbered 4 rows slide to the left while the odd numbered rows slide to the right. You can assume that the applet is 160 by 160 pixels. Each row should be offset from its usual position by the amount `getFrameNumber() % 40`. Hints: Anything you draw outside the boundaries of the applet will be invisible, so you can draw more than 8 squares in a row. You can use negative values of `x` in `g.fillRect(x,y,w,h)`. (Before trying to do this exercise, it would be a good idea to look at a working applet, which can be found in the on-line version of this book.)

Your applet will extend the non-standard class, *SimpleAnimationApplet2*, which was 5 introduced in Section 3.8. The compiled class files, `SimpleAnimationApplet2.class` and `SimpleAnimationApplet2$1.class`, must be in the same directory as your Web-page source file along with the compiled class file for your own class. These files are produced when you compile *SimpleAnimationApplet2.java*. Assuming that the name of your class is `SlidingCheckerboard`, then the source file for the Web page should contain the lines:

```
<applet code="SlidingCheckerboard.class" width=160 height=160> 6
</applet>
```

Quiz on Chapter 3¹

1. What is an *algorithm*?²

2. Explain briefly what is meant by “pseudocode” and how is it useful in the development of algorithms.³

3. What is a *block statement*? How are block statements used in Java programs?⁴

4. What is the main difference between a *while* loop and a *do..while* loop?⁵

5. What does it mean to *prime* a loop?⁶

6. Explain what is meant by an *animation* and how a computer displays an animation.⁷

7. Write a *for* loop that will print out all the multiples of 3 from 3 to 36, that is: 3 6 9 12 15 18 21 24 27 30 33 36.⁸

8. Fill in the following *main()* routine so that it will ask the user to enter an integer, read the user’s response, and tell the user whether the number entered is even or odd. (You can use `TextIO.getInt()` to read the integer. Recall that an integer *n* is even if `n % 2 == 0`.)⁹

```
public static void main(String[] args) {
    // Fill in the body of this subroutine!
```

```
}10
```

9. Suppose that *s1* and *s2* are variables of type *String*, whose values are expected to be string representations of values of type *int*. Write a code segment that will compute and print the integer sum of those values, or will print an error message if the values cannot successfully be converted into integers. (Use a *try..catch* statement.)¹¹

10. Show the exact output that would be produced by the following *main()* routine:¹²

```
public static void main(String[] args) {13
    int N;
    N = 1;
    while (N <= 32) {
        N = 2 * N;
        System.out.println(N);
    }
}
```

11. Show the exact output produced by the following *main()* routine:¹⁴

```
public static void main(String[] args) {15
    int x,y;
    x = 5;
    y = 1;
    while (x > 0) {
        x = x - 1;
        y = y * x;
        System.out.println(y);
    }
}
```

12. What output is produced by the following program segment? **Why?** (Recall that `name.charAt(i)` is the i-th character in the string, `name`.)

```
String name;  
int i;  
boolean startWord;  
  
name = "Richard M. Nixon";  
startWord = true;  
for (i = 0; i < name.length(); i++) {  
    if (startWord)  
        System.out.println(name.charAt(i));  
    if (name.charAt(i) == ' ')  
        startWord = true;  
    else  
        startWord = false;  
}
```

3

Chapter 4

Programming in the Large I: Subroutines¹

ONE WAY TO BREAK UP A COMPLEX PROGRAM into manageable pieces is to use *subroutines*.² A subroutine consists of the instructions for carrying out a certain task, grouped together and given a name. Elsewhere in the program, that name can be used as a stand-in for the whole set of instructions. As a computer executes a program, whenever it encounters a subroutine name, it executes all the instructions necessary to carry out the task associated with that subroutine.

Subroutines can be used over and over, at different places in the program. A subroutine³ can even be used inside another subroutine. This allows you to write simple subroutines and then use them to help write more complex subroutines, which can then be used in turn in other subroutines. In this way, very complex programs can be built up step-by-step, where each step in the construction is reasonably simple.

As mentioned in Section 3.8, subroutines in Java can be either static or non-static. This⁴ chapter covers static subroutines only. Non-static subroutines, which are used in true object-oriented programming, will be covered in the next chapter.

4.1 Black Boxes⁵

A SUBROUTINE CONSISTS OF INSTRUCTIONS for performing some task, chunked together and given a name. “Chunking” allows you to deal with a potentially very complicated task as a single concept. Instead of worrying about the many, many steps that the computer might have to go through to perform that task, you just need to remember the name of the subroutine. Whenever you want your program to perform the task, you just call the subroutine. Subroutines are a major tool for dealing with complexity.⁶

A subroutine is sometimes said to be a “black box” because you can’t see what’s “inside”⁷ it (or, to be more precise, you usually don’t **want** to see inside it, because then you would have to deal with all the complexity that the subroutine is meant to hide). Of course, a black box that has no way of interacting with the rest of the world would be pretty useless. A black box needs some kind of *interface* with the rest of the world, which allows some interaction between what’s inside the box and what’s outside. A physical black box might have buttons on the outside that you can push, dials that you can set, and slots that can be used for passing information back and forth. Since we are trying to hide complexity, not create it, we have the first rule of black boxes:

The interface of a black box should be fairly straightforward, well-defined, and easy to understand. 1

Are there any examples of black boxes in the real world? Yes; in fact, you are surrounded by them. Your television, your car, your VCR, your refrigerator. . . . You can turn your television on and off, change channels, and set the volume by using elements of the television's interface—dials, remote control, don't forget to plug in the power—without understanding anything about how the thing actually works. The same goes for a VCR, although if the stories are true about how hard people find it to set the time on a VCR, then maybe the VCR violates the simple interface rule. 2

Now, a black box does have an inside—the code in a subroutine that actually performs the task, all the electronics inside your television set. The inside of a black box is called its *implementation*. The second rule of black boxes is that: 3

To use a black box, you shouldn't need to know anything about its implementation; all you need to know is its interface. 4

In fact, it should be possible to **change** the implementation, as long as the behavior of the box, as seen from the outside, remains unchanged. For example, when the insides of TV sets went from using vacuum tubes to using transistors, the users of the sets didn't even need to know about it—or even know what it means. Similarly, it should be possible to rewrite the inside of a subroutine, to use more efficient code, for example, without affecting the programs that use that subroutine. 5

Of course, to have a black box, someone must have designed and built the implementation in the first place. The black box idea works to the advantage of the implementor as well as of the user of the black box. After all, the black box might be used in an unlimited number of different situations. The implementor of the black box doesn't need to know about any of that. The implementor just needs to make sure that the box performs its assigned task and interfaces correctly with the rest of the world. This is the third rule of black boxes: 6

The implementor of a black box should not need to know anything about the larger systems in which the box will be used. 7

In a way, a black box divides the world into two parts: the inside (implementation) and the outside. The interface is at the boundary, connecting those two parts. 8

* * * 9

By the way, you should **not** think of an interface as just the physical connection between the box and the rest of the world. The interface also includes a *specification* of what the box does and how it can be controlled by using the elements of the physical interface. It's not enough to say that a TV set has a power switch; you need to specify that the power switch is used to turn the TV on and off! 10

To put this in computer science terms, the interface of a subroutine has a semantic as well as a syntactic component. The syntactic part of the interface tells you just what you have to type in order to call the subroutine. The semantic component specifies exactly what task the subroutine will accomplish. To write a legal program, you need to know the syntactic specification of the subroutine. To understand the purpose of the subroutine and to use it effectively, you need to know the subroutine's semantic specification. I will refer to both parts of the interface—syntactic and semantic—collectively as the *contract* of the subroutine. 11

The contract of a subroutine says, essentially, “Here is what you have to do to use me, and here is what I will do for you, guaranteed.” When you write a subroutine, the comments that you write for the subroutine should make the contract very clear. (I should admit that in practice, subroutines’ contracts are often inadequately specified, much to the regret and annoyance of the programmers who have to use them.)

For the rest of this chapter, I turn from general ideas about black boxes and subroutines in general to the specifics of writing and using subroutines in Java. But keep the general ideas and principles in mind. They are the reasons that subroutines exist in the first place, and they are your guidelines for using them. This should be especially clear in Section 4.6, where I will discuss subroutines as a tool in program development.

* * * 3

You should keep in mind that subroutines are not the only example of black boxes in programming. For example, a class is also a black box. We’ll see that a class can have a “public” part, representing its interface, and a “private” part that is entirely inside its hidden implementation. All the principles of black boxes apply to classes as well as to subroutines.

4.2 Static Subroutines and Static Variables 5

EVERY SUBROUTINE IN JAVA must be defined inside some class. This makes Java rather unusual among programming languages, since most languages allow free-floating, independent subroutines. One purpose of a class is to group together related subroutines and variables. Perhaps the designers of Java felt that everything must be related to something. As a less philosophical motivation, Java’s designers wanted to place firm controls on the ways things are named, since a Java program potentially has access to a huge number of subroutines created by many different programmers. The fact that those subroutines are grouped into named classes (and classes are grouped into named “packages”) helps control the confusion that might result from so many different names.

A subroutine that is a member of a class is often called a *method*, and “method” is the term that most people prefer for subroutines in Java. I will start using the term “method” occasionally; however, I will continue to prefer the more general term “subroutine” for static subroutines. I will use the term “method” most often to refer to non-static subroutines, which belong to objects rather than to classes. This chapter will deal with static subroutines almost exclusively. We’ll turn to non-static methods and object-oriented programming in the next chapter.

4.2.1 Subroutine Definitions 8

A subroutine definition in Java takes the form: 9

```
<modifiers> <return-type> <subroutine-name> ( <parameter-list> ) { 10
    <statements>
}
```

It will take us a while—most of the chapter—to get through what all this means in detail. Of course, you’ve already seen examples of subroutines in previous chapters, such as the `main()` routine of a program and the `paint()` routine of an applet. So you are familiar with the general format.

The *<statements>* between the braces, { and }, in a subroutine definition make up the *body* of the subroutine. These statements are the inside, or implementation part, of the “black box”,

as discussed in the previous section. They are the instructions that the computer executes when the method is called. Subroutines can contain any of the statements discussed in Chapter 2 and Chapter 3. 1

The *modifiers* that can occur at the beginning of a subroutine definition are words that set certain characteristics of the subroutine, such as whether it is static or not. The modifiers that you’ve seen so far are “**static**” and “**public**”. There are only about a half-dozen possible modifiers altogether. 2

If the subroutine is a function, whose job is to compute some value, then the *return-type* is used to specify the type of value that is returned by the function. We’ll be looking at functions and return types in some detail in Section 4.4. If the subroutine is not a function, then the *return-type* is replaced by the special value **void**, which indicates that no value is returned. The term “void” is meant to indicate that the return value is empty or non-existent. 3

Finally, we come to the *parameter-list* of the method. Parameters are part of the interface of a subroutine. They represent information that is passed into the subroutine from outside, to be used by the subroutine’s internal computations. For a concrete example, imagine a class named *Television* that includes a method named **changeChannel()**. The immediate question is: What channel should it change to? A parameter can be used to answer this question. Since the channel number is an integer, the type of the parameter would be **int**, and the declaration of the **changeChannel()** method might look like 4

```
public void changeChannel(int channelNum) { ... } 5
```

This declaration specifies that **changeChannel()** has a parameter named **channelNum** of type **int**. However, **channelNum** does not yet have any particular value. A value for **channelNum** is provided when the subroutine is called; for example: **changeChannel(17)**; 6

The parameter list in a subroutine can be empty, or it can consist of one or more parameter declarations of the form *type* *parameter-name*. If there are several declarations, they are separated by commas. Note that each declaration can name only one parameter. For example, if you want two parameters of type **double**, you have to say “**double x, double y**”, rather than “**double x, y**”. 7

Parameters are covered in more detail in the next section. 8

Here are a few examples of subroutine definitions, leaving out the statements that define what the subroutines do: 9

```
public static void playGame() { 10
    // "public" and "static" are modifiers; "void" is the
    // return-type; "playGame" is the subroutine-name;
    // the parameter-list is empty.
    . . . // Statements that define what playGame does go here.
}

int getNextN(int N) {
    // There are no modifiers; "int" in the return-type
    // "getNextN" is the subroutine-name; the parameter-list
    // includes one parameter whose name is "N" and whose
    // type is "int".
    . . . // Statements that define what getNextN does go here.
}

static boolean lessThan(double x, double y) {
    // "static" is a modifier; "boolean" is the
    // return-type; "lessThan" is the subroutine-name; the
```

```
// parameter-list includes two parameters whose names are
// "x" and "y", and the type of each of these parameters
// is "double".
. . . // Statements that define what lessThan does go here.
}
```

In the second example given here, `getNextN` is a non-static method, since its definition does not include the modifier “`static`”—and so it’s not an example that we should be looking at in this chapter! The other modifier shown in the examples is “`public`”. This modifier indicates that the method can be called from anywhere in a program, even from outside the class where the method is defined. There is another modifier, “`private`”, which indicates that the method can be called **only** from inside the same class. The modifiers `public` and `private` are called *access specifiers*. If no access specifier is given for a method, then by default, that method can be called from anywhere in the “package” that contains the class, but not from outside that package. (Packages were introduced in Subsection 2.6.4, and you’ll learn more about them later in this chapter, in Section 4.5.) There is one other access modifier, `protected`, which will only become relevant when we turn to object-oriented programming in Chapter 5.

Note, by the way, that the `main()` routine of a program follows the usual syntax rules for a subroutine. In

```
public static void main(String[] args) { ... }
```

the modifiers are `public` and `static`, the return type is `void`, the subroutine name is `main`, and the parameter list is “`String[] args`”. The only question might be about “`String[]`”, which has to be a type if it is to match the syntax of a parameter list. In fact, `String[]` represents a so-called “array type”, so the syntax is valid. We will cover arrays in Chapter 7. (The parameter, `args`, represents information provided to the program when the `main()` routine is called by the system. In case you know the term, the information consists of any “command-line arguments” specified in the command that the user typed to run the program.)

You’ve already had some experience with filling in the implementation of a subroutine. In this chapter, you’ll learn all about writing your own complete subroutine definitions, including the interface part.

4.2.2 Calling Subroutines

When you define a subroutine, all you are doing is telling the computer that the subroutine exists and what it does. The subroutine doesn’t actually get executed until it is called. (This is true even for the `main()` routine in a class—even though **you** don’t call it, it is called by the system when the system runs your program.) For example, the `playGame()` method given as an example above could be called using the following subroutine call statement:

```
playGame();
```

This statement could occur anywhere in the same class that includes the definition of `playGame()`, whether in a `main()` method or in some other subroutine. Since `playGame()` is a `public` method, it can also be called from other classes, but in that case, you have to tell the computer which class it comes from. Since `playGame()` is a `static` method, its full name includes the name of the class in which it is defined. Let’s say, for example, that `playGame()` is defined in a class named `Poker`. Then to call `playGame()` from **outside** the `Poker` class, you would have to say

```
Poker.playGame();
```

The use of the class name here tells the computer which class to look in to find the method. It also lets you distinguish between `Poker.playGame()` and other potential `playGame()` methods defined in other classes, such as `Roulette.playGame()` or `Blackjack.playGame()`.

More generally, a **subroutine call statement** for a static subroutine takes the form

```
<subroutine-name>(<parameters>);
```

if the subroutine that is being called is in the same class, or

```
<class-name>.<subroutine-name>(<parameters>);
```

if the subroutine is defined elsewhere, in a different class. (Non-static methods belong to objects rather than classes, and they are called using object names instead of class names. More on that later.) Note that the parameter list can be empty, as in the `playGame()` example, but the parentheses must be there even if there is nothing between them.

4.2.3 Subroutines in Programs

It's time to give an example of what a complete program looks like, when it includes other subroutines in addition to the `main()` routine. Let's write a program that plays a guessing game with the user. The computer will choose a random number between 1 and 100, and the user will try to guess it. The computer tells the user whether the guess is high or low or correct. If the user gets the number after six guesses or fewer, the user wins the game. After each game, the user has the option of continuing with another game.

Since playing one game can be thought of as a single, coherent task, it makes sense to write a subroutine that will play one guessing game with the user. The `main()` routine will use a loop to call the `playGame()` subroutine over and over, as many times as the user wants to play. We approach the problem of designing the `playGame()` subroutine the same way we write a `main()` routine: Start with an outline of the algorithm and apply stepwise refinement. Here is a short pseudocode algorithm for a guessing game program:

```
Pick a random number
while the game is not over:
    Get the user's guess
    Tell the user whether the guess is high, low, or correct.
```

The test for whether the game is over is complicated, since the game ends if either the user makes a correct guess or the number of guesses is six. As in many cases, the easiest thing to do is to use a “`while (true)`” loop and use `break` to end the loop whenever we find a reason to do so. Also, if we are going to end the game after six guesses, we'll have to keep track of the number of guesses that the user has made. Filling out the algorithm gives:

```
Let computersNumber be a random number between 1 and 100
Let guessCount = 0
while (true):
    Get the user's guess
    Count the guess by adding 1 to guess count
    if the user's guess equals computersNumber:
        Tell the user he won
        break out of the loop
    if the number of guesses is 6:
        Tell the user he lost
        break out of the loop
    if the user's guess is less than computersNumber:
```



```

    Tell the user the guess was low
else if the user's guess is higher than computersNumber:
    Tell the user the guess was high

```

1

With variable declarations added and translated into Java, this becomes the definition of the `playGame()` routine. A random integer between 1 and 100 can be computed as `(int)(100 * Math.random()) + 1`. I've cleaned up the interaction with the user to make it flow better.

```

static void playGame() {
    int computersNumber; // A random number picked by the computer.
    int usersGuess;      // A number entered by user as a guess.
    int guessCount;      // Number of guesses the user has made.
    computersNumber = (int)(100 * Math.random()) + 1;
        // The value assigned to computersNumber is a randomly
        // chosen integer between 1 and 100, inclusive.
    guessCount = 0;
    TextIO.putln();
    TextIO.put("What is your first guess? ");
    while (true) {
        usersGuess = TextIO.getInt(); // Get the user's guess.
        guessCount++;
        if (usersGuess == computersNumber) {
            TextIO.putln("You got it in " + guessCount
                + " guesses! My number was " + computersNumber);
            break; // The game is over; the user has won.
        }
        if (guessCount == 6) {
            TextIO.putln("You didn't get the number in 6 guesses.");
            TextIO.putln("You lose. My number was " + computersNumber);
            break; // The game is over; the user has lost.
        }
        // If we get to this point, the game continues.
        // Tell the user if the guess was too high or too low.
        if (usersGuess < computersNumber)
            TextIO.put("That's too low. Try again: ");
        else if (usersGuess > computersNumber)
            TextIO.put("That's too high. Try again: ");
    }
    TextIO.putln();
} // end of playGame()

```

3

Now, where exactly should you put this? It should be part of the same class as the `main()` routine, but **not** inside the `main` routine. It is not legal to have one subroutine physically nested inside another. The `main()` routine will **call** `playGame()`, but not contain it physically. You can put the definition of `playGame()` either before or after the `main()` routine. Java is not very picky about having the members of a class in any particular order.

4

It's pretty easy to write the `main` routine. You've done things like this before. Here's what the complete program looks like (except that a serious program needs more comments than I've included here).

5

```

public class GuessingGame {

    public static void main(String[] args) {
        TextIO.putln("Let's play a game. I'll pick a number between");
        TextIO.putln("1 and 100, and you try to guess it.");
    }
}

```

6


```

boolean playAgain;
do {
    playGame(); // call subroutine to play one game
    TextIO.put("Would you like to play again? ");
    playAgain = TextIO.getlnBoolean();
} while (playAgain);
TextIO.putln("Thanks for playing. Goodbye.");
} // end of main()

static void playGame() {
    int computersNumber; // A random number picked by the computer.
    int usersGuess;      // A number entered by user as a guess.
    int guessCount;      // Number of guesses the user has made.
    computersNumber = (int)(100 * Math.random()) + 1;
    // The value assigned to computersNumber is a randomly
    // chosen integer between 1 and 100, inclusive.
    guessCount = 0;
    TextIO.putln();
    TextIO.put("What is your first guess? ");
    while (true) {
        usersGuess = TextIO.getInt(); // Get the user's guess.
        guessCount++;
        if (usersGuess == computersNumber) {
            TextIO.putln("You got it in " + guessCount
                + " guesses! My number was " + computersNumber);
            break; // The game is over; the user has won.
        }
        if (guessCount == 6) {
            TextIO.putln("You didn't get the number in 6 guesses.");
            TextIO.putln("You lose. My number was " + computersNumber);
            break; // The game is over; the user has lost.
        }
        // If we get to this point, the game continues.
        // Tell the user if the guess was too high or too low.
        if (usersGuess < computersNumber)
            TextIO.put("That's too low. Try again: ");
        else if (usersGuess > computersNumber)
            TextIO.put("That's too high. Try again: ");
    }
    TextIO.putln();
} // end of playGame()
} // end of class GuessingGame

```

Take some time to read the program carefully and figure out how it works. And try to convince yourself that even in this relatively simple case, breaking up the program into two methods makes the program easier to understand and probably made it easier to write each piece.

4.2.4 Member Variables ⁴

A class can include other things besides subroutines. In particular, it can also include variable declarations. Of course, you can declare variables **inside** subroutines. Those are called *local variables*. However, you can also have variables that are not part of any subroutine. To

distinguish such variables from local variables, we call them *member variables*, since they are members of a class. 1

Just as with subroutines, member variables can be either static or non-static. In this chapter, we'll stick to static variables. A static member variable belongs to the class itself, and it exists as long as the class exists. Memory is allocated for the variable when the class is first loaded by the Java interpreter. Any assignment statement that assigns a value to the variable changes the content of that memory, no matter where that assignment statement is located in the program. Any time the variable is used in an expression, the value is fetched from that same memory, no matter where the expression is located in the program. This means that the value of a static member variable can be set in one subroutine and used in another subroutine. Static member variables are “shared” by all the static subroutines in the class. A local variable in a subroutine, on the other hand, exists only while that subroutine is being executed, and is completely inaccessible from outside that one subroutine. 2

The declaration of a member variable looks just like the declaration of a local variable except for two things: The member variable is declared outside any subroutine (although it still has to be inside a class), and the declaration can be marked with modifiers such as **static**, **public**, and **private**. Since we are only working with static member variables for now, every declaration of a member variable in this chapter will include the modifier **static**. They might also be marked as **public** or **private**. For example: 3

```
static String userName; 4
public static int numberOfPlayers; 5
private static double velocity, time; 6
```

A static member variable that is not declared to be **private** can be accessed from outside the class where it is defined, as well as inside. When it is used in some other class, it must be referred to with a compound identifier of the form `<class-name>.<variable-name>`. For example, the *System* class contains the public static member variable named `out`, and you use this variable in your own classes by referring to `System.out`. If `numberOfPlayers` is a public static member variable in a class named `Poker`, then subroutines in the `Poker` class would refer to it simply as `numberOfPlayers`, while subroutines in another class would refer to it as `Poker.numberOfPlayers`. 7

As an example, let's add a static member variable to the `GuessingGame` class that we wrote earlier in this section. This variable will be used to keep track of how many games the user wins. We'll call the variable `gamesWon` and declare it with the statement “`static int gamesWon;`”. In the `playGame()` routine, we add 1 to `gamesWon` if the user wins the game. At the end of the `main()` routine, we print out the value of `gamesWon`. It would be impossible to do the same thing with a local variable, since we need access to the same variable from both subroutines. 8

When you declare a local variable in a subroutine, you have to assign a value to that variable before you can do anything with it. Member variables, on the other hand are automatically initialized with a default value. For numeric variables, the default value is zero. For **boolean** variables, the default is **false**. And for **char** variables, it's the unprintable character that has Unicode code number zero. (For objects, such as `Strings`, the default initial value is a special value called `null`, which we won't encounter officially until later.) 9

Since it is of type **int**, the static member variable `gamesWon` automatically gets assigned an initial value of zero. This happens to be the correct initial value for a variable that is being used as a counter. You can, of course, assign a different value to the variable at the beginning of the `main()` routine if you are not satisfied with the default initial value. 10

Here's a revised version of `GuessingGame.java` that includes the `gamesWon` variable. The changes from the above version are shown in *italic*:

```
public class GuessingGame2 {
    static int gamesWon;        // The number of games won by
                                // the user.

    public static void main(String[] args) {
        gamesWon = 0; // This is actually redundant, since 0 is
                       // the default initial value.
        TextIO.putln("Let's play a game. I'll pick a number between");
        TextIO.putln("1 and 100, and you try to guess it.");
        boolean playAgain;
        do {
            playGame(); // call subroutine to play one game
            TextIO.put("Would you like to play again? ");
            playAgain = TextIO.getlnBoolean();
        } while (playAgain);
        TextIO.putln();
        TextIO.putln("You won " + gamesWon + " games.");
        TextIO.putln("Thanks for playing. Goodbye.");
    } // end of main()

    static void playGame() {
        int computersNumber; // A random number picked by the computer.
        int usersGuess;      // A number entered by user as a guess.
        int guessCount;      // Number of guesses the user has made.
        computersNumber = (int)(100 * Math.random()) + 1;
                           // The value assigned to computersNumber is a randomly
                           // chosen integer between 1 and 100, inclusive.
        guessCount = 0;
        TextIO.putln();
        TextIO.put("What is your first guess? ");
        while (true) {
            usersGuess = TextIO.getInt(); // Get the user's guess.
            guessCount++;
            if (usersGuess == computersNumber) {
                TextIO.putln("You got it in " + guessCount
                             + " guesses! My number was " + computersNumber);
                gamesWon++; // Count this game by incrementing gamesWon.
                break;      // The game is over; the user has won.
            }
            if (guessCount == 6) {
                TextIO.putln("You didn't get the number in 6 guesses.");
                TextIO.putln("You lose. My number was " + computersNumber);
                break; // The game is over; the user has lost.
            }
            // If we get to this point, the game continues.
            // Tell the user if the guess was too high or too low.
            if (usersGuess < computersNumber)
                TextIO.put("That's too low. Try again: ");
            else if (usersGuess > computersNumber)
                TextIO.put("That's too high. Try again: ");
        }
    }
}
```

```

        TextIO.putln();
    } // end of playGame()
} // end of class GuessingGame2

```

4.3 Parameters

IF A SUBROUTINE IS A BLACK BOX, then a parameter provides a mechanism for passing information from the outside world into the box. Parameters are part of the interface of a subroutine. They allow you to customize the behavior of a subroutine to adapt it to a particular situation.

As an analogy, consider a thermostat—a black box whose task it is to keep your house at a certain temperature. The thermostat has a parameter, namely the dial that is used to set the desired temperature. The thermostat always performs the same task: maintaining a constant temperature. However, the exact task that it performs—that is, **which** temperature it maintains—is customized by the setting on its dial.

4.3.1 Using Parameters

As an example, let's go back to the “3N+1” problem that was discussed in Subsection 3.2.2. (Recall that a 3N+1 sequence is computed according to the rule, “if N is odd, multiply by 3 and add 1; if N is even, divide by 2; continue until N is equal to 1.” For example, starting from N=3 we get the sequence: 3, 10, 5, 16, 8, 4, 2, 1.) Suppose that we want to write a subroutine to print out such sequences. The subroutine will always perform the same task: Print out a 3N+1 sequence. But the exact sequence it prints out depends on the starting value of N. So, the starting value of N would be a parameter to the subroutine. The subroutine could be written like this:

```

/**
 * This subroutine prints a 3N+1 sequence to standard output, using
 * startingValue as the initial value of N. It also prints the number
 * of terms in the sequence. The value of the parameter, startingValue,
 * must be a positive integer.
 */
static void print3NSequence(int startingValue) {
    int N;        // One of the terms in the sequence.
    int count;    // The number of terms.

    N = startingValue; // The first term is whatever value
                       // is passed to the subroutine as
                       // a parameter.

    int count = 1; // We have one term, the starting value, so far.

    System.out.println("The 3N+1 sequence starting from " + N);
    System.out.println();
    System.out.println(N); // print initial term of sequence

    while (N > 1) {
        if (N % 2 == 1) // is N odd?
            N = 3 * N + 1;
        else
            N = N / 2;
    }
}

```

```

        count++;    // count this term
        System.out.println(N); // print this term
    }

    System.out.println();
    System.out.println("There were " + count + " terms in the sequence.");
} // end print3NSequence

```

The parameter list of this subroutine, “(int startingValue)”, specifies that the subroutine has one parameter, of type **int**. Within the body of the subroutine, the parameter name can be used in the same way as a variable name. However, the parameter gets its initial value from **outside** the subroutine. When the subroutine is called, a value must be provided for this parameter in the subroutine call statement. This value will be assigned to the parameter, **startingValue**, before the body of the subroutine is executed. For example, the subroutine could be called using the subroutine call statement “**print3NSequence(17);**”. When the computer executes this statement, the computer assigns the value 17 to **startingValue** and then executes the statements in the subroutine. This prints the $3N+1$ sequence starting from 17. If **K** is a variable of type **int**, then when the computer executes the subroutine call statement “**print3NSequence(K);**”, it will take the value of the variable **K**, assign that value to **startingValue**, and execute the body of the subroutine.

The class that contains **print3NSequence** can contain a **main()** routine (or other subroutines) that call **print3NSequence**. For example, here is a **main()** program that prints out $3N+1$ sequences for various starting values specified by the user:

```

public static void main(String[] args) {
    TextIO.putln("This program will print out 3N+1 sequences");
    TextIO.putln("for starting values that you specify.");
    TextIO.putln();
    int K; // Input from user; loop ends when K < 0.
    do {
        TextIO.putln("Enter a starting value;")
        TextIO.put("To end the program, enter 0: ");
        K = TextIO.getInt(); // Get starting value from user.
        if (K > 0) // Print sequence, but only if K is > 0.
            print3NSequence(K);
    } while (K > 0); // Continue only if K > 0.
} // end main

```

Remember that before you can use this program, the definitions of **main** and of **print3NSequence** must both be wrapped inside a class definition.

4.3.2 Formal and Actual Parameters

Note that the term “parameter” is used to refer to two different, but related, concepts. There are parameters that are used in the definitions of subroutines, such as **startingValue** in the above example. And there are parameters that are used in subroutine call statements, such as the **K** in the statement “**print3NSequence(K);**”. Parameters in a subroutine definition are called **formal parameters** or **dummy parameters**. The parameters that are passed to a subroutine when it is called are called **actual parameters** or **arguments**. When a subroutine is called, the actual parameters in the subroutine call statement are evaluated and the values are assigned to the formal parameters in the subroutine’s definition. Then the body of the subroutine is executed.

A formal parameter must be a **name**, that is, a simple identifier. A formal parameter is very much like a variable, and—like a variable—it has a specified type such as **int**, **boolean**, or *String*. An actual parameter is a **value**, and so it can be specified by any expression, provided that the expression computes a value of the correct type. The type of the actual parameter must be one that could legally be assigned to the formal parameter with an assignment statement. For example, if the formal parameter is of type **double**, then it would be legal to pass an **int** as the actual parameter since **ints** can legally be assigned to **doubles**. When you call a subroutine, you must provide one actual parameter for each formal parameter in the subroutine’s definition. Consider, for example, a subroutine

```
static void doTask(int N, double x, boolean test) {
    // statements to perform the task go here
}
```

This subroutine might be called with the statement

```
doTask(17, Math.sqrt(z+1), z >= 10);
```

When the computer executes this statement, it has essentially the same effect as the block of statements:

```
{
    int N;          // Allocate memory locations for the formal parameters.
    double x;
    boolean test;
    N = 17;          // Assign 17 to the first formal parameter, N.
    x = Math.sqrt(z+1); // Compute Math.sqrt(z+1), and assign it to
                      // the second formal parameter, x.
    test = (z >= 10); // Evaluate "z >= 10" and assign the resulting
                      // true/false value to the third formal
                      // parameter, test.
    // statements to perform the task go here
}
```

(There are a few technical differences between this and “doTask(17,Math.sqrt(z+1),z>=10);”—besides the amount of typing—because of questions about scope of variables and what happens when several variables or parameters have the same name.)

Beginning programming students often find parameters to be surprisingly confusing. Calling a subroutine that already exists is not a problem—the idea of providing information to the subroutine in a parameter is clear enough. Writing the subroutine definition is another matter. A common mistake is to assign values to the formal parameters at the beginning of the subroutine, or to ask the user to input their values. This represents a fundamental misunderstanding. When the statements in the subroutine are executed, the formal parameters will already have values. The values come from the subroutine call statement. Remember that a subroutine is not independent. It is called by some other routine, and it is the calling routine’s responsibility to provide appropriate values for the parameters.

4.3.3 Overloading

In order to call a subroutine legally, you need to know its name, you need to know how many formal parameters it has, and you need to know the type of each parameter. This information is called the subroutine’s *signature*. The signature of the subroutine `doTask`, used as an example above, can be expressed as as: `doTask(int,double,boolean)`. Note that the signature does

not include the names of the parameters; in fact, if you just want to **use** the subroutine, you don't even need to know what the formal parameter names are, so the names are not part of the interface. 1

Java is somewhat unusual in that it allows two different subroutines in the same class to have the same name, provided that their signatures are different. (The language C++ on which Java is based also has this feature.) When this happens, we say that the name of the subroutine is **overloaded** because it has several different meanings. The computer doesn't get the subroutines mixed up. It can tell which one you want to call by the number and types of the actual parameters that you provide in the subroutine call statement. You have already seen overloading used in the *TextIO* class. This class includes many different methods named **putln**, for example. These methods all have different signatures, such as: 2

<code>putln(int)</code>	<code>putln(double)</code> 3
<code>putln(String)</code>	<code>putln(char)</code>
<code>putln(boolean)</code>	<code>putln()</code>

The computer knows which of these subroutines you want to use based on the type of the actual parameter that you provide. `TextIO.putln(17)` calls the subroutine with signature `putln(int)`, while `TextIO.putln("Hello")` calls the subroutine with signature `putln(String)`. Of course all these different subroutines are semantically related, which is why it is acceptable programming style to use the same name for them all. But as far as the computer is concerned, printing out an **int** is very different from printing out a *String*, which is different from printing out a **boolean**, and so forth—so that each of these operations requires a different method. 4

Note, by the way, that the signature does **not** include the subroutine's return type. It is illegal to have two subroutines in the same class that have the same signature but that have different return types. For example, it would be a syntax error for a class to contain two methods defined as: 5

```
int    getln() { ... } 6
double getln() { ... }
```

So it should be no surprise that in the *TextIO* class, the methods for reading different types are not all named `getln()`. In a given class, there can only be one routine that has the name `getln` and has no parameters. So, the input routines in *TextIO* are distinguished by having different names, such as `getlnInt()` and `getlnDouble()`. 7

Java 5.0 introduced another complication: It is possible to have a single subroutine that takes a variable number of actual parameters. You have already used subroutines that do this—the formatted output routines `System.out.printf` and `TextIO.printf`. When you call these subroutines, the number of parameters in the subroutine call can be arbitrarily large, so it would be impossible to have different subroutines to handle each case. Unfortunately, writing the definition of such a subroutine requires some knowledge of arrays, which will not be covered until Chapter 7. When we get to that chapter, you'll learn how to write subroutines with a variable number of parameters. For now, we will ignore this complication. 8

4.3.4 Subroutine Examples 9

Let's do a few examples of writing small subroutines to perform assigned tasks. Of course, this is only one side of programming with subroutines. The task performed by a subroutine is always a subtask in a larger program. The art of designing those programs—of deciding how to 10

break them up into subtasks—is the other side of programming with subroutines. We’ll return 1 to the question of program design in Section 4.6.

As a first example, let’s write a subroutine to compute and print out all the divisors of a 2 given positive integer. The integer will be a parameter to the subroutine. Remember that the syntax of any subroutine is:

```
<modifiers> <return-type> <subroutine-name> ( <parameter-list> ) { 3  
    <statements>  
}
```

Writing a subroutine always means filling out this format. In this case, the statement of the 4 problem tells us that there is one parameter, of type **int**, and it tells us what the statements in the body of the subroutine should do. Since we are only working with static subroutines for now, we’ll need to use **static** as a modifier. We could add an access modifier (**public** or **private**), but in the absence of any instructions, I’ll leave it out. Since we are not told to return a value, the return type is **void**. Since no names are specified, we’ll have to make up names for the formal parameter and for the subroutine itself. I’ll use **N** for the parameter and **printDivisors** for the subroutine name. The subroutine will look like

```
static void printDivisors( int N ) { 5  
    <statements>  
}
```

and all we have left to do is to write the statements that make up the body of the routine. This 6 is not difficult. Just remember that you have to write the body assuming that **N** already has a value! The algorithm is: “For each possible divisor **D** in the range from 1 to **N**, if **D** evenly divides **N**, then print **D**.” Written in Java, this becomes:

```
/**  
 * Print all the divisors of N.  
 * We assume that N is a positive integer.  
 */  
  
static void printDivisors( int N ) {  
    int D;    // One of the possible divisors of N.  
    System.out.println("The divisors of " + N + " are:");  
    for ( D = 1; D <= N; D++ ) {  
        if ( N % D == 0 )  
            System.out.println(D);  
    }  
} 7
```

I’ve added a comment before the subroutine definition indicating the contract of the 8 subroutine—that is, what it does and what assumptions it makes. The contract includes the assumption that **N** is a positive integer. It is up to the caller of the subroutine to make sure that this assumption is satisfied.

As a second short example, consider the problem: Write a subroutine named **printRow**. It 9 should have a parameter **ch** of type **char** and a parameter **N** of type **int**. The subroutine should print out a line of text containing **N** copies of the character **ch**.

Here, we are told the name of the subroutine and the names of the two parameters, so we 10 don’t have much choice about the first line of the subroutine definition. The task in this case is pretty simple, so the body of the subroutine is easy to write. The complete subroutine is given by


```

/**
 * Write one line of output containing N copies of the
 * character ch.  If N <= 0, an empty line is output.
 */

static void printRow( char ch, int N ) {
    int i; // Loop-control variable for counting off the copies.
    for ( i = 1; i <= N; i++ ) {
        System.out.print( ch );
    }
    System.out.println();
}

```

1

Note that in this case, the contract makes no assumption about N, but it makes it clear what will happen in all cases, including the unexpected case that $N < 0$. 2

Finally, let's do an example that shows how one subroutine can build on another. Let's 3 write a subroutine that takes a *String* as a parameter. For each character in the string, it will print a line of output containing 25 copies of that character. It should use the `printRow()` subroutine to produce the output.

Again, we get to choose a name for the subroutine and a name for the parameter. I'll call 4 the subroutine `printRowsFromString` and the parameter `str`. The algorithm is pretty clear: For each position `i` in the string `str`, call `printRow(str.charAt(i),25)` to print one line of the output. So, we get:

```

/**
 * For each character in str, write a line of output
 * containing 25 copies of that character.
 */

static void printRowsFromString( String str ) {
    int i; // Loop-control variable for counting off the chars.
    for ( i = 0; i < str.length(); i++ ) {
        printRow( str.charAt(i), 25 );
    }
}

```

5

We could use `printRowsFromString` in a `main()` routine such as 6

```

public static void main(String[] args) {
    String inputLine; // Line of text input by user.
    TextIO.put("Enter a line of text: ");
    inputLine = TextIO.getln();
    TextIO.putln();
    printRowsFromString( inputLine );
}

```

7

Of course, the three routines, `main()`, `printRowsFromString()`, and `printRow()`, would 8 have to be collected together inside the same class. The program is rather useless, but it does demonstrate the use of subroutines. You'll find the program in the file *RowsOfChars.java*, if you want to take a look.

4.3.5 Throwing Exceptions 9

I have been talking about the “contract” of a subroutine. The contract says what the subroutine 10 will do, provided that the caller of the subroutine provides acceptable values for subroutine's

parameters. The question arises, though, what should the subroutine do when the caller violates the contract by providing bad parameter values? 1

We’ve already seen that some subroutines respond to bad parameter values by throwing exceptions. (See Section 3.7.) For example, the contract of the built-in subroutine `Double.parseDouble` says that the parameter should be a string representation of a number of type **double**; if this is true, then the subroutine will convert the string into the equivalent numeric value. If the caller violates the contract by passing an invalid string as the actual parameter, the subroutine responds by throwing an exception of type *NumberFormatException*. 2

Many subroutines throw *IllegalArgumentException* in response to bad parameter values. You might want to take this response in your own subroutines. This can be done with a **throw statement**. An exception is an object, and in order to throw an exception, you must create an exception object. You won’t officially learn how to do this until Chapter 5, but for now, you can use the following syntax for a **throw** statement that throws an *IllegalArgumentException*: 3

```
throw new IllegalArgumentException( <error-message> ); 4
```

where *<error-message>* is a string that describes the error that has been detected. (The word “new” in this statement is what creates the object.) To use this statement in a subroutine, you would check whether the values of the parameters are legal. If not, you would throw the exception. For example, consider the `print3NSequence` subroutine from the beginning of this section. The parameter of `print3NSequence` is supposed to be a positive integer. We can modify the subroutine definition to make it throw an exception when this condition is violated: 5

```
static void print3NSequence(int startingValue) { 6
    if (startingValue <= 0) // The contract is violated! 7
        throw new IllegalArgumentException( "Starting value must be positive." ); 8
    .
    . // (The rest of the subroutine is the same as before.) 9
    .
}
```

If the start value is bad, the computer executes the **throw** statement. This will immediately terminate the subroutine, without executing the rest of the body of the subroutine. Furthermore, the program as a whole will crash unless the exception is “caught” and handled elsewhere in the program by a `try..catch` statement, as discussed in Section 3.7. 10

4.3.6 Global and Local Variables 11

I’ll finish this section on parameters by noting that we now have three different sorts of variables that can be used inside a subroutine: local variables declared in the subroutine, formal parameter names, and static member variables that are declared outside the subroutine but inside the same class as the subroutine. 12

Local variables have no connection to the outside world; they are purely part of the internal working of the subroutine. Parameters are used to “drop” values into the subroutine when it is called, but once the subroutine starts executing, parameters act much like local variables. Changes made inside a subroutine to a formal parameter have no effect on the rest of the program (at least if the type of the parameter is one of the primitive types—things are more complicated in the case of objects, as we’ll see later). 13

Things are different when a subroutine uses a variable that is defined outside the subroutine. That variable exists independently of the subroutine, and it is accessible to other parts of the program, as well as to the subroutine. Such a variable is said to be **global** to the subroutine, 14

as opposed to the local variables defined inside the subroutine. The scope of a global variable includes the entire class in which it is defined. Changes made to a global variable can have effects that extend outside the subroutine where the changes are made. You’ve seen how this works in the last example in the previous section, where the value of the global variable, `gamesWon`, is computed inside a subroutine and is used in the `main()` routine.

It’s not always bad to use global variables in subroutines, but you should realize that the global variable then has to be considered part of the subroutine’s interface. The subroutine uses the global variable to communicate with the rest of the program. This is a kind of sneaky, back-door communication that is less visible than communication done through parameters, and it risks violating the rule that the interface of a black box should be straightforward and easy to understand. So before you use a global variable in a subroutine, you should consider whether it’s really necessary.

I don’t advise you to take an absolute stand against using global variables inside subroutines. There is at least one good reason to do it: If you think of the class as a whole as being a kind of black box, it can be very reasonable to let the subroutines inside that box be a little sneaky about communicating with each other, if that will make the class as a whole look simpler from the outside.

4.4 Return Values

A SUBROUTINE THAT RETURNS A VALUE is called a *function*. A given function can only return a value of a specified type, called the *return type* of the function. A function call generally occurs in a position where the computer is expecting to find a value, such as the right side of an assignment statement, as an actual parameter in a subroutine call, or in the middle of some larger expression. A boolean-valued function can even be used as the test condition in an `if`, `while`, `for` or `do..while` statement.

(It is also legal to use a function call as a stand-alone statement, just as if it were a regular subroutine. In this case, the computer ignores the value computed by the subroutine. Sometimes this makes sense. For example, the function `TextIO.getln()`, with a return type of *String*, reads and returns a line of input typed in by the user. Usually, the line that is returned is assigned to a variable to be used later in the program, as in the statement “`name = TextIO.getln();`”. However, this function is also useful as a subroutine call statement “`TextIO.getln();`”, which still reads all input up to and including the next carriage return. Since the return value is not assigned to a variable or used in an expression, it is simply discarded. So, the effect of the subroutine call is to read **and discard** some input. Sometimes, discarding unwanted input is exactly what you need to do.)

4.4.1 The return statement

You’ve already seen how functions such as `Math.sqrt()` and `TextIO.getInt()` can be used. What you haven’t seen is how to write functions of your own. A function takes the same form as a regular subroutine, except that you have to specify the value that is to be returned by the subroutine. This is done with a *return statement*, which has the following syntax:

```
return <expression>;
```

Such a **return** statement can only occur inside the definition of a function, and the type of the *<expression>* must match the return type that was specified for the function. (More exactly, it must be legal to assign the expression to a variable whose type is specified by the

return type.) When the computer executes this **return** statement, it evaluates the expression, terminates execution of the function, and uses the value of the expression as the returned value of the function. 1

For example, consider the function definition 2

```
static double pythagoras(double x, double y) {
    // Computes the length of the hypotenuse of a right
    // triangle, where the sides of the triangle are x and y.
    return Math.sqrt( x*x + y*y );
}
```

3

Suppose the computer executes the statement “`totalLength = 17 + pythagoras(12,5);`”. When it gets to the term `pythagoras(12,5)`, it assigns the actual parameters 12 and 5 to the formal parameters `x` and `y` in the function. In the body of the function, it evaluates `Math.sqrt(12.0*12.0 + 5.0*5.0)`, which works out to 13.0. This value is “returned” by the function, so the 13.0 essentially replaces the function call in the statement “`totalLength = 17 + pythagoras(12,5);`”. The return value is added to 17, and the result, 30.0, is stored in the variable, `totalLength`. The effect is the same as if the statement had been “`totalLength = 17 + 13.0;`”. 4

Note that a **return** statement does not have to be the last statement in the function definition. At any point in the function where you know the value that you want to return, you can return it. Returning a value will end the function immediately, skipping any subsequent statements in the function. However, it must be the case that the function definitely does return some value, no matter what path the execution of the function takes through the code. 5

You can use a **return** statement inside an ordinary subroutine, one with declared return type “`void`”. Since a void subroutine does not return a value, the **return** statement does not include an expression; it simply takes the form “`return;`”. The effect of this statement is to terminate execution of the subroutine and return control back to the point in the program from which the subroutine was called. This can be convenient if you want to terminate execution somewhere in the middle of the subroutine, but **return** statements are optional in non-function subroutines. In a function, on the other hand, a return statement, with expression, is always required. 6

4.4.2 Function Examples 7

Here is a very simple function that could be used in a program to compute $3N+1$ sequences. (The $3N+1$ sequence problem is one we’ve looked at several times already, including in the previous section). Given one term in a $3N+1$ sequence, this function computes the next term of the sequence: 8

```
static int nextN(int currentN) {
    if (currentN % 2 == 1)    // test if current N is odd
        return 3*currentN + 1; // if so, return this value
    else
        return currentN / 2;   // if not, return this instead
}
```

9

This function has two **return** statements. Exactly one of the two **return** statements is executed to give the value of the function. Some people prefer to use a single **return** statement at the very end of the function when possible. This allows the reader to find the **return** statement easily. You might choose to write `nextN()` like this, for example: 10

```
static int nextN(int currentN) {
    int answer; // answer will be the value returned
    if (currentN % 2 == 1) // test if current N is odd
        answer = 3*currentN+1; // if so, this is the answer
    else
        answer = currentN / 2; // if not, this is the answer
    return answer; // (Don't forget to return the answer!)
}
```

2

Here is a subroutine that uses this `nextN` function. In this case, the improvement from the version of this subroutine in Section 4.3 is not great, but if `nextN()` were a long function that performed a complex computation, then it would make a lot of sense to hide that complexity inside a function:

```
static void print3NSequence(int startingValue) {
    int N; // One of the terms in the sequence.
    int count; // The number of terms found.

    N = startingValue; // Start the sequence with startingValue.
    count = 1;

    TextIO.putln("The 3N+1 sequence starting from " + N);
    TextIO.putln();
    TextIO.putln(N); // print initial term of sequence

    while (N > 1) {
        N = nextN(N); // Compute next term, using the function nextN.
        count++; // Count this term.
        TextIO.putln(N); // Print this term.
    }

    TextIO.putln();
    TextIO.putln("There were " + count + " terms in the sequence.");
}
```

4

* * * 1

Here are a few more examples of functions. The first one computes a letter grade corresponding to a given numerical grade, on a typical grading scale:

```
/**
 * Returns the letter grade corresponding to the numerical
 * grade that is passed to this function as a parameter.
 */
```

```
static char letterGrade(int numGrade) {
    if (numGrade >= 90)
        return 'A'; // 90 or above gets an A
    else if (numGrade >= 80)
        return 'B'; // 80 to 89 gets a B
    else if (numGrade >= 65)
        return 'C'; // 65 to 79 gets a C
    else if (numGrade >= 50)
        return 'D'; // 50 to 64 gets a D
    else
```

6

```

        return 'F';    // anything else gets an F 1
    } // end of function letterGrade 3

```

The type of the return value of `letterGrade()` is **char**. Functions can return values of any type at all. Here's a function whose return value is of type **boolean**. It demonstrates some interesting programming points, so you should read the comments: ⁴

```

/** 5
 * The function returns true if N is a prime number. A prime number
 * is an integer greater than 1 that is not divisible by any positive
 * integer, except itself and 1. If N has any divisor, D, in the range
 * 1 < D < N, then it has a divisor in the range 2 to Math.sqrt(N), namely
 * either D itself or N/D. So we only test possible divisors from 2 to
 * Math.sqrt(N).
 */
static boolean isPrime(int N) {
    int divisor; // A number we will test to see whether it evenly divides N.
    if (N <= 1)
        return false; // No number <= 1 is a prime.
    int maxToTry; // The largest divisor that we need to test.
    maxToTry = (int)Math.sqrt(N);
    // We will try to divide N by numbers between 2 and maxToTry.
    // If N is not evenly divisible by any of these numbers, then
    // N is prime. (Note that since Math.sqrt(N) is defined to
    // return a value of type double, the value must be typecast
    // to type int before it can be assigned to maxToTry.)
    for (divisor = 2; divisor <= maxToTry; divisor++) {
        if ( N % divisor == 0 ) // Test if divisor evenly divides N.
            return false;     // If so, we know N is not prime.
                                // No need to continue testing!
    }
    // If we get to this point, N must be prime. Otherwise,
    // the function would already have been terminated by
    // a return statement in the previous loop.
    return true; // Yes, N is prime.
} // end of function isPrime 2

```

Finally, here is a function with return type *String*. This function has a *String* as parameter. ⁶ The returned value is a reversed copy of the parameter. For example, the reverse of “Hello World” is “dlroW olleH”. The algorithm for computing the reverse of a string, `str`, is to start with an empty string and then to append each character from `str`, starting from the last character of `str` and working backwards to the first:

```

static String reverse(String str) { 7
    String copy; // The reversed copy.
    int i;       // One of the positions in str,
                //           from str.length() - 1 down to 0.
    copy = "";   // Start with an empty string.
    for ( i = str.length() - 1; i >= 0; i-- ) {

```

```

        // Append i-th char of str to copy. 1
        copy = copy + str.charAt(i);
    }
    return copy;
}

```

A *palindrome* is a string that reads the same backwards and forwards, such as “radar”. The `reverse()` function could be used to check whether a string, `word`, is a palindrome by testing “if (`word.equals(reverse(word))`)”.

By the way, a typical beginner’s error in writing functions is to print out the answer, instead of returning it. This represents a fundamental misunderstanding. The task of a function is to compute a value and return it to the point in the program where the function was called. That’s where the value is used. Maybe it will be printed out. Maybe it will be assigned to a variable. Maybe it will be used in an expression. But it’s not for the function to decide.

4.4.3 3N+1 Revisited 4

I’ll finish this section with a complete new version of the 3N+1 program. This will give me a chance to show the function `nextN()`, which was defined above, used in a complete program. I’ll also take the opportunity to improve the program by getting it to print the terms of the sequence in columns, with five terms on each line. This will make the output more presentable. This idea is this: Keep track of how many terms have been printed on the current line; when that number gets up to 5, start a new line of output. To make the terms line up into neat columns, I use formatted output.

```

/**
 * A program that computes and displays several 3N+1 sequences. Starting
 * values for the sequences are input by the user. Terms in the sequence
 * are printed in columns, with five terms on each line of output.
 * After a sequence has been displayed, the number of terms in that
 * sequence is reported to the user.
 */
public class ThreeN2 {

    public static void main(String[] args) {

        TextIO.putln("This program will print out 3N+1 sequences");
        TextIO.putln("for starting values that you specify.");
        TextIO.putln();

        int K;    // Starting point for sequence, specified by the user.
        do {
            TextIO.putln("Enter a starting value;");
            TextIO.put("To end the program, enter 0: ");
            K = TextIO.getInt();    // get starting value from user
            if (K > 0)                // print sequence, but only if K is > 0
                print3NSequence(K);
        } while (K > 0);            // continue only if K > 0

    } // end main

    /**
     * print3NSequence prints a 3N+1 sequence to standard output, using

```

```

* startingValue as the initial value of N. It also prints the number
* of terms in the sequence. The value of the parameter, startingValue,
* must be a positive integer.
*/
static void print3NSequence(int startingValue) {
    int N;          // One of the terms in the sequence.
    int count;      // The number of terms found.
    int onLine;     // The number of terms that have been output
                   // so far on the current line.

    N = startingValue; // Start the sequence with startingValue;
    count = 1;         // We have one term so far.

    TextIO.putln("The 3N+1 sequence starting from " + N);
    TextIO.putln();
    TextIO.put(N, 8); // Print initial term, using 8 characters.
    onLine = 1;       // There's now 1 term on current output line.

    while (N > 1) {
        N = nextN(N); // compute next term
        count++;      // count this term
        if (onLine == 5) { // If current output line is full
            TextIO.putln(); // ...then output a carriage return
            onLine = 0;     // ...and note that there are no terms
                           // on the new line.
        }
        TextIO.printf("%8d", N); // Print this term in an 8-char column.
        onLine++;              // Add 1 to the number of terms on this line.
    }

    TextIO.putln(); // end current line of output
    TextIO.putln(); // and then add a blank line
    TextIO.putln("There were " + count + " terms in the sequence.");
} // end of Print3NSequence

/**
 * nextN computes and returns the next term in a 3N+1 sequence,
 * given that the current term is currentN.
 */
static int nextN(int currentN) {
    if (currentN % 2 == 1)
        return 3 * currentN + 1;
    else
        return currentN / 2;
} // end of nextN()
} // end of class ThreeN2

```

You should read this program carefully and try to understand how it works. (Try using 27 for the starting value!) 3

4.5 APIs, Packages, and Javadoc¹

AS COMPUTERS AND THEIR USER INTERFACES have become easier to use, they have also become more complex for programmers to deal with. You can write programs for a simple console-style user interface using just a few subroutines that write output to the console and read the user's typed replies. A modern graphical user interface, with windows, buttons, scroll bars, menus, text-input boxes, and so on, might make things easier for the user, but it forces the programmer to cope with a hugely expanded array of possibilities. The programmer sees this increased complexity in the form of great numbers of subroutines that are provided for managing the user interface, as well as for other purposes.

4.5.1 Toolboxes³

Someone who wants to program for Macintosh computers—and to produce programs that look and behave the way users expect them to—must deal with the Macintosh Toolbox, a collection of well over a thousand different subroutines. There are routines for opening and closing windows, for drawing geometric figures and text to windows, for adding buttons to windows, and for responding to mouse clicks on the window. There are other routines for creating menus and for reacting to user selections from menus. Aside from the user interface, there are routines for opening files and reading data from them, for communicating over a network, for sending output to a printer, for handling communication between programs, and in general for doing all the standard things that a computer has to do. Microsoft Windows provides its own set of subroutines for programmers to use, and they are quite a bit different from the subroutines used on the Mac. Linux has several different GUI toolboxes for the programmer to choose from.

The analogy of a “toolbox” is a good one to keep in mind. Every programming project involves a mixture of innovation and reuse of existing tools. A programmer is given a set of tools to work with, starting with the set of basic tools that are built into the language: things like variables, assignment statements, if statements, and loops. To these, the programmer can add existing toolboxes full of routines that have already been written for performing certain tasks. These tools, if they are well-designed, can be used as true black boxes: They can be called to perform their assigned tasks without worrying about the particular steps they go through to accomplish those tasks. The innovative part of programming is to take all these tools and apply them to some particular project or problem (word-processing, keeping track of bank accounts, processing image data from a space probe, Web browsing, computer games, . . .). This is called *applications programming*.

A software toolbox is a kind of black box, and it presents a certain interface to the programmer. This interface is a specification of what routines are in the toolbox, what parameters they use, and what tasks they perform. This information constitutes the **API**, or **Applications Programming Interface**, associated with the toolbox. The Macintosh API is a specification of all the routines available in the Macintosh Toolbox. A company that makes some hardware device—say a card for connecting a computer to a network—might publish an API for that device consisting of a list of routines that programmers can call in order to communicate with and control the device. Scientists who write a set of routines for doing some kind of complex computation—such as solving “differential equations,” say—would provide an API to allow others to use those routines without understanding the details of the computations they perform.

The Java programming language is supplemented by a large, standard API. You've seen part of this API already, in the form of mathematical subroutines such as `Math.sqrt()`, the *String* data type and its associated routines, and the `System.out.print()` routines. The standard Java API includes routines for working with graphical user interfaces, for network communication, for reading and writing files, and more. It's tempting to think of these routines as being built into the Java language, but they are technically subroutines that have been written and made available for use in Java programs.

Java is platform-independent. That is, the same program can run on platforms as diverse as Macintosh, Windows, Linux, and others. The same Java API must work on all these platforms. But notice that it is the **interface** that is platform-independent; the **implementation** varies from one platform to another. A Java system on a particular computer includes implementations of all the standard API routines. A Java program includes only **calls** to those routines. When the Java interpreter executes a program and encounters a call to one of the standard routines, it will pull up and execute the implementation of that routine which is appropriate for the particular platform on which it is running. This is a very powerful idea. It means that you only need to learn one API to program for a wide variety of platforms.

4.5.2 Java's Standard Packages

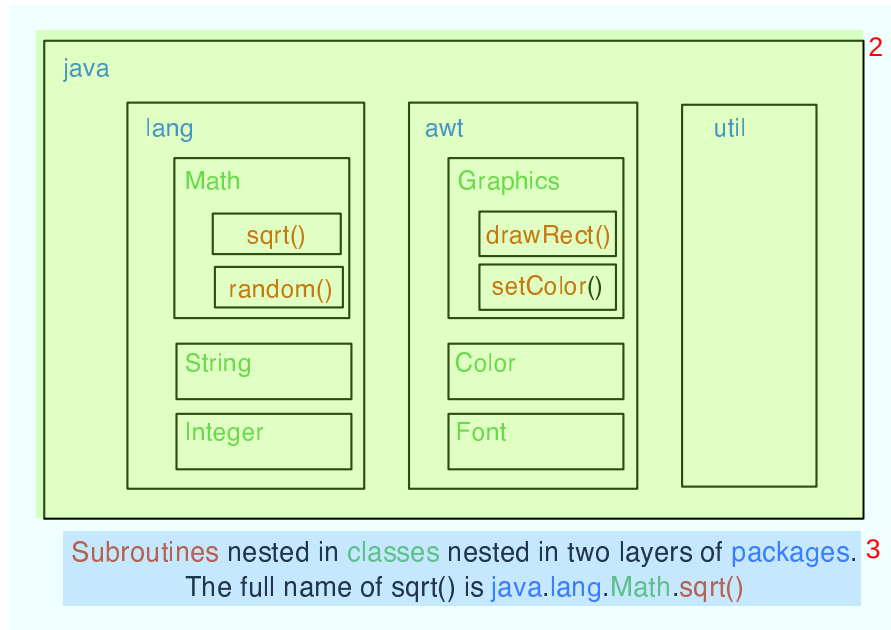
Like all subroutines in Java, the routines in the standard API are grouped into classes. To provide larger-scale organization, classes in Java can be grouped into **packages**, which were introduced briefly in Subsection 2.6.4. You can have even higher levels of grouping, since packages can also contain other packages. In fact, the entire standard Java API is implemented in several packages. One of these, which is named "java", contains several non-GUI packages as well as the original AWT graphics user interface classes. Another package, "javax", was added in Java version 1.2 and contains the classes used by the Swing graphical user interface and other additions to the API.

A package can contain both classes and other packages. A package that is contained in another package is sometimes called a "sub-package." Both the java package and the javax package contain sub-packages. One of the sub-packages of java, for example, is called "awt". Since `awt` is contained within `java`, its full name is actually `java.awt`. This package contains classes that represent GUI components such as buttons and menus in the AWT, the older of the two Java GUI toolboxes, which is no longer widely used. However, `java.awt` also contains a number of classes that form the foundation for all GUI programming, such as the `Graphics` class which provides routines for drawing on the screen, the `Color` class which represents colors, and the `Font` class which represents the fonts that are used to display characters on the screen. Since these classes are contained in the package `java.awt`, their full names are actually `java.awt.Graphics`, `java.awt.Color`, and `java.awt.Font`. (I hope that by now you've gotten the hang of how this naming thing works in Java.) Similarly, `javax` contains a sub-package named `javax.swing`, which includes such classes as `javax.swing.JButton`, `javax.swing.JMenu`, and `javax.swing.JFrame`. The GUI classes in `javax.swing`, together with the foundational classes in `java.awt`, are all part of the API that makes it possible to program graphical user interfaces in Java.

The `java` package includes several other sub-packages, such as `java.io`, which provides facilities for input/output, `java.net`, which deals with network communication, and `java.util`, which provides a variety of "utility" classes. The most basic package is called `java.lang`. This package contains fundamental classes such as *String*, *Math*, *Integer*, and *Double*.

It might be helpful to look at a graphical representation of the levels of nesting in the

java package, its sub-packages, the classes in those sub-packages, and the subroutines in those classes. This is not a complete picture, since it shows only a very few of the many items in each element:



The official documentation for the standard Java 5.0 API lists 165 different packages, including sub-packages, and it lists 3278 classes in these packages. Many of these are rather obscure or very specialized, but you might want to browse through the documentation to see what is available. As I write this, the documentation for the complete API can be found at

<http://java.sun.com/j2se/1.5.0/docs/api/index.html>

Even an expert programmer won't be familiar with the entire API, or even a majority of it. In this book, you'll only encounter several dozen classes, and those will be sufficient for writing a wide variety of programs.

4.5.3 Using Classes from Packages

Let's say that you want to use the class `java.awt.Color` in a program that you are writing. Like any class, `java.awt.Color` is a type, which means that you can use it to declare variables and parameters and to specify the return type of a function. One way to do this is to use the full name of the class as the name of the type. For example, suppose that you want to declare a variable named `rectColor` of type `java.awt.Color`. You could say:

```
java.awt.Color rectColor;
```

This is just an ordinary variable declaration of the form “`<type-name> <variable-name>;`”. Of course, using the full name of every class can get tiresome, so Java makes it possible to avoid using the full name of a class by *importing* the class. If you put

```
import java.awt.Color;
```

at the beginning of a Java source code file, then, in the rest of the file, you can abbreviate the full name `java.awt.Color` to just the simple name of the class, `Color`. Note that the `import`

line comes at the start of a file and is not inside any class. Although it is sometimes referred 1
to as a statement, it is more properly called an *import directive* since it is not a statement
in the usual sense. Using this `import` directive would allow you to say

```
Color rectColor; 2
```

to declare the variable. Note that the only effect of the `import` directive is to allow you to use 3
simple class names instead of full “package.class” names; you aren’t really importing anything
substantial. If you leave out the `import` directive, you can still access the class—you just have
to use its full name. There is a shortcut for importing all the classes from a given package. You
can import all the classes from `java.awt` by saying

```
import java.awt.*; 4
```

The “*” is a *wildcard* that matches every class in the package. (However, it does not match 5
sub-packages; you **cannot** import the entire contents of all the sub-packages of the `java` package
by saying `import java.*`.)

Some programmers think that using a wildcard in an `import` statement is bad style, since 6
it can make a large number of class names available that you are not going to use and might
not even know about. They think it is better to explicitly import each individual class that
you want to use. In my own programming, I often use wildcards to import all the classes from
the most relevant packages, and use individual imports when I am using just one or two classes
from a given package.

In fact, any Java program that uses a graphical user interface is likely to use many 7
classes from the `java.awt` and `java.swing` packages as well as from another package named
`java.awt.event`, and I usually begin such programs with

```
import java.awt.*; 8  
import java.awt.event.*; 9  
import javax.swing.*; 10
```

A program that works with networking might include the line “`import java.net.*`,” while 11
one that reads or writes files might use “`import java.io.*`.” (But when you start importing
lots of packages in this way, you have to be careful about one thing: It’s possible for two classes
that are in different packages to have the same name. For example, both the `java.awt` package
and the `java.util` package contain classes named `List`. If you import both `java.awt.*` and
`java.util.*`, the simple name `List` will be ambiguous. If you try to declare a variable of type
`List`, you will get a compiler error message about an ambiguous class name. The solution is
simple: Use the full name of the class, either `java.awt.List` or `java.util.List`. Another
solution, of course, is to use `import` to import the individual classes you need, instead of
importing entire packages.)

Because the package `java.lang` is so fundamental, all the classes in `java.lang` are **auto-** 12
atically imported into every program. It’s as if every program began with the statement
“`import java.lang.*`.” This is why we have been able to use the class name *String* instead
of `java.lang.String`, and `Math.sqrt()` instead of `java.lang.Math.sqrt()`. It would still,
however, be perfectly legal to use the longer forms of the names.

Programmers can create new packages. Suppose that you want some classes that you are 13
writing to be in a package named `utilities`. Then the source code file that defines those
classes must begin with the line

```
package utilities; 14
```

This would come even before any `import` directive in that file. Furthermore, as mentioned in Subsection 2.6.4, the source code file would be placed in a folder with the same name as the package. A class that is in a package automatically has access to other classes in the same package; that is, a class doesn't have to import the package in which it is defined. 1

In projects that define large numbers of classes, it makes sense to organize those classes into packages. It also makes sense for programmers to create new packages as toolboxes that provide functionality and API's for dealing with areas not covered in the standard Java API. (And in fact such "toolmaking" programmers often have more prestige than the applications programmers who use their tools.) 2

However, I will not be creating any packages in this textbook. For the purposes of this book, you need to know about packages mainly so that you will be able to import the standard packages. These packages are always available to the programs that you write. You might wonder where the standard classes are actually located. Again, that can depend to some extent on the version of Java that you are using, but in the standard Java 5.0, they are stored in *jar files* in a subdirectory of the main Java installation directory. A jar (or "Java archive") file is a single file that can contain many classes. Most of the standard classes can be found in a jar file named `classes.jar`. In fact, Java programs are generally distributed in the form of jar files, instead of as individual class files. 3

Although we won't be creating packages explicitly, **every** class is actually part of a package. If a class is not specifically placed in a package, then it is put in something called the *default package*, which has no name. All the examples that you see in this book are in the default package. 4

4.5.4 Javadoc 5

To use an API effectively, you need good documentation for it. The documentation for most Java APIs is prepared using a system called *Javadoc*. For example, this system is used to prepare the documentation for Java's standard packages. And almost everyone who creates a toolbox in Java publishes Javadoc documentation for it. 6

Javadoc documentation is prepared from special comments that are placed in the Java source code file. Recall that one type of Java comment begins with `/*` and ends with `*/`. A Javadoc comment takes the same form, but it begins with `/**` rather than simply `/*`. You have already seen comments of this form in some of the examples in this book, such as this subroutine from Section 4.3: 7

```
/** 8
 * This subroutine prints a 3N+1 sequence to standard output, using 9
 * startingValue as the initial value of N. It also prints the number 10
 * of terms in the sequence. The value of the parameter, startingValue, 11
 * must be a positive integer. 12
 */ 13

static void print3NSequence(int startingValue) { ... 14
```

Note that the Javadoc comment is placed just **before** the subroutine that it is commenting on. This rule is always followed. You can have Javadoc comments for subroutines, for member variables, and for classes. The Javadoc comment always immediately precedes the thing it is commenting on. 15

Like any comment, a Javadoc comment is ignored by the computer when the file is compiled. But there is a tool called `javadoc` that reads Java source code files, extracts any Javadoc 16

comments that it finds, and creates a set of Web pages containing the comments in a nicely formatted, interlinked form. By default, `javadoc` will only collect information about public classes, subroutines, and member variables, but it allows the option of creating documentation for non-public things as well. If `javadoc` doesn't find any Javadoc comment for something, it will construct one, but the comment will contain only basic information such as the name and type of a member variable or the name, return type, and parameter list of a subroutine. This is **syntactic** information. To add information about semantics and pragmatics, you have to write a Javadoc comment.

As an example, you can look at the documentation Web page for *TextIO*. The documentation page was created by applying the `javadoc` tool to the source code file, *TextIO.java*. If you have downloaded the on-line version of this book, the documentation can be found in the `TextIO_Javadoc` directory, or you can find a link to it in the on-line version of this section.

In a Javadoc comment, the `*`'s at the start of each line are optional. The `javadoc` tool will remove them. In addition to normal text, the comment can contain certain special codes. For one thing, the comment can contain **HTML mark-up** commands. HTML is the language that is used to create web pages, and Javadoc comments are meant to be shown on web pages. The `javadoc` tool will copy any HTML commands in the comments to the web pages that it creates. You'll learn some basic HTML in Section 6.2, but as an example, you can add `<p>` to indicate the start of a new paragraph. (Generally, in the absence of HTML commands, blank lines and extra spaces in the comment are ignored.)

In addition to HTML commands, Javadoc comments can include **doc tags**, which are processed as commands by the `javadoc` tool. A doc tag has a name that begins with the character `@`. I will only discuss three tags: `@param`, `@return`, and `@throws`. These tags are used in Javadoc comments for subroutines to provide information about its parameters, its return value, and the exceptions that it might throw. These tags are always placed at the end of the comment, after any description of the subroutine itself. The syntax for using them is:

```
@param <parameter-name> <description-of-parameter>
```

```
@return <description-of-return-value>
```

```
@throws <exception-class-name> <description-of-exception>
```

The `<descriptions>` can extend over several lines. The description ends at the next tag or at the end of the comment. You can include a `@param` tag for every parameter of the subroutine and a `@throws` for as many types of exception as you want to document. You should have a `@return` tag only for a non-void subroutine. These tags do not have to be given in any particular order.

Here is an example that doesn't do anything exciting but that does use all three types of doc tag:

```
/**
 * This subroutine computes the area of a rectangle, given its width
 * and its height. The length and the width should be positive numbers.
 * @param width the length of one side of the rectangle
 * @param height the length the second side of the rectangle
 * @return the area of the rectangle
 * @throws IllegalArgumentException if either the width or the height
 *         is a negative number.
 */
public static double areaOfRectangle( double length, double width ) {
    if ( width < 0 || height < 0 )
        throw new IllegalArgumentException("Sides must have positive length.");
```



```
double area;  
area = width * height;  
return area;  
}
```

I will use Javadoc comments for some of my examples. I encourage you to use them in your own code, even if you don't plan to generate Web page documentation of your work, since it's a standard format that other Java programmers will be familiar with.

If you do want to create Web-page documentation, you need to run the `javadoc` tool. This tool is available as a command in the Java Development Kit that was discussed in Section 2.6. You can use `javadoc` in a command line interface similarly to the way that the `javac` and `java` commands are used. Javadoc can also be applied in the Eclipse integrated development environment that was also discussed in Section 2.6: Just right-click the class or package that you want to document in the Package Explorer, select "Export," and select "Javadoc" in the window that pops up. I won't go into any of the details here; see the documentation.

4.6 More on Program Design

UNDERSTANDING HOW PROGRAMS WORK is one thing. Designing a program to perform some particular task is another thing altogether. In Section 3.2, I discussed how pseudocode and stepwise refinement can be used to methodically develop an algorithm. We can now see how subroutines can fit into the process.

Stepwise refinement is inherently a top-down process, but the process does have a "bottom," that is, a point at which you stop refining the pseudocode algorithm and translate what you have directly into proper programming language. In the absence of subroutines, the process would not bottom out until you get down to the level of assignment statements and very primitive input/output operations. But if you have subroutines lying around to perform certain useful tasks, you can stop refining as soon as you've managed to express your algorithm in terms of those tasks.

This allows you to add a bottom-up element to the top-down approach of stepwise refinement. Given a problem, you might start by writing some subroutines that perform tasks relevant to the problem domain. The subroutines become a toolbox of ready-made tools that you can integrate into your algorithm as you develop it. (Alternatively, you might be able to buy or find a software toolbox written by someone else, containing subroutines that you can use in your project as black boxes.)

Subroutines can also be helpful even in a strict top-down approach. As you refine your algorithm, you are free at any point to take any sub-task in the algorithm and make it into a subroutine. Developing that subroutine then becomes a separate problem, which you can work on separately. Your main algorithm will merely call the subroutine. This, of course, is just a way of breaking your problem down into separate, smaller problems. It is still a top-down approach because the top-down analysis of the problem tells you what subroutines to write. In the bottom-up approach, you start by writing or obtaining subroutines that are relevant to the problem domain, and you build your solution to the problem on top of that foundation of subroutines.

4.6.1 Preconditions and Postconditions

When working with subroutines as building blocks, it is important to be clear about how a subroutine interacts with the rest of the program. This interaction is specified by the *contract*

of the subroutine, as discussed in Section 4.1. A convenient way to express the contract of a subroutine is in terms of *preconditions* and *postconditions*.

The precondition of a subroutine is something that must be true when the subroutine is called, if the subroutine is to work correctly. For example, for the built-in function `Math.sqrt(x)`, a precondition is that the parameter, `x`, is greater than or equal to zero, since it is not possible to take the square root of a negative number. In terms of a contract, a precondition represents an obligation of the *caller* of the subroutine. If you call a subroutine without meeting its precondition, then there is no reason to expect it to work properly. The program might crash or give incorrect results, but you can only blame yourself, not the subroutine.

A postcondition of a subroutine represents the other side of the contract. It is something that will be true after the subroutine has run (assuming that its preconditions were met—and that there are no bugs in the subroutine). The postcondition of the function `Math.sqrt()` is that the square of the value that is returned by this function is equal to the parameter that is provided when the subroutine is called. Of course, this will only be true if the precondition—that the parameter is greater than or equal to zero—is met. A postcondition of the built-in subroutine `System.out.print()` is that the value of the parameter has been displayed on the screen.

Preconditions most often give restrictions on the acceptable values of parameters, as in the example of `Math.sqrt(x)`. However, they can also refer to global variables that are used in the subroutine. The postcondition of a subroutine specifies the task that it performs. For a function, the postcondition should specify the value that the function returns.

Subroutines are often described by comments that explicitly specify their preconditions and postconditions. When you are given a pre-written subroutine, a statement of its preconditions and postconditions tells you how to use it and what it does. When you are assigned to write a subroutine, the preconditions and postconditions give you an exact specification of what the subroutine is expected to do. I will use this approach in the example that constitutes the rest of this section. The comments are given in the form of Javadoc comments, but I will explicitly label the preconditions and postconditions. (Many computer scientists think that new doc tags `@precondition` and `@postcondition` should be added to the Javadoc system for explicit labeling of preconditions and postconditions, but that has not yet been done.)

4.6.2 A Design Example

Let's work through an example of program design using subroutines. In this example, we will use prewritten subroutines as building blocks and we will also design new subroutines that we need to complete the project.

Suppose that I have found an already-written class called `Mosaic`. This class allows a program to work with a window that displays little colored rectangles arranged in rows and columns. The window can be opened, closed, and otherwise manipulated with static member subroutines defined in the `Mosaic` class. In fact, the class defines a toolbox or API that can be used for working with such windows. Here are some of the available routines in the API, with Javadoc-style comments:

```
/**
 * Opens a "mosaic" window on the screen.
 *
 * Precondition:  The parameters rows, cols, w, and h are positive integers.
 * Postcondition: A window is open on the screen that can display rows and
 *                columns of colored rectangles. Each rectangle is w pixels
```



```

*           wide and h pixels high. The number of rows is given by
*           the first parameter and the number of columns by the
*           second. Initially, all rectangles are black. 1
* Note: The rows are numbered from 0 to rows - 1, and the columns are
* numbered from 0 to cols - 1.
*/
public static void open(int rows, int cols, int w, int h) 2

/** 3
* Sets the color of one of the rectangles in the window. 4
*
* Precondition: row and col are in the valid range of row and column numbers, 5
*               and r, g, and b are in the range 0 to 255, inclusive. 6
* Postcondition: The color of the rectangle in row number row and column 7
*               number col has been set to the color specified by r, g, 8
*               and b. r gives the amount of red in the color with 0 9
*               representing no red and 255 representing the maximum 10
*               possible amount of red. The larger the value of r, the 11
*               more red in the color. g and b work similarly for the 12
*               green and blue color components. 13
*/
public static void setColor(int row, int col, int r, int g, int b) 14

/** 15
* Gets the red component of the color of one of the rectangles. 16
*
* Precondition: row and col are in the valid range of row and column numbers. 17
* Postcondition: The red component of the color of the specified rectangle is 18
*               returned as an integer in the range 0 to 255 inclusive. 19
*/
public static int getRed(int row, int col) 20

/** 21
* Like getRed, but returns the green component of the color. 22
*/
public static int getGreen(int row, int col) 23

/** 24
* Like getRed, but returns the blue component of the color. 25
*/
public static int getBlue(int row, int col) 26

/**
* Tests whether the mosaic window is currently open. 27
*
* Precondition: None.
* Postcondition: The return value is true if the window is open when this 28
*               function is called, and it is false if the window is 29
*               closed.
*/
public static boolean isOpen() 30

/**

```

```

* Inserts a delay in the program (to regulate the speed at which the colors 1
* are changed, for example). 2
*
* Precondition: milliseconds is a positive integer. 3
* Postcondition: The program has paused for at least the specified number 4
*                 of milliseconds, where one second is equal to 1000 5
*                 milliseconds. 6
*/
public static void delay(int milliseconds) 7

```

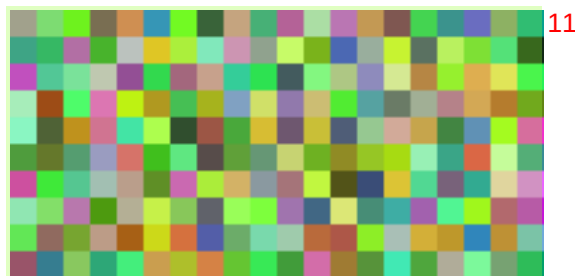
Remember that these subroutines are members of the `Mosaic` class, so when they are called 8 from outside `Mosaic`, the name of the class must be included as part of the name of the routine. For example, we'll have to use the name `Mosaic.isOpen()` rather than simply `isOpen()`.

```

* * * 9

```

My idea is to use the `Mosaic` class as the basis for a neat animation. I want to fill the 10 window with randomly colored squares, and then randomly change the colors in a loop that continues as long as the window is open. “Randomly change the colors” could mean a lot of different things, but after thinking for a while, I decide it would be interesting to have a “disturbance” that wanders randomly around the window, changing the color of each square that it encounters. Here’s a picture showing what the contents of the window might look like at one point in time:



With basic routines for manipulating the window as a foundation, I can turn to the specific 12 problem at hand. A basic outline for my program is

```

Open a Mosaic window 13
Fill window with random colors; 14
Move around, changing squares at random. 15

```

Filling the window with random colors seems like a nice coherent task that I can work on 16 separately, so let's decide to write a separate subroutine to do it. The third step can be expanded a bit more, into the steps: Start in the middle of the window, then keep moving to a new square and changing the color of that square. This should continue as long as the mosaic window is still open. Thus we can refine the algorithm to:

```

Open a Mosaic window 17
Fill window with random colors; 18
Set the current position to the middle square in the window; 19
As long as the mosaic window is open: 20
    Randomly change color of the square at the current position; 21
    Move current position up, down, left, or right, at random; 22

```

I need to represent the current position in some way. That can be done with two **int** variables **currentRow** and **currentColumn** that hold the row number and the column number of the square where the disturbance is currently located. I'll use 10 rows and 20 columns of squares in my mosaic, so setting the current position to be in the center means setting **currentRow** to 5 and **currentColumn** to 10. I already have a subroutine, **Mosaic.open()**, to open the window, and I have a function, **Mosaic.isOpen()**, to test whether the window is open. To keep the main routine simple, I decide that I will write two more subroutines of my own to carry out the two tasks in the while loop. The algorithm can then be written in Java as:

```
Mosaic.open(10,20,10,10)
fillWithRandomColors();
currentRow = 5;           // Middle row, halfway down the window.
currentColumn = 10;       // Middle column.
while ( Mosaic.isOpen() ) {
    changeToRandomColor(currentRow, currentColumn);
    randomMove();
}
```

With the proper wrapper, this is essentially the **main()** routine of my program. It turns out I have to make one small modification: To prevent the animation from running too fast, the line “**Mosaic.delay(20);**” is added to the **while** loop.

The **main()** routine is taken care of, but to complete the program, I still have to write the subroutines **fillWithRandomColors()**, **changeToRandomColor(int,int)**, and **randomMove()**. Writing each of these subroutines is a separate, small task. The **fillWithRandomColors()** routine is defined by the postcondition that “each of the rectangles in the mosaic has been changed to a random color.” Pseudocode for an algorithm to accomplish this task can be given as:

```
For each row:
    For each column:
        set the square in that row and column to a random color
```

“For each row” and “for each column” can be implemented as for loops. We've already planned to write a subroutine **changeToRandomColor** that can be used to set the color. (The possibility of reusing subroutines in several places is one of the big payoffs of using them!) So, **fillWithRandomColors()** can be written in proper Java as:

```
static void fillWithRandomColors() {
    for (int row = 0; row < 10; row++)
        for (int column = 0; column < 20; column++)
            changeToRandomColor(row,column);
}
```

Turning to the **changeToRandomColor** subroutine, we already have a method in the **Mosaic** class, **Mosaic.setColor()**, that can be used to change the color of a square. If we want a random color, we just have to choose random values for **r**, **g**, and **b**. According to the precondition of the **Mosaic.setColor()** subroutine, these random values must be integers in the range from 0 to 255. A formula for randomly selecting such an integer is “**(int)(256*Math.random())**”. So the random color subroutine becomes:

```
static void changeToRandomColor(int rowNum, int colNum) {
    int red = (int)(256*Math.random());
    int green = (int)(256*Math.random());
    int blue = (int)(256*Math.random());
```

```

        mosaic.setColor(rowNum,colNum,red,green,blue); 1
    }

```

Finally, consider the `randomMove` subroutine, which is supposed to randomly move the disturbance up, down, left, or right. To make a random choice among four directions, we can choose a random integer in the range 0 to 3. If the integer is 0, move in one direction; if it is 1, move in another direction; and so on. The position of the disturbance is given by the variables `currentRow` and `currentColumn`. To “move up” means to subtract 1 from `currentRow`. This leaves open the question of what to do if `currentRow` becomes -1, which would put the disturbance above the window. Rather than let this happen, I decide to move the disturbance to the opposite edge of the applet by setting `currentRow` to 9. (Remember that the 10 rows are numbered from 0 to 9.) Moving the disturbance down, left, or right is handled similarly. If we use a `switch` statement to decide which direction to move, the code for `randomMove` becomes:

```

int directionNum;
directionNum = (int)(4*Math.random());
switch (directionNum) {
    case 0: // move up
        currentRow--;
        if (currentRow < 0) // CurrentRow is outside the mosaic;
            currentRow = 9; // move it to the opposite edge.
        break;
    case 1: // move right
        currentColumn++;
        if (currentColumn >= 20)
            currentColumn = 0;
        break;
    case 2: // move down
        currentRow++;
        if (currentRow >= 10)
            currentRow = 0;
        break;
    case 3: // move left
        currentColumn--;
        if (currentColumn < 0)
            currentColumn = 19;
        break;
}

```

4.6.3 The Program ⁴

Putting this all together, we get the following complete program. Note that I’ve added Javadoc-style comments for the class itself and for each of the subroutines. The variables `currentRow` and `currentColumn` are defined as static members of the class, rather than local variables, because each of them is used in several different subroutines. This program actually depends on two other classes, `Mosaic` and another class called `MosaicCanvas` that is used by `Mosaic`. If you want to compile and run this program, both of these classes must be available to the program. ⁵

```

/**
 * This program opens a window full of randomly colored squares. A "disturbance"
 * moves randomly around in the window, randomly changing the color of each
 * square that it visits. The program runs until the user closes the window.
 */

public class RandomMosaicWalk {

    static int currentRow;    // Row currently containing the disturbance.
    static int currentColumn; // Column currently containing disturbance.

    /**
     * The main program creates the window, fills it with random colors,
     * and then moves the disturbances in a random walk around the window
     * as long as the window is open.
     */
    public static void main(String[] args) {
        Mosaic.open(10,20,10,10);
        fillWithRandomColors();
        currentRow = 5;    // start at center of window
        currentColumn = 10;
        while (Mosaic.isOpen()) {
            changeToRandomColor(currentRow, currentColumn);
            randomMove();
            Mosaic.delay(20);
        }
    } // end main

    /**
     * Fills the window with randomly colored squares.
     * Precondition: The mosaic window is open.
     * Postcondition: Each square has been set to a random color.
     */
    static void fillWithRandomColors() {
        for (int row=0; row < 10; row++) {
            for (int column=0; column < 20; column++) {
                changeToRandomColor(row, column);
            }
        }
    } // end fillWithRandomColors

    /**
     * Changes one square to a new randomly selected color.
     * Precondition: The specified rowNum and colNum are in the valid range
     *                of row and column numbers.
     * Postcondition: The square in the specified row and column has
     *                been set to a random color.
     * @param rowNum the row number of the square, counting rows down
     *                from 0 at the top
     * @param colNum the column number of the square, counting columns over
     *                from 0 at the left
     */
    static void changeToRandomColor(int rowNum, int colNum) {
        int red = (int)(256*Math.random());    // Choose random levels in range
        int green = (int)(256*Math.random());    //      0 to 255 for red, green,
        int blue = (int)(256*Math.random());    //      and blue color components.
    }
}

```

```

        Mosaic.setColor(rowNum,colNum,red,green,blue);
    } // end of changeToRandomColor()

    /**
     * Move the disturbance.
     * Precondition:   The global variables currentRow and currentColumn
     *                are within the legal range of row and column numbers.
     * Postcondition: currentRow or currentColumn is changed to one of the
     *                neighboring positions in the grid -- up, down, left, or
     *                right from the current position. If this moves the
     *                position outside of the grid, then it is moved to the
     *                opposite edge of the grid.
     */
    static void randomMove() {
        int directionNum; // Randomly set to 0, 1, 2, or 3 to choose direction.
        directionNum = (int)(4*Math.random());
        switch (directionNum) {
            case 0: // move up
                currentRow--;
                if (currentRow < 0)
                    currentRow = 9;
                break;
            case 1: // move right
                currentColumn++;
                if (currentColumn >= 20)
                    currentColumn = 0;
                break;
            case 2: // move down
                currentRow++;
                if (currentRow >= 10)
                    currentRow = 0;
                break;
            case 3: // move left
                currentColumn--;
                if (currentColumn < 0)
                    currentColumn = 19;
                break;
        }
    } // end randomMove
} // end class RandomMosaicWalk

```

4.7 The Truth About Declarations ²

NAMES ARE FUNDAMENTAL TO PROGRAMMING, as I said a few chapters ago. There are a lot of details involved in declaring and using names. I have been avoiding some of those details. In this section, I'll reveal most of the truth (although still not the full truth) about declaring and using variables in Java. The material in the subsections "Initialization in Declarations" and "Named Constants" is particularly important, since I will be using it regularly in future chapters. ³

4.7.1 Initialization in Declarations ²

When a variable declaration is executed, memory is allocated for the variable. This memory ³ must be initialized to contain some definite value before the variable can be used in an expression. In the case of a local variable, the declaration is often followed closely by an assignment statement that does the initialization. For example,

```
int count;    // Declare a variable named count. 4
count = 0;    // Give count its initial value.
```

However, the truth about declaration statements is that it is legal to include the initialization ⁵ of the variable in the declaration statement. The two statements above can therefore be abbreviated as

```
int count = 0; // Declare count and give it an initial value. 6
```

The computer still executes this statement in two steps: Declare the variable `count`, then assign ⁷ the value 0 to the newly created variable. The initial value does not have to be a constant. It can be any expression. It is legal to initialize several variables in one declaration statement. For example,

```
char firstInitial = 'D', secondInitial = 'E'; 1
int x, y = 1;    // OK, but only y has been initialized!
int N = 3, M = N+2; // OK, N is initialized
                  // before its value is used.
```

This feature is especially common in `for` loops, since it makes it possible to declare a loop control ⁸ variable at the same point in the loop where it is initialized. Since the loop control variable generally has nothing to do with the rest of the program outside the loop, it's reasonable to have its declaration in the part of the program where it's actually used. For example:

```
for ( int i = 0; i < 10; i++ ) { 9
    System.out.println(i);
}
```

Again, you should remember that this is simply an abbreviation for the following, where I've ¹⁰ added an extra pair of braces to show that `i` is considered to be local to the `for` statement and no longer exists after the `for` loop ends:

```
{ 11
    int i;
    for ( i = 0; i < 10; i++ ) {
        System.out.println(i);
    }
}
```

(You might recall, by the way, that for “for-each” loops, the special type of `for` statement ¹² that is used with enumerated types, declaring the variable in the `for` is **required**. See Subsection 3.4.4.)

A member variable can also be initialized at the point where it is declared, just as for a ¹³ local variable. For example:

```
public class Bank { 14
    static double interestRate = 0.05;
    static int maxWithdrawal = 200;
```



```

    .
    . // More variables and subroutines.
    .
}

```

A static member variable is created as soon as the class is loaded by the Java interpreter, and the initialization is also done at that time. In the case of member variables, this is not simply an abbreviation for a declaration followed by an assignment statement. Declaration statements are the only type of statement that can occur outside of a subroutine. Assignment statements cannot, so the following is illegal:

```

public class Bank {
    static double interestRate;
    interestRate = 0.05; // ILLEGAL:
    .                  // Can't be outside a subroutine!
    .
    .
}

```

Because of this, declarations of member variables often include initial values. In fact, as mentioned in Subsection 4.2.4, if no initial value is provided for a member variable, then a default initial value is used. For example, when declaring an integer member variable, `count`, “`static int count;`” is equivalent to “`static int count = 0;`”.

4.7.2 Named Constants

Sometimes, the value of a variable is not supposed to change after it is initialized. For example, in the above example where `interestRate` is initialized to the value 0.05, it’s quite possible that that is meant to be the value throughout the entire program. In this case, the programmer is probably defining the variable, `interestRate`, to give a meaningful name to the otherwise meaningless number, 0.05. It’s easier to understand what’s going on when a program says “`principal += principal*interestRate;`” rather than “`principal += principal*0.05;`”.

In Java, the modifier “`final`” can be applied to a variable declaration to ensure that the value stored in the variable cannot be changed after the variable has been initialized. For example, if the member variable `interestRate` is declared with

```
final static double interestRate = 0.05;
```

then it would be impossible for the value of `interestRate` to change anywhere else in the program. Any assignment statement that tries to assign a value to `interestRate` will be rejected by the computer as a syntax error when the program is compiled.

It is legal to apply the `final` modifier to local variables and even to formal parameters, but it is most useful for member variables. I will often refer to a static member variable that is declared to be `final` as a *named constant*, since its value remains constant for the whole time that the program is running. The readability of a program can be greatly enhanced by using named constants to give meaningful names to important quantities in the program. A recommended style rule for named constants is to give them names that consist entirely of upper case letters, with underscore characters to separate words if necessary. For example, the preferred style for the interest rate constant would be

```
final static double INTEREST_RATE = 0.05;
```

This is the style that is generally used in Java’s standard classes, which define many named constants. For example, we have already seen that the *Math* class contains a variable `Math.PI`. This variable is declared in the *Math* class as a “`public final static`” variable of type `double`.

Similarly, the `Color` class contains named constants such as `Color.RED` and `Color.YELLOW` 1 which are public final static variables of type `Color`. Many named constants are created just to give meaningful names to be used as parameters in subroutine calls. For example, the standard class named `Font` contains named constants `Font.PLAIN`, `Font.BOLD`, and `Font.ITALIC`. These constants are used for specifying different styles of text when calling various subroutines in the `Font` class.

Enumerated type constants (See Subsection 2.3.3.) are also examples of named constants. 2 The enumerated type definition

```
enum Alignment { LEFT, RIGHT, CENTER } 3
```

defines the constants `Alignment.LEFT`, `Alignment.RIGHT`, and `Alignment.CENTER`. Technically, 4 *Alignment* is a class, and the three constants are public final static members of that class. Defining the enumerated type is similar to defining three constants of type, say, `int`:

```
public static final int ALIGNMENT_LEFT = 0; 5
public static final int ALIGNMENT_RIGHT = 1; 6
public static final int ALIGNMENT_CENTER = 2; 7
```

In fact, this is how things were generally done before the introduction of enumerated types in 8 Java 5.0, and it is what is done with the constants `Font.PLAIN`, `Font.BOLD`, and `Font.ITALIC` mentioned above. Using the integer constants, you could define a variable of type `int` and assign it the values `ALIGNMENT_LEFT`, `ALIGNMENT_RIGHT`, or `ALIGNMENT_CENTER` to represent different types of alignment. The only problem with this is that the computer has no way of knowing that you intend the value of the variable to represent an alignment, and it will not raise any objection if the value that is assigned to the variable is not one of the three valid alignment values.

With the enumerated type, on the other hand, the only values that can be assigned to 9 a variable of type *Alignment* are the constant values that are listed in the definition of the enumerated type. Any attempt to assign an invalid value to the variable is a syntax error which the computer will detect when the program is compiled. This extra safety is one of the major advantages of enumerated types.

* * * 10

Curiously enough, one of the major reasons to use named constants is that it's easy to 11 change the value of a named constant. Of course, the value can't change while the program is running. But between runs of the program, it's easy to change the value in the source code and recompile the program. Consider the interest rate example. It's quite possible that the value of the interest rate is used many times throughout the program. Suppose that the bank changes the interest rate and the program has to be modified. If the literal number 0.05 were used throughout the program, the programmer would have to track down each place where the interest rate is used in the program and change the rate to the new value. (This is made even harder by the fact that the number 0.05 might occur in the program with other meanings besides the interest rate, as well as by the fact that someone might have used 0.025 to represent half the interest rate.) On the other hand, if the named constant `INTEREST_RATE` is declared and used consistently throughout the program, then only the single line where the constant is initialized needs to be changed.

As an extended example, I will give a new version of the `RandomMosaicWalk` program from 12 the previous section. This version uses named constants to represent the number of rows in the mosaic, the number of columns, and the size of each little square. The three constants are declared as `final static` member variables with the lines:

```
final static int ROWS = 30;           // Number of rows in mosaic. 1
final static int COLUMNS = 30;      // Number of columns in mosaic.
final static int SQUARE_SIZE = 15;  // Size of each square in mosaic.
```

The rest of the program is carefully modified to use the named constants. For example, in the new version of the program, the Mosaic window is opened with the statement

```
Mosaic.open(ROWS, COLUMNS, SQUARE_SIZE, SQUARE_SIZE); 3
```

Sometimes, it's not easy to find all the places where a named constant needs to be used. If you don't use the named constant consistently, you've more or less defeated the purpose. It's always a good idea to run a program using several different values for any named constants, to test that it works properly in all cases.

Here is the complete new program, `RandomMosaicWalk2`, with all modifications from the previous version shown in *italic*. I've left out some of the comments to save space.

```
public class RandomMosaicWalk2 { 6
    final static int ROWS = 30;           // Number of rows in mosaic.
    final static int COLUMNS = 30;      // Number of columns in mosaic.
    final static int SQUARE_SIZE = 15;  // Size of each square in mosaic.

    static int currentRow;    // Row currently containing the disturbance.
    static int currentColumn; // Column currently containing disturbance.

    public static void main(String[] args) {
        Mosaic.open( ROWS, COLUMNS, SQUARE_SIZE, SQUARE_SIZE );
        fillWithRandomColors();
        currentRow = ROWS / 2;    // start at center of window
        currentColumn = COLUMNS / 2;
        while (Mosaic.isOpen()) {
            changeToRandomColor(currentRow, currentColumn);
            randomMove();
            Mosaic.delay(20);
        }
    } // end main

    static void fillWithRandomColors() {
        for (int row=0; row < ROWS; row++) {
            for (int column=0; column < COLUMNS; column++) {
                changeToRandomColor(row, column);
            }
        }
    } // end fillWithRandomColors

    static void changeToRandomColor(int rowNum, int colNum) {
        int red = (int)(256*Math.random());    // Choose random levels in range
        int green = (int)(256*Math.random()); //      0 to 255 for red, green,
        int blue = (int)(256*Math.random());  //      and blue color components.
        Mosaic.setColor(rowNum,colNum,red,green,blue);
    } // end changeToRandomColor

    static void randomMove() {
        int directionNum; // Randomly set to 0, 1, 2, or 3 to choose direction.
        directionNum = (int)(4*Math.random());
        switch (directionNum) {
            case 0: // move up
```

```

        currentRow--;
        if (currentRow < 0)
            currentRow = ROWS - 1;
        break;
    case 1: // move right
        currentColumn++;
        if (currentColumn >= COLUMNS)
            currentColumn = 0;
        break;
    case 2: // move down
        currentRow++;
        if (currentRow >= ROWS)
            currentRow = 0;
        break;
    case 3: // move left
        currentColumn--;
        if (currentColumn < 0)
            currentColumn = COLUMNS - 1;
        break;
    }
} // end randomMove
} // end class RandomMosaicWalk2

```

1

4.7.3 Naming and Scope Rules ²

When a variable declaration is executed, memory is allocated for that variable. The variable name can be used in at least some part of the program source code to refer to that memory or to the data that is stored in the memory. The portion of the program source code where the variable name is valid is called the **scope** of the variable. Similarly, we can refer to the scope of subroutine names and formal parameter names. ³

For static member subroutines, scope is straightforward. The scope of a static subroutine is the entire source code of the class in which it is defined. That is, it is possible to call the subroutine from any point in the class, including at a point in the source code before the point where the definition of the subroutine appears. It is even possible to call a subroutine from within itself. This is an example of something called “recursion,” a fairly advanced topic that we will return to later. ⁴

For a variable that is declared as a static member variable in a class, the situation is similar, but with one complication. It is legal to have a local variable or a formal parameter that has the same name as a member variable. In that case, within the scope of the local variable or parameter, the member variable is **hidden**. Consider, for example, a class named `Game` that has the form: ⁵

```

public class Game {
    static int count; // member variable

    static void playGame() {
        int count; // local variable
        .
        . // Some statements to define playGame()
        .
    }
}

```

6

```
.
. // More variables and subroutines. 1
.
```

```
} // end Game 2
```

In the statements that make up the body of the `playGame()` subroutine, the name “`count`”³ refers to the local variable. In the rest of the `Game` class, “`count`” refers to the member variable, unless hidden by other local variables or parameters named `count`. However, there is one further complication. The member variable named `count` can also be referred to by the full name `Game.count`. Usually, the full name is only used outside the class where `count` is defined. However, there is no rule against using it inside the class. The full name, `Game.count`, can be used inside the `playGame()` subroutine to refer to the member variable. So, the full scope rule is that the scope of a static member variable includes the entire class in which it is defined, but where the simple name of the member variable is hidden by a local variable or formal parameter name, the member variable must be referred to by its full name of the form $\langle \text{className} \rangle . \langle \text{variableName} \rangle$. (Scope rules for non-static members are similar to those for static members, except that, as we shall see, non-static members cannot be used in static subroutines.)

The scope of a formal parameter of a subroutine is the block that makes up the body of the subroutine. The scope of a local variable extends from the declaration statement that defines the variable to the end of the block in which the declaration occurs. As noted above, it is possible to declare a loop control variable of a `for` loop in the `for` statement, as in “`for (int i=0; i < 10; i++)`”. The scope of such a declaration is considered as a special case: It is valid only within the `for` statement and does not extend to the remainder of the block that contains the `for` statement.⁴

It is not legal to redefine the name of a formal parameter or local variable within its scope, even in a nested block. For example, this is not allowed:⁵

```
void badSub(int y) {
    int x;
    while (y > 0) {
        int x; // ERROR: x is already defined.
        .
        .
        .
    }
}
```

In many languages, this would be legal; the declaration of `x` in the `while` loop would hide the original declaration. It is not legal in Java; however, once the block in which a variable is declared ends, its name does become available for reuse in Java. For example:⁷

```
void goodSub(int y) {
    while (y > 10) {
        int x;
        .
        .
        .
        // The scope of x ends here.
    }
    while (y > 0) {
```

```
    int x;  // OK: Previous declaration of x has expired. 1
    .
    .
    .
}
}
```

You might wonder whether local variable names can hide subroutine names. This can't 2 happen, for a reason that might be surprising. There is no rule that variables and subroutines have to have different names. The computer can always tell whether a name refers to a variable or to a subroutine, because a subroutine name is always followed by a left parenthesis. It's perfectly legal to have a variable called `count` and a subroutine called `count` in the same class. (This is one reason why I often write subroutine names with parentheses, as when I talk about the `main()` routine. It's a good idea to think of the parentheses as part of the name.) Even more is true: It's legal to reuse class names to name variables and subroutines. The syntax rules of Java guarantee that the computer can always tell when a name is being used as a class name. A class name is a type, and so it can be used to declare variables and formal parameters and to specify the return type of a function. This means that you could legally have a class called `Insanity` in which you declare a function

```
static Insanity Insanity( Insanity Insanity ) { ... } 3
```

The first `Insanity` is the return type of the function. The second is the function name, the 4 third is the type of the formal parameter, and the fourth is a formal parameter name. However, please remember that not everything that is possible is a good idea!

Exercises for Chapter 4¹

1. To “capitalize” a string means to change the first letter of each word in the string to upper case (if it is not already upper case). For example, a capitalized version of “Now is the time to act!” is “Now Is The Time To Act!”. Write a subroutine named `printCapitalized` that will print a capitalized version of a string to standard output. The string to be printed should be a parameter to the subroutine. Test your subroutine with a `main()` routine that gets a line of input from the user and applies the subroutine to it.

Note that a letter is the first letter of a word if it is not immediately preceded in the string by another letter. Recall that there is a standard **boolean**-valued function `Character.isLetter(char)` that can be used to test whether its parameter is a letter. There is another standard **char**-valued function, `Character.toUpperCase(char)`, that returns a capitalized version of the single character passed to it as a parameter. That is, if the parameter is a letter, it returns the upper-case version. If the parameter is not a letter, it just returns a copy of the parameter.

2. The hexadecimal digits are the ordinary, base-10 digits '0' through '9' plus the letters 'A' through 'F'. In the hexadecimal system, these digits represent the values 0 through 15, respectively. Write a function named `hexValue` that uses a `switch` statement to find the hexadecimal value of a given character. The character is a parameter to the function, and its hexadecimal value is the return value of the function. You should count lower case letters 'a' through 'f' as having the same value as the corresponding upper case letters. If the parameter is not one of the legal hexadecimal digits, return -1 as the value of the function.

A hexadecimal integer is a sequence of hexadecimal digits, such as 34A7, FF8, 174204, or FADE. If `str` is a string containing a hexadecimal integer, then the corresponding base-10 integer can be computed as follows:

```
value = 0;
for ( i = 0; i < str.length(); i++ )
    value = value*16 + hexValue( str.charAt(i) );
```

Of course, this is not valid if `str` contains any characters that are not hexadecimal digits. Write a program that reads a string from the user. If all the characters in the string are hexadecimal digits, print out the corresponding base-10 value. If not, print out an error message.

3. Write a function that simulates rolling a pair of dice until the total on the dice comes up to be a given number. The number that you are rolling for is a parameter to the function. The number of times you have to roll the dice is the return value of the function. The parameter should be one of the possible totals: 2, 3, ..., 12. The function should throw an *IllegalArgumentException* if this is not the case. Use your function in a program that computes and prints the number of rolls it takes to get snake eyes. (Snake eyes means that the total showing on the dice is 2.)

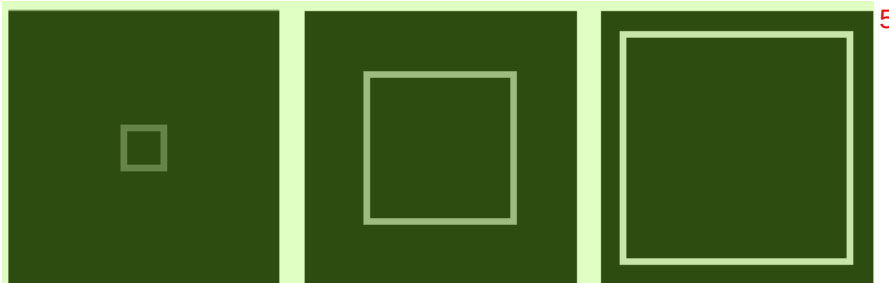
4. This exercise builds on Exercise 4.3. Every time you roll the dice repeatedly, trying to get a given total, the number of rolls it takes can be different. The question naturally arises, what's the average number of rolls to get a given total? Write a function that performs the experiment of rolling to get a given total 10000 times. The desired total is

a parameter to the subroutine. The average number of rolls is the return value. Each individual experiment should be done by calling the function you wrote for Exercise 4.3. Now, write a main program that will call your function once for each of the possible totals (2, 3, ..., 12). It should make a table of the results, something like:

Total On Dice	Average Number of Rolls
2	35.8382
3	18.0607
.	.
.	.

5. The sample program *RandomMosaicWalk.java* from Section 4.6 shows a “disturbance” that wanders around a grid of colored squares. When the disturbance visits a square, the color of that square is changed. The applet at the bottom of Section 4.7 in the on-line version of this book shows a variation on this idea. In this applet, all the squares start out with the default color, black. Every time the disturbance visits a square, a small amount is added to the red component of the color of that square. Write a subroutine that will add 25 to the red component of one of the squares in the mosaic. The row and column numbers of the square should be passed as parameters to the subroutine. Recall that you can discover the current red component of the square in row *r* and column *c* with the function call `Mosaic.getRed(r,c)`. Use your subroutine as a substitute for the `changeToRandomColor()` subroutine in the program *RandomMosaicWalk2.java*. (This is the improved version of the program from Section 4.7 that uses named constants for the number of rows, number of columns, and square size.) Set the number of rows and the number of columns to 80. Set the square size to 5.

6. For this exercise, you will write another program based on the non-standard `Mosaic` class that was presented in Section 4.6. While the program does not do anything particularly interesting, it’s interesting as a programming problem. An applet that does the same thing as the program can be seen in the on-line version of this book. Here is a picture showing what it looks like at several different times:



The program will show a rectangle that grows from the center of the applet to the edges, getting brighter as it grows. The rectangle is made up of the little squares of the mosaic. You should first write a subroutine that draws a rectangle on a `Mosaic` window. More specifically, write a subroutine named `rectangle` such that the subroutine call statement

```
rectangle(top,left,height,width,r,g,b);
```

will call `Mosaic.setColor(row,col,r,g,b)` for each little square that lies on the outline of a rectangle. The topmost row of the rectangle is specified by `top`. The number of rows in the rectangle is specified by `height` (so the bottommost row is `top+height-1`). The leftmost column of the rectangle is specified by `left`. The number of columns in the rectangle is specified by `width` (so the rightmost column is `left+width-1`.) 1

The animation loops through the same sequence of steps over and over. In each step, a rectangle is drawn in gray (that is, with all three color components having the same value). There is a pause of 200 milliseconds so the user can see the rectangle. Then the very same rectangle is drawn in black, effectively erasing the gray rectangle. Finally, the variables giving the top row, left column, size, and color level of the rectangle are adjusted to get ready for the next step. In the applet, the color level starts at 50 and increases by 10 after each step. When the rectangle gets to the outer edge of the applet, the loop ends. The animation then starts again at the beginning of the loop. You might want to make a subroutine that does one loop through all the steps of the animation. 2

The `main()` routine simply opens a Mosaic window and then does the animation loop over and over until the user closes the window. There is a 1000 millisecond delay between one animation loop and the next. Use a Mosaic window that has 41 rows and 41 columns. (I advise you **not** to use named constants for the numbers of rows and columns, since the problem is complicated enough already.) 3

Quiz on Chapter 4 ¹

1. A “black box” has an interface and an implementation. Explain what is meant by the terms *interface* and *implementation*. ²

2. A subroutine is said to have a *contract*. What is meant by the contract of a subroutine? ³
When you want to use a subroutine, why is it important to understand its contract? The contract has both “syntactic” and “semantic” aspects. What is the syntactic aspect? What is the semantic aspect?

3. Briefly explain how subroutines can be a useful tool in the top-down design of programs. ⁴

4. Discuss the concept of *parameters*. What are parameters for? What is the difference between *formal parameters* and *actual parameters*? ⁵

5. Give two different reasons for using named constants (declared with the **final** modifier). ⁶

6. What is an API? Give an example. ⁷

7. Write a subroutine named “stars” that will output a line of stars to standard output. (A star is the character “*”). The number of stars should be given as a parameter to the subroutine. Use a *for* loop. For example, the command “stars(20)” would output ⁸

```
*****
```

⁹

8. Write a `main()` routine that uses the subroutine that you wrote for Question 7 to output 10 lines of stars with 1 star in the first line, 2 stars in the second line, and so on, as shown below. ¹⁰

```
*
**
***
****
*****
*****
*****
*****
*****
*****
*****
```

¹¹

9. Write a function named `countChars` that has a *String* and a **char** as parameters. The function should count the number of times the character occurs in the string, and it should return the result as the value of the function. ¹²

10. Write a subroutine with three parameters of type *int*. The subroutine should determine which of its parameters is smallest. The value of the smallest parameter should be returned as the value of the subroutine. ¹³

Chapter 5

Programming in the Large II: Objects and Classes¹

WHEREAS A SUBROUTINE represents a single task, an object can encapsulate both data (in the form of instance variables) and a number of different tasks or “behaviors” related to that data (in the form of instance methods). Therefore objects provide another, more sophisticated type of structure that can be used to help manage the complexity of large programs.²

This chapter covers the creation and use of objects in Java. Section 5.5 covers the central ideas of object-oriented programming: inheritance and polymorphism. However, in this textbook, we will generally use these ideas in a limited form, by creating independent classes and building on existing classes rather than by designing entire hierarchies of classes from scratch. Section 5.6 and Section 5.7 cover some of the many details of object oriented programming in Java. Although these details are used occasionally later in the book, you might want to skim through them now and return to them later when they are actually needed.³

5.1 Objects, Instance Methods, and Instance Variables⁴

OBJECT-ORIENTED PROGRAMMING (OOP) represents an attempt to make programs more closely model the way people think about and deal with the world. In the older styles of programming, a programmer who is faced with some problem must identify a computing task that needs to be performed in order to solve the problem. Programming then consists of finding a sequence of instructions that will accomplish that task. But at the heart of object-oriented programming, instead of tasks we find objects—entities that have behaviors, that hold information, and that can interact with one another. Programming consists of designing a set of objects that somehow model the problem at hand. Software objects in the program can represent real or abstract entities in the problem domain. This is supposed to make the design of the program more natural and hence easier to get right and easier to understand.⁵

To some extent, OOP is just a change in point of view. We can think of an object in standard programming terms as nothing more than a set of variables together with some subroutines for manipulating those variables. In fact, it is possible to use object-oriented techniques in any programming language. However, there is a big difference between a language that makes OOP possible and one that actively supports it. An object-oriented programming language such as Java includes a number of features that make it very different from a standard language. In order to make effective use of those features, you have to “orient” your thinking correctly.⁶

5.1.1 Objects, Classes, and Instances ¹

Objects are closely related to classes. We have already been working with classes for several chapters, and we have seen that a class can contain variables and subroutines. If an object is also a collection of variables and subroutines, how do they differ from classes? And why does it require a different type of thinking to understand and use them effectively? In the one section where we worked with objects rather than classes, Section 3.8, it didn't seem to make much difference: We just left the word “**static**” out of the subroutine definitions! ²

I have said that classes “describe” objects, or more exactly that the non-static portions of classes describe objects. But it's probably not very clear what this means. The more usual terminology is to say that objects *belong to* classes, but this might not be much clearer. (There is a real shortage of English words to properly distinguish all the concepts involved. An object certainly doesn't “belong” to a class in the same way that a member variable “belongs” to a class.) From the point of view of programming, it is more exact to say that classes are used to create objects. A class is a kind of factory for constructing objects. The non-static parts of the class specify, or describe, what variables and subroutines the objects will contain. This is part of the explanation of how objects differ from classes: Objects are created and destroyed as the program runs, and there can be many objects with the same structure, if they are created using the same class. ³

Consider a simple class whose job is to group together a few static member variables. For example, the following class could be used to store information about the person who is using the program: ⁴

```
class UserData {  
    static String name;  
    static int age;  
}
```

⁵

In a program that uses this class, there is only one copy of each of the variables `UserData.name` and `UserData.age`. There can only be one “user,” since we only have memory space to store data about one user. The class, `UserData`, and the variables it contains exist as long as the program runs. Now, consider a similar class that includes non-static variables: ⁶

```
class PlayerData {  
    String name;  
    int age;  
}
```

⁷

In this case, there is no such variable as `PlayerData.name` or `PlayerData.age`, since `name` and `age` are not static members of `PlayerData`. So, there is nothing much in the class at all—except the potential to create objects. But, it's a lot of potential, since it can be used to create any number of objects! Each object will have its **own** variables called `name` and `age`. There can be many “players” because we can make new objects to represent new players on demand. A program might use this class to store information about multiple players in a game. Each player has a name and an age. When a player joins the game, a new `PlayerData` object can be created to represent that player. If a player leaves the game, the `PlayerData` object that represents that player can be destroyed. A system of objects in the program is being used to dynamically model what is happening in the game. You can't do this with “static” variables! ⁸

In Section 3.8, we worked with applets, which are objects. The reason they didn't seem to be any different from classes is because we were only working with one applet in each class that we looked at. But one class can be used to make many applets. Think of an applet that scrolls ⁹

a message across a Web page. There could be several such applets on the same page, all created from the same class. If the scrolling message in the applet is stored in a non-static variable, then each applet will have its own variable, and each applet can show a different message. The situation is even clearer if you think about windows, which, like applets, are objects. As a program runs, many windows might be opened and closed, but all those windows can belong to the same class. Here again, we have a dynamic situation where multiple objects are created and destroyed as a program runs.

* * * 2

An object that belongs to a class is said to be an *instance* of that class. The variables that the object contains are called *instance variables*. The subroutines that the object contains are called *instance methods*. (Recall that in the context of object-oriented programming, *method* is a synonym for “subroutine”. From now on, since we are doing object-oriented programming, I will prefer the term “method.”) For example, if the `PlayerData` class, as defined above, is used to create an object, then that object is an instance of the `PlayerData` class, and `name` and `age` are instance variables in the object. It is important to remember that the class of an object determines the **types** of the instance variables; however, the actual data is contained inside the individual objects, not the class. Thus, each object has its own set of data.

An applet that scrolls a message across a Web page might include a subroutine named `scroll()`. Since the applet is an object, this subroutine is an instance method of the applet. The source code for the method is in the class that is used to create the applet. Still, it’s better to think of the instance method as belonging to the object, not to the class. The non-static subroutines in the class merely specify the instance methods that every object created from the class will contain. The `scroll()` methods in two different applets do the same thing in the sense that they both scroll messages across the screen. But there is a real difference between the two `scroll()` methods. The messages that they scroll can be different. You might say that the method definition in the class specifies what type of behavior the objects will have, but the specific behavior can vary from object to object, depending on the values of their instance variables.

As you can see, the static and the non-static portions of a class are very different things and serve very different purposes. Many classes contain only static members, or only non-static. However, it is possible to mix static and non-static members in a single class, and we’ll see a few examples later in this chapter where it is reasonable to do so. You should distinguish between the **source code** for the class, and the **class itself**. The source code determines both the class and the objects that are created from that class. The “static” definitions in the source code specify the things that are part of the class itself, whereas the non-static definitions in the source code specify things that will become part of every instance object that is created from the class. By the way, static member variables and static member subroutines in a class are sometimes called *class variables* and *class methods*, since they belong to the class itself, rather than to instances of that class.

5.1.2 Fundamentals of Objects 6

So far, I’ve been talking mostly in generalities, and I haven’t given you much idea what you have to put in a program if you want to work with objects. Let’s look at a specific example to see how it works. Consider this extremely simplified version of a `Student` class, which could be used to store information about students taking a course:

```

public class Student {
    public String name; // Student's name.
    public double test1, test2, test3; // Grades on three tests.

    public double getAverage() { // compute average test grade
        return (test1 + test2 + test3) / 3;
    }
} // end of class Student

```

None of the members of this class are declared to be **static**, so the class exists only for creating objects. This class definition says that any object that is an instance of the **Student** class will include instance variables named **name**, **test1**, **test2**, and **test3**, and it will include an instance method named **getAverage()**. The names and tests in different objects will generally have different values. When called for a particular student, the method **getAverage()** will compute an average using **that student's** test grades. Different students can have different averages. (Again, this is what it means to say that an instance method belongs to an individual object, not to the class.)

In Java, a class is a **type**, similar to the built-in types such as **int** and **boolean**. So, a class name can be used to specify the type of a variable in a declaration statement, the type of a formal parameter, or the return type of a function. For example, a program could define a variable named **std** of type **Student** with the statement

```
Student std;
```

However, declaring a variable does **not** create an object! This is an important point, which is related to this Very Important Fact:

**In Java, no variable can ever hold an object.
A variable can only hold a reference to an object.**

You should think of objects as floating around independently in the computer's memory. In fact, there is a special portion of memory called the **heap** where objects live. Instead of holding an object itself, a variable holds the information necessary to find the object in memory. This information is called a **reference** or **pointer** to the object. In effect, a reference to an object is the address of the memory location where the object is stored. When you use a variable of class type, the computer uses the reference in the variable to find the actual object.

In a program, objects are created using an operator called **new**, which creates an object and returns a reference to that object. For example, assuming that **std** is a variable of type **Student**, declared as above, the assignment statement

```
std = new Student();
```

would create a new object which is an instance of the class **Student**, and it would store a reference to that object in the variable **std**. The value of the variable is a reference to the object, not the object itself. It is not quite true, then, to say that the object is the "value of the variable **std**" (though sometimes it is hard to avoid using this terminology). It is certainly **not at all true** to say that the object is "stored in the variable **std**." The proper terminology is that "the variable **std** *refers to* the object," and I will try to stick to that terminology as much as possible.

So, suppose that the variable **std** refers to an object belonging to the class **Student**. That object has instance variables **name**, **test1**, **test2**, and **test3**. These instance variables can

be referred to as `std.name`, `std.test1`, `std.test2`, and `std.test3`. This follows the usual naming convention that when B is part of A, then the full name of B is A.B. For example, a program might include the lines

```
System.out.println("Hello, " + std.name + ". Your test grades are:");
System.out.println(std.test1);
System.out.println(std.test2);
System.out.println(std.test3);
```

This would output the name and test grades from the object to which `std` refers. Similarly, `std` can be used to call the `getAverage()` instance method in the object by saying `std.getAverage()`. To print out the student's average, you could say:

```
System.out.println( "Your average is " + std.getAverage() );
```

More generally, you could use `std.name` any place where a variable of type *String* is legal. You can use it in expressions. You can assign a value to it. You can even use it to call subroutines from the *String* class. For example, `std.name.length()` is the number of characters in the student's name.

It is possible for a variable like `std`, whose type is given by a class, to refer to no object at all. We say in this case that `std` holds a *null reference*. The null reference is written in Java as “`null`”. You can store a null reference in the variable `std` by saying

```
std = null;
```

and you could test whether the value of `std` is null by testing

```
if (std == null) . . .
```

If the value of a variable is `null`, then it is, of course, illegal to refer to instance variables or instance methods through that variable—since there is no object, and hence no instance variables to refer to. For example, if the value of the variable `std` is `null`, then it would be illegal to refer to `std.test1`. If your program attempts to use a null reference illegally like this, the result is an error called a *null pointer exception*.

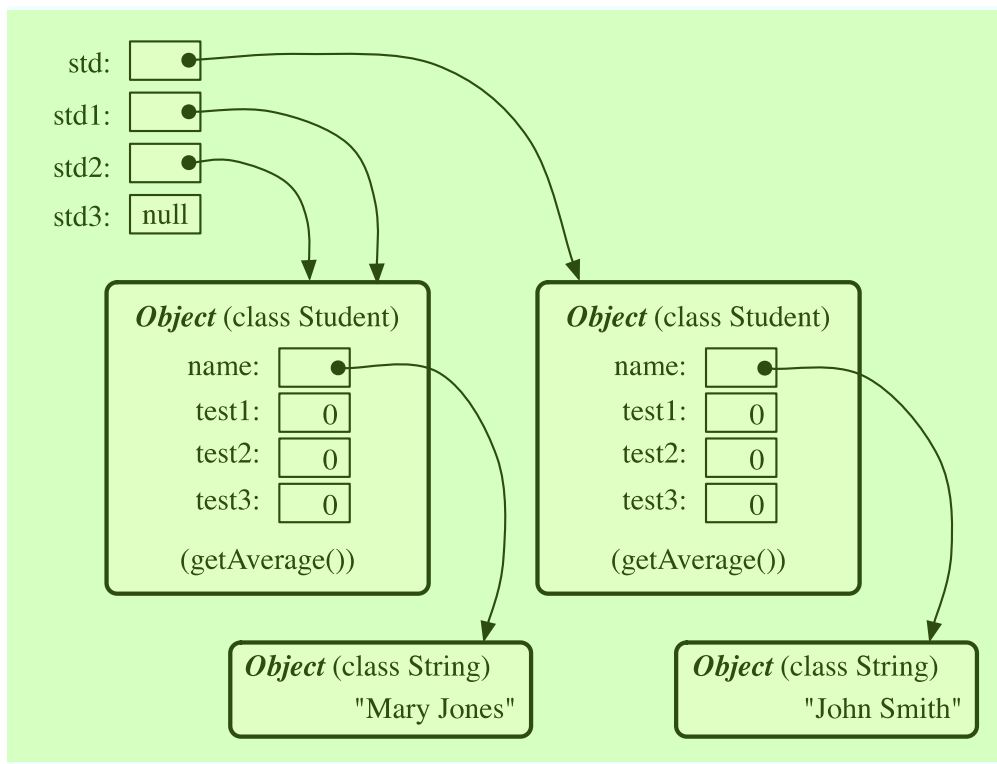
Let's look at a sequence of statements that work with objects:

```
Student std, std1,      // Declare four variables of
    std2, std3;         //   type Student.
std = new Student();    // Create a new object belonging
                        //   to the class Student, and
                        //   store a reference to that
                        //   object in the variable std.
std1 = new Student();   // Create a second Student object
                        //   and store a reference to
                        //   it in the variable std1.
std2 = std1;            // Copy the reference value in std1
                        //   into the variable std2.
std3 = null;            // Store a null reference in the
                        //   variable std3.

std.name = "John Smith"; // Set values of some instance variables.
std1.name = "Mary Jones";

// (Other instance variables have default
//   initial values of zero.)
```

After the computer executes these statements, the situation in the computer's memory looks like this:



This picture shows variables as little boxes, labeled with the names of the variables. Objects are shown as boxes with round corners. When a variable contains a reference to an object, the value of that variable is shown as an arrow pointing to the object. The variable `std3`, with a value of `null`, doesn't point anywhere. The arrows from `std1` and `std2` both point to the same object. This illustrates a Very Important Point:

When one object variable is assigned to another, only a reference is copied. The object referred to is not copied.

When the assignment "`std2 = std1;`" was executed, no new object was created. Instead, `std2` was set to refer to the very same object that `std1` refers to. This has some consequences that might be surprising. For example, `std1.name` and `std2.name` are two different names for the same variable, namely the instance variable in the object that both `std1` and `std2` refer to. After the string "Mary Jones" is assigned to the variable `std1.name`, it is also true that the value of `std2.name` is "Mary Jones". There is a potential for a lot of confusion here, but you can help protect yourself from it if you keep telling yourself, "The object is not in the variable. The variable just holds a pointer to the object."

You can test objects for equality and inequality using the operators `==` and `!=`, but here again, the semantics are different from what you are used to. When you make a test "`if (std1 == std2)`", you are testing whether the values stored in `std1` and `std2` are the same. But the values are references to objects, not objects. So, you are testing whether `std1` and `std2` refer to the same object, that is, whether they point to the same location

in memory. This is fine, if its what you want to do. But sometimes, what you want to check is whether the instance variables in the objects have the same values. To do that, you would need to ask whether “`std1.test1 == std2.test1 && std1.test2 == std2.test2 && std1.test3 == std2.test3 && std1.name.equals(std2.name)`”.

I’ve remarked previously that `Strings` are objects, and I’ve shown the strings “Mary Jones” and “John Smith” as objects in the above illustration. A variable of type `String` can only hold a reference to a string, not the string itself. It could also hold the value `null`, meaning that it does not refer to any string at all. This explains why using the `==` operator to test strings for equality is not a good idea. Suppose that `greeting` is a variable of type `String`, and that the string it refers to is “Hello”. Then would the test `greeting == "Hello"` be true? Well, maybe, maybe not. The variable `greeting` and the `String` literal “Hello” each refer to a string that contains the characters H-e-l-l-o. But the strings could still be different objects, that just happen to contain the same characters. The function `greeting.equals("Hello")` tests whether `greeting` and “Hello” contain the same characters, which is almost certainly the question you want to ask. The expression `greeting == "Hello"` tests whether `greeting` and “Hello” contain the same characters **stored in the same memory location**.

* * *

The fact that variables hold references to objects, not objects themselves, has a couple of other consequences that you should be aware of. They follow logically, if you just keep in mind the basic fact that the object is not stored in the variable. The object is somewhere else; the variable points to it.

Suppose that a variable that refers to an object is declared to be `final`. This means that the value stored in the variable can never be changed, once the variable has been initialized. The value stored in the variable is a reference to the object. So the variable will continue to refer to the same object as long as the variable exists. However, this does not prevent the data **in the object** from changing. The variable is `final`, not the object. It’s perfectly legal to say

```
final Student stu = new Student();

stu.name = "John Doe"; // Change data in the object;
                       // The value stored in stu is not changed!
                       // It still refers to the same object.
```

Next, suppose that `obj` is a variable that refers to an object. Let’s consider what happens when `obj` is passed as an actual parameter to a subroutine. The value of `obj` is assigned to a formal parameter in the subroutine, and the subroutine is executed. The subroutine has no power to change the value stored in the variable, `obj`. It only has a copy of that value. However, that value is a reference to an object. Since the subroutine has a reference to the object, it can change the data stored in the object. After the subroutine ends, `obj` still points to the same object, but the data stored **in the object** might have changed. Suppose `x` is a variable of type `int` and `stu` is a variable of type `Student`. Compare:

<pre>void dontChange(int z) { z = 42; } The lines: x = 17; dontChange(x); System.out.println(x);</pre>	<pre>void change(Student s) { s.name = "Fred"; } The lines: stu.name = "Jane"; change(stu); System.out.println(stu.name);</pre>
--	---

output the value 17.

The value of `x` is not changed by the subroutine, which is equivalent to

```
z = x;
z = 42;
```

output the value "Fred". 1

The value of `stu` is not changed, but `stu.name` is. This is equivalent to

```
s = stu;
s.name = "Fred";
```

5.1.3 Getters and Setters 2

When writing new classes, it's a good idea to pay attention to the issue of access control. Recall 3 that making a member of a class **public** makes it accessible from anywhere, including from other classes. On the other hand, a **private** member can only be used in the class where it is defined.

In the opinion of many programmers, almost all member variables should be declared 4 **private**. This gives you complete control over what can be done with the variable. Even if the variable itself is private, you can allow other classes to find out what its value is by providing a **public** *accessor method* that returns the value of the variable. For example, if your class contains a **private** member variable, `title`, of type *String*, you can provide a method

```
public String getTitle() { 5
    return title;
}
```

that returns the value of `title`. By convention, the name of an accessor method for a variable 6 is obtained by capitalizing the name of variable and adding “get” in front of the name. So, for the variable `title`, we get an accessor method named “get” + “Title”, or `getTitle()`. Because of this naming convention, accessor methods are more often referred to as *getter methods*. A getter method provides “read access” to a variable.

You might also want to allow “write access” to a **private** variable. That is, you might 7 want to make it possible for other classes to specify a new value for the variable. This is done with a *setter method*. (If you don't like simple, Anglo-Saxon words, you can use the fancier term *mutator method*.) The name of a setter method should consist of “set” followed by a capitalized copy of the variable's name, and it should have a parameter with the same type as the variable. A setter method for the variable `title` could be written

```
public void setTitle( String newTitle ) { 8
    title = newTitle;
}
```

It is actually very common to provide both a getter and a setter method for a private 9 member variable. Since this allows other classes both to see and to change the value of the variable, you might wonder why not just make the variable **public**? The reason is that getters and setters are not restricted to simply reading and writing the variable's value. In fact, they can take any action at all. For example, a getter method might keep track of the number of times that the variable has been accessed:

```
public String getTitle() {
    titleAccessCount++; // Increment member variable titleAccessCount.
    return title;
} 10
```

and a setter method might check that the value that is being assigned to the variable is legal: 11

```
public void setTitle( String newTitle ) {  
    if ( newTitle == null )    // Don't allow null strings as titles!  
        title = "(Untitled)"; // Use an appropriate default value instead.  
    else  
        title = newTitle;  
}
```

Even if you can't think of any extra chores to do in a getter or setter method, you might change your mind in the future when you redesign and improve your class. If you've used a getter and setter from the beginning, you can make the modification to your class without affecting any of the classes that use your class. The **private** member variable is not part of the public interface of your class; only the **public** getter and setter methods are. If you **haven't** used `get` and `set` from the beginning, you'll have to contact everyone who uses your class and tell them, "Sorry guys, you'll have to track down every use that you've made of this variable and change your code to use my new `get` and `set` methods instead."

A couple of final notes: Some advanced aspects of Java rely on the naming convention for getter and setter methods, so it's a good idea to follow the convention rigorously. And though I've been talking about using getter and setter methods for a variable, you can define `get` and `set` methods even if there is no variable. A getter and/or setter method defines a *property* of the class, that might or might not correspond to a variable. For example, if a class includes a **public void** instance method with signature `setValue(double)`, then the class has a "property" named `value` of type **double**, and it has this property whether or not the class has a member variable named `value`.

5.2 Constructors and Object Initialization

OBJECT TYPES IN JAVA are very different from the primitive types. Simply declaring a variable whose type is given as a class does not automatically create an object of that class. Objects must be explicitly *constructed*. For the computer, the process of constructing an object means, first, finding some unused memory in the heap that can be used to hold the object and, second, filling in the object's instance variables. As a programmer, you don't care where in memory the object is stored, but you will usually want to exercise some control over what initial values are stored in a new object's instance variables. In many cases, you will also want to do more complicated initialization or bookkeeping every time an object is created.

5.2.1 Initializing Instance Variables

An instance variable can be assigned an initial value in its declaration, just like any other variable. For example, consider a class named `PairOfDice`. An object of this class will represent a pair of dice. It will contain two instance variables to represent the numbers showing on the dice and an instance method for rolling the dice:

```
public class PairOfDice {  
  
    public int die1 = 3;    // Number showing on the first die.  
    public int die2 = 4;    // Number showing on the second die.  
  
    public void roll() {  
        // Roll the dice by setting each of the dice to be  
        // a random number between 1 and 6.  
        die1 = (int)(Math.random()*6) + 1;  
    }  
}
```

```

        die2 = (int)(Math.random()*6) + 1; 1
    }
} // end class PairOfDice

```

The instance variables `die1` and `die2` are initialized to the values 3 and 4 respectively. These 2 initializations are executed whenever a `PairOfDice` object is constructed. It's important to understand when and how this happens. There can be many `PairOfDice` objects. Each time one is created, it gets its own instance variables, and the assignments “`die1 = 3`” and “`die2 = 4`” are executed to fill in the values of those variables. To make this clearer, consider a variation of the `PairOfDice` class:

```

public class PairOfDice { 3
    public int die1 = (int)(Math.random()*6) + 1;
    public int die2 = (int)(Math.random()*6) + 1;

    public void roll() {
        die1 = (int)(Math.random()*6) + 1;
        die2 = (int)(Math.random()*6) + 1;
    }
} // end class PairOfDice

```

Here, the dice are initialized to random values, as if a new pair of dice were being thrown onto 4 the gaming table. Since the initialization is executed for each new object, a set of random initial values will be computed for each new pair of dice. Different pairs of dice can have different initial values. For initialization of **static** member variables, of course, the situation is quite different. There is only one copy of a **static** variable, and initialization of that variable is executed just once, when the class is first loaded.

If you don't provide any initial value for an instance variable, a default initial value is provided 5 automatically. Instance variables of numerical type (**int**, **double**, etc.) are automatically initialized to zero if you provide no other values; **boolean** variables are initialized to **false**; and **char** variables, to the Unicode character with code number zero. An instance variable can also be a variable of object type. For such variables, the default initial value is **null**. (In particular, since **Strings** are objects, the default initial value for *String* variables is **null**.)

5.2.2 Constructors 6

Objects are created with the operator, **new**. For example, a program that wants to use a 7 `PairOfDice` object could say:

```

PairOfDice dice; // Declare a variable of type PairOfDice. 8

dice = new PairOfDice(); // Construct a new object and store a 9
                        // reference to it in the variable.

```

In this example, “`new PairOfDice()`” is an expression that allocates memory for the object, 10 initializes the object's instance variables, and then returns a reference to the object. This reference is the value of the expression, and that value is stored by the assignment statement in the variable, `dice`, so that after the assignment statement is executed, `dice` refers to the newly created object. Part of this expression, “`PairOfDice()`”, looks like a subroutine call, and that is no accident. It is, in fact, a call to a special type of subroutine called a **constructor**. This might puzzle you, since there is no such subroutine in the class definition. However, every class has at least one constructor. If the programmer doesn't write a constructor definition in a class,

then the system will provide a **default constructor** for that class. This default constructor does nothing beyond the basics: allocate memory and initialize instance variables. If you want more than that to happen when an object is created, you can include one or more constructors in the class definition. 1

The definition of a constructor looks much like the definition of any other subroutine, with three exceptions. A constructor does not have any return type (not even `void`). The name of the constructor must be the same as the name of the class in which it is defined. The only modifiers that can be used on a constructor definition are the access modifiers `public`, `private`, and `protected`. (In particular, a constructor can't be declared `static`.) 2

However, a constructor does have a subroutine body of the usual form, a block of statements. There are no restrictions on what statements can be used. And it can have a list of formal parameters. In fact, the ability to include parameters is one of the main reasons for using constructors. The parameters can provide data to be used in the construction of the object. For example, a constructor for the `PairOfDice` class could provide the values that are initially showing on the dice. Here is what the class would look like in that case: 3

```
public class PairOfDice {
    public int die1;    // Number showing on the first die.
    public int die2;    // Number showing on the second die.

    public PairOfDice(int val1, int val2) {
        // Constructor.  Creates a pair of dice that
        // are initially showing the values val1 and val2.
        die1 = val1;    // Assign specified values
        die2 = val2;    //           to the instance variables.
    }

    public void roll() {
        // Roll the dice by setting each of the dice to be
        // a random number between 1 and 6.
        die1 = (int)(Math.random()*6) + 1;
        die2 = (int)(Math.random()*6) + 1;
    }
} // end class PairOfDice
```

4

The constructor is declared as “`public PairOfDice(int val1, int val2) ...`”, with no return type and with the same name as the name of the class. This is how the Java compiler recognizes a constructor. The constructor has two parameters, and values for these parameters must be provided when the constructor is called. For example, the expression “`new PairOfDice(3,4)`” would create a `PairOfDice` object in which the values of the instance variables `die1` and `die2` are initially 3 and 4. Of course, in a program, the value returned by the constructor should be used in some way, as in 5

```
PairOfDice dice;           // Declare a variable of type PairOfDice.
dice = new PairOfDice(1,1); // Let dice refer to a new PairOfDice
                           //   object that initially shows 1, 1.
```

6

Now that we've added a constructor to the `PairOfDice` class, we can no longer create an object by saying “`new PairOfDice()`”! The system provides a default constructor for a class **only** if the class definition does not already include a constructor, so there is only one constructor in the class, and it requires two actual parameters. However, this is not a big 7

problem, since we can add a second constructor to the class, one that has no parameters. In fact, you can have as many different constructors as you want, as long as their signatures are different, that is, as long as they have different numbers or types of formal parameters. In the `PairOfDice` class, we might have a constructor with no parameters which produces a pair of dice showing random numbers:

```
public class PairOfDice {

    public int die1;    // Number showing on the first die.
    public int die2;    // Number showing on the second die.

    public PairOfDice() {
        // Constructor. Rolls the dice, so that they initially
        // show some random values.
        roll(); // Call the roll() method to roll the dice.
    }

    public PairOfDice(int val1, int val2) {
        // Constructor. Creates a pair of dice that
        // are initially showing the values val1 and val2.
        die1 = val1; // Assign specified values
        die2 = val2; //           to the instance variables.
    }

    public void roll() {
        // Roll the dice by setting each of the dice to be
        // a random number between 1 and 6.
        die1 = (int)(Math.random()*6) + 1;
        die2 = (int)(Math.random()*6) + 1;
    }

} // end class PairOfDice
```

Now we have the option of constructing a `PairOfDice` object either with “`new PairOfDice()`” or with “`new PairOfDice(x,y)`”, where `x` and `y` are **int**-valued expressions.

This class, once it is written, can be used in any program that needs to work with one or more pairs of dice. None of those programs will ever have to use the obscure incantation “`(int)(Math.random()*6)+1`”, because it’s done inside the `PairOfDice` class. And the programmer, having once gotten the dice-rolling thing straight will never have to worry about it again. Here, for example, is a main program that uses the `PairOfDice` class to count how many times two pairs of dice are rolled before the two pairs come up showing the same value. This illustrates once again that you can create several instances of the same class:

```
public class RollTwoPairs {

    public static void main(String[] args) {

        PairOfDice firstDice; // Refers to the first pair of dice.
        firstDice = new PairOfDice();

        PairOfDice secondDice; // Refers to the second pair of dice.
        secondDice = new PairOfDice();

        int countRolls; // Counts how many times the two pairs of
                        //      dice have been rolled.

        int total1;     // Total showing on first pair of dice.
        int total2;     // Total showing on second pair of dice.
```

```

        countRolls = 0;
        do { // Roll the two pairs of dice until totals are the same.
            firstDice.roll(); // Roll the first pair of dice.
            total1 = firstDice.die1 + firstDice.die2; // Get total.
            System.out.println("First pair comes up " + total1);

            secondDice.roll(); // Roll the second pair of dice.
            total2 = secondDice.die1 + secondDice.die2; // Get total.
            System.out.println("Second pair comes up " + total2);

            countRolls++; // Count this roll.

            System.out.println(); // Blank line.
        } while (total1 != total2);

        System.out.println("It took " + countRolls
                           + " rolls until the totals were the same.");
    } // end main()
} // end class RollTwoPairs

```

* * *

Constructors are subroutines, but they are subroutines of a special type. They are certainly not instance methods, since they don't belong to objects. Since they are responsible for creating objects, they exist before any objects have been created. They are more like **static** member subroutines, but they are not and cannot be declared to be **static**. In fact, according to the Java language specification, they are technically not members of the class at all! In particular, constructors are **not** referred to as "methods."

Unlike other subroutines, a constructor can only be called using the **new** operator, in an expression that has the form

```
new <class-name> ( <parameter-list> )
```

where the *<parameter-list>* is possibly empty. I call this an expression because it computes and returns a value, namely a reference to the object that is constructed. Most often, you will store the returned reference in a variable, but it is also legal to use a constructor call in other ways, for example as a parameter in a subroutine call or as part of a more complex expression. Of course, if you don't save the reference in a variable, you won't have any way of referring to the object that was just created.

A constructor call is more complicated than an ordinary subroutine or function call. It is helpful to understand the exact steps that the computer goes through to execute a constructor call:

1. First, the computer gets a block of unused memory in the heap, large enough to hold an object of the specified type.
2. It initializes the instance variables of the object. If the declaration of an instance variable specifies an initial value, then that value is computed and stored in the instance variable. Otherwise, the default initial value is used.
3. The actual parameters in the constructor, if any, are evaluated, and the values are assigned to the formal parameters of the constructor.

4. The statements in the body of the constructor, if any, are executed. ¹

5. A reference to the object is returned as the value of the constructor call. ²

The end result of this is that you have a reference to a newly constructed object. You can ³ use this reference to get at the instance variables in that object or to call its instance methods.

* * *

For another example, let's rewrite the `Student` class that was used in Section 1. I'll add a ⁴ constructor, and I'll also take the opportunity to make the instance variable, `name`, private.

```
public class Student { 5
    private String name;           // Student's name.
    public double test1, test2, test3; // Grades on three tests.

    Student(String theName) {
        // Constructor for Student objects;
        // provides a name for the Student.
        name = theName;
    }

    public String getName() {
        // Getter method for reading the value of the private
        // instance variable, name.
        return name;
    }

    public double getAverage() {
        // Compute average test grade.
        return (test1 + test2 + test3) / 3;
    }
} // end of class Student
```

An object of type `Student` contains information about some particular student. The constructor in this class has a parameter of type *String*, which specifies the name of that student. ⁶ Objects of type `Student` can be created with statements such as:

```
std = new Student("John Smith"); 7
std1 = new Student("Mary Jones");
```

In the original version of this class, the value of `name` had to be assigned by a program after ⁸ it created the object of type `Student`. There was no guarantee that the programmer would always remember to set the `name` properly. In the new version of the class, there is no way to create a `Student` object except by calling the constructor, and that constructor automatically sets the `name`. The programmer's life is made easier, and whole hordes of frustrating bugs are squashed before they even have a chance to be born.

Another type of guarantee is provided by the `private` modifier. ⁹ Since the instance variable, `name`, is `private`, there is no way for any part of the program outside the `Student` class to get at the `name` directly. The program sets the value of `name`, indirectly, when it calls the constructor. I've provided a function, `getName()`, that can be used from outside the class to find out the `name` of the student. But I haven't provided any setter method or other way to change the name. Once a student object is created, it keeps the same name as long as it exists.

5.2.3 Garbage Collection ¹

So far, this section has been about creating objects. What about destroying them? In Java, ² the destruction of objects takes place automatically.

An object exists in the heap, and it can be accessed only through variables that hold ³ references to the object. What should be done with an object if there are no variables that refer to it? Such things can happen. Consider the following two statements (though in reality, you'd never do anything like this):

```
Student std = new Student("John Smith"); 4  
std = null;
```

In the first line, a reference to a newly created `Student` object is stored in the variable `std`. ⁵ But in the next line, the value of `std` is changed, and the reference to the `Student` object is gone. In fact, there are now no references whatsoever to that object stored in any variable. So there is no way for the program ever to use the object again. It might as well not exist. In fact, the memory occupied by the object should be reclaimed to be used for another purpose.

Java uses a procedure called *garbage collection* to reclaim memory occupied by objects ⁶ that are no longer accessible to a program. It is the responsibility of the system, not the programmer, to keep track of which objects are “garbage.” In the above example, it was very easy to see that the `Student` object had become garbage. Usually, it's much harder. If an object has been used for a while, there might be several references to the object stored in several variables. The object doesn't become garbage until all those references have been dropped.

In many other programming languages, it's the programmer's responsibility to delete the ⁷ garbage. Unfortunately, keeping track of memory usage is very error-prone, and many serious program bugs are caused by such errors. A programmer might accidentally delete an object even though there are still references to that object. This is called a *dangling pointer error*, and it leads to problems when the program tries to access an object that is no longer there. Another type of error is a *memory leak*, where a programmer neglects to delete objects that are no longer in use. This can lead to filling memory with objects that are completely inaccessible, and the program might run out of memory even though, in fact, large amounts of memory are being wasted.

Because Java uses garbage collection, such errors are simply impossible. Garbage collection ⁸ is an old idea and has been used in some programming languages since the 1960s. You might wonder why all languages don't use garbage collection. In the past, it was considered too slow and wasteful. However, research into garbage collection techniques combined with the incredible speed of modern computers have combined to make garbage collection feasible. Programmers should rejoice.

5.3 Programming with Objects ⁹

THERE ARE SEVERAL WAYS in which object-oriented concepts can be applied to the process ¹⁰ of designing and writing programs. The broadest of these is *object-oriented analysis and design* which applies an object-oriented methodology to the earliest stages of program development, during which the overall design of a program is created. Here, the idea is to identify things in the problem domain that can be modeled as objects. On another level, object-oriented programming encourages programmers to produce *generalized software components* that can be used in a wide variety of programming projects.

Of course, for the most part, you will experience “generalized software components” by using the standard classes that come along with Java. We begin this section by looking at some built-in classes that are used for creating objects. At the end of the section, we will get back to generalities.

5.3.1 Some Built-in Classes

Although the focus of object-oriented programming is generally on the design and implementation of new classes, it’s important not to forget that the designers of Java have already provided a large number of reusable classes. Some of these classes are meant to be extended to produce new classes, while others can be used directly to create useful objects. A true mastery of Java requires familiarity with a large number of built-in classes—something that takes a lot of time and experience to develop. In the next chapter, we will begin the study of Java’s GUI classes, and you will encounter other built-in classes throughout the remainder of this book. But let’s take a moment to look at a few built-in classes that you might find useful.

A string can be built up from smaller pieces using the `+` operator, but this is not very efficient. If `str` is a *String* and `ch` is a character, then executing the command “`str = str + ch;`” involves creating a whole new string that is a copy of `str`, with the value of `ch` appended onto the end. Copying the string takes some time. Building up a long string letter by letter would require a surprising amount of processing. The class *StringBuffer* makes it possible to be efficient about building up a long string from a number of smaller pieces. To do this, you must make an object belonging to the *StringBuffer* class. For example:

```
StringBuffer buffer = new StringBuffer();
```

(This statement both declares the variable `buffer` and initializes it to refer to a newly created *StringBuffer* object. Combining declaration with initialization was covered in Subsection 4.7.1 and works for objects just as it does for primitive types.)

Like a *String*, a *StringBuffer* contains a sequence of characters. However, it is possible to add new characters onto the end of a *StringBuffer* without making a copy of the data that it already contains. If `x` is a value of any type and `buffer` is the variable defined above, then the command `buffer.append(x)` will add `x`, converted into a string representation, onto the end of the data that was already in the buffer. This command actually modifies the buffer, rather than making a copy, and that can be done efficiently. A long string can be built up in a *StringBuffer* using a sequence of `append()` commands. When the string is complete, the function `buffer.toString()` will return a copy of the string in the buffer as an ordinary value of type *String*. The *StringBuffer* class is in the standard package `java.lang`, so you can use its simple name without importing it.

A number of useful classes are collected in the package `java.util`. For example, this package contains classes for working with collections of objects. We will study these collection classes in Chapter 10. Another class in this package, `java.util.Date`, is used to represent times. When a *Date* object is constructed without parameters, the result represents the current date and time, so an easy way to display this information is:

```
System.out.println( new Date() );
```

Of course, to use the *Date* class in this way, you must make it available by importing it with one of the statements “`import java.util.Date;`” or “`import java.util.*;`” at the beginning of your program. (See Subsection 4.5.3 for a discussion of packages and `import`.)

I will also mention the class `java.util.Random`. An object belonging to this class is a *source* of random numbers (or, more precisely pseudorandom numbers). The standard function

`Math.random()` uses one of these objects behind the scenes to generate its random numbers. 1
An object of type `Random` can generate random integers, as well as random real numbers. If `randGen` is created with the command:

```
Random randGen = new Random(); 2
```

and if `N` is a positive integer, then `randGen.nextInt(N)` generates a random integer in the range 3
from 0 to `N-1`. For example, this makes it a little easier to roll a pair of dice. Instead of saying “`die1 = (int)(6*Math.random()+1);`”, one can say “`die1 = randGen.nextInt(6)+1;`”. (Since you also have to import the class `java.util.Random` and create the `Random` object, you might not agree that it is actually easier.) An object of type `Random` can also be used to generate so-called Gaussian distributed random real numbers.

The main point here, again, is that many problems have already been solved, and the 4
solutions are available in Java’s standard classes. If you are faced with a task that looks like it should be fairly common, it might be worth looking through a Java reference to see whether someone has already written a class that you can use.

5.3.2 Wrapper Classes and Autoboxing 5

We have already encountered the classes *Double* and *Integer* in Subsection 2.5.7. These classes 6
contain the `static` methods `Double.parseDouble` and `Integer.parseInt` that are used to convert strings to numerical values. We have also encountered the *Character* class in some examples, and static methods such as `Character.isLetter`, which can be used to test whether a given value of type `char` is a letter. There is a similar class for each of the other primitive types, *Long*, *Short*, *Byte*, *Float*, and *Boolean*. These classes are called *wrapper classes*. Although they contain useful `static` members, they have another use as well: They are used for creating objects that represent primitive type values.

Remember that the primitive types are not classes, and values of primitive type are not 7
objects. However, sometimes it’s useful to treat a primitive value as if it were an object. You can’t do that literally, but you can “wrap” the primitive type value in an object belonging to one of the wrapper classes.

For example, an object of type *Double* contains a single instance variable, of type **double**. 8
The object is a wrapper for the **double** value. For example, you can create an object that wraps the **double** value `6.0221415e23` with

```
Double d = new Double(6.0221415e23); 9
```

The value of `d` contains the same information as the value of type **double**, but it is an object. 10
If you want to retrieve the **double** value that is wrapped in the object, you can call the function `d.doubleValue()`. Similarly, you can wrap an **int** in an object of type *Integer*, a **boolean** value in an object of type *Boolean*, and so on. (As an example of where this would be useful, the collection classes that will be studied in Chapter 10 can only hold objects. If you want to add a primitive type value to a collection, it has to be put into a wrapper object first.)

In Java 5.0, wrapper classes have become easier to use. Java 5.0 introduced automatic 11
conversion between a primitive type and the corresponding wrapper class. For example, if you use a value of type **int** in a context that requires an object of type *Integer*, the **int** will automatically be wrapped in an *Integer* object. For example, you can say

```
Integer answer = 42; 12
```

and the computer will silently read this as if it were 13


```
Integer answer = new Integer(42); 1
```

This is called *autoboxing*. It works in the other direction, too. For example, if `d` refers to an object of type `Double`, you can use `d` in a numerical expression such as `2*d`. The **double** value inside `d` is automatically *unboxed* and multiplied by 2. Autoboxing and unboxing also apply to subroutine calls. For example, you can pass an actual parameter of type **int** to a subroutine that has a formal parameter of type `Integer`. In fact, autoboxing and unboxing make it possible in many circumstances to ignore the difference between primitive types and objects.

```
* * * 3
```

The wrapper classes contain a few other things that deserve to be mentioned. `Integer`, for example, contains constants `Integer.MIN_VALUE` and `Integer.MAX_VALUE`, which are equal to the largest and smallest possible values of type **int**, that is, to -2147483648 and 2147483647 respectively. It's certainly easier to remember the names than the numerical values. There are similar named constants in `Long`, `Short`, and `Byte`. `Double` and `Float` also have constants named `MIN_VALUE` and `MAX_VALUE`. `MAX_VALUE` still gives the largest number that can be represented in the given type, but `MIN_VALUE` represents the smallest possible **positive** value. For type **double**, `Double.MIN_VALUE` is 4.9 times 10^{-324} . Since **double** values have only a finite accuracy, they can't get arbitrarily close to zero. This is the closest they can get without actually being equal to zero.

The class `Double` deserves special mention, since **doubles** are so much more complicated than integers. The encoding of real numbers into values of type **double** has room for a few special values that are not real numbers at all in the mathematical sense. These values are given by named constants in class `Double`: `Double.POSITIVE_INFINITY`, `Double.NEGATIVE_INFINITY`, and `Double.NaN`. The infinite values can occur as the values of certain mathematical expressions. For example, dividing a positive number by zero will give the result `Double.POSITIVE_INFINITY`. (It's even more complicated than this, actually, because the **double** type includes a value called "negative zero", written `-0.0`. Dividing a positive number by negative zero gives `Double.NEGATIVE_INFINITY`.) You also get `Double.POSITIVE_INFINITY` whenever the mathematical value of an expression is greater than `Double.MAX_VALUE`. For example, `1e200*1e200` is considered to be infinite. The value `Double.NaN` is even more interesting. "NaN" stands for *Not a Number*, and it represents an undefined value such as the square root of a negative number or the result of dividing zero by zero. Because of the existence of `Double.NaN`, no mathematical operation on real numbers will ever throw an exception; it simply gives `Double.NaN` as the result.

You can test whether a value, `x`, of type **double** is infinite or undefined by calling the boolean-valued static functions `Double.isInfinite(x)` and `Double.isNaN(x)`. (It's especially important to use `Double.isNaN()` to test for undefined values, because `Double.NaN` has really weird behavior when used with relational operators such as `==`. In fact, the values of `x == Double.NaN` and `x != Double.NaN` are **both false**, no matter what the value of `x`, so you really can't use these expressions to test whether `x` is `Double.NaN`.)

5.3.3 The class "Object" 7

We have already seen that one of the major features of object-oriented programming is the ability to create subclasses of a class. The subclass inherits all the properties or behaviors of the class, but can modify and add to what it inherits. In Section 5.5, you'll learn how to create subclasses. What you don't know yet is that **every** class in Java (with just one exception) is a subclass of some other class. If you create a class and don't explicitly make it a subclass of

some other class, then it automatically becomes a subclass of the special class named *Object*. (*Object* is the one class that is not a subclass of any other class.)

Class *Object* defines several instance methods that are inherited by every other class. These methods can be used with any object whatsoever. I will mention just one of them here. You will encounter more of them later in the book.

The instance method `toString()` in class *Object* returns a value of type *String* that is supposed to be a string representation of the object. You've already used this method implicitly, any time you've printed out an object or concatenated an object onto a string. When you use an object in a context that requires a string, the object is automatically converted to type *String* by calling its `toString()` method.

The version of `toString()` that is defined in *Object* just returns the name of the class that the object belongs to, concatenated with a code number called the *hash code* of the object; this is not very useful. When you create a class, you can write a new `toString()` method for it, which will replace the inherited version. For example, we might add the following method to any of the *PairOfDice* classes from the previous section:

```
public String toString() {
    // Return a String representation of a pair of dice, where die1
    // and die2 are instance variables containing the numbers that are
    // showing on the two dice.
    if (die1 == die2)
        return "double " + die1;
    else
        return die1 + " and " + die2;
}
```

If `dice` refers to a *PairOfDice* object, then `dice.toString()` will return strings such as “3 and 6”, “5 and 1”, and “double 2”, depending on the numbers showing on the dice. This method would be used automatically to convert `dice` to type *String* in a statement such as

```
System.out.println( "The dice came up " + dice );
```

so this statement might output, “The dice came up 5 and 1” or “The dice came up double 2”. You'll see another example of a `toString()` method in the next section.

5.3.4 Object-oriented Analysis and Design

Every programmer builds up a stock of techniques and expertise expressed as snippets of code that can be reused in new programs using the tried-and-true method of cut-and-paste: The old code is physically copied into the new program and then edited to customize it as necessary. The problem is that the editing is error-prone and time-consuming, and the whole enterprise is dependent on the programmer's ability to pull out that particular piece of code from last year's project that looks like it might be made to fit. (On the level of a corporation that wants to save money by not reinventing the wheel for each new project, just keeping track of all the old wheels becomes a major task.)

Well-designed classes are software components that can be reused without editing. A well-designed class is not carefully crafted to do a particular job in a particular program. Instead, it is crafted to model some particular type of object or a single coherent concept. Since objects and concepts can recur in many problems, a well-designed class is likely to be reusable without modification in a variety of projects.

Furthermore, in an object-oriented programming language, it is possible to make **subclasses** 1 of an existing class. This makes classes even more reusable. If a class needs to be customized, a subclass can be created, and additions or modifications can be made in the subclass without making any changes to the original class. This can be done even if the programmer doesn't have access to the source code of the class and doesn't know any details of its internal, hidden implementation.

* * * 2

The *PairOfDice* class in the previous section is already an example of a generalized software 3 component, although one that could certainly be improved. The class represents a single, coherent concept, "a pair of dice." The instance variables hold the data relevant to the state of the dice, that is, the number showing on each of the dice. The instance method represents the behavior of a pair of dice, that is, the ability to be rolled. This class would be reusable in many different programming projects.

On the other hand, the *Student* class from the previous section is not very reusable. It 4 seems to be crafted to represent students in a particular course where the grade will be based on three tests. If there are more tests or quizzes or papers, it's useless. If there are two people in the class who have the same name, we are in trouble (one reason why numerical student ID's are often used). Admittedly, it's much more difficult to develop a general-purpose student class than a general-purpose pair-of-dice class. But this particular *Student* class is good mostly as an example in a programming textbook.

* * * 5

A large programming project goes through a number of stages, starting with **specification** 6 of the problem to be solved, followed by **analysis** of the problem and **design** of a program to solve it. Then comes **coding**, in which the program's design is expressed in some actual programming language. This is followed by **testing** and **debugging** of the program. After that comes a long period of **maintenance**, which means fixing any new problems that are found in the program and modifying it to adapt it to changing requirements. Together, these stages form what is called the **software life cycle**. (In the real world, the ideal of consecutive stages is seldom if ever achieved. During the analysis stage, it might turn out that the specifications are incomplete or inconsistent. A problem found during testing requires at least a brief return to the coding stage. If the problem is serious enough, it might even require a new design. Maintenance usually involves redoing some of the work from previous stages....)

Large, complex programming projects are only likely to succeed if a careful, systematic 7 approach is adopted during all stages of the software life cycle. The systematic approach to programming, using accepted principles of good design, is called **software engineering**. The software engineer tries to efficiently construct programs that verifiably meet their specifications and that are easy to modify if necessary. There is a wide range of "methodologies" that can be applied to help in the systematic design of programs. (Most of these methodologies seem to involve drawing little boxes to represent program components, with labeled arrows to represent relationships among the boxes.)

We have been discussing object orientation in programming languages, which is relevant to 8 the coding stage of program development. But there are also object-oriented methodologies for analysis and design. The question in this stage of the software life cycle is, How can one discover or invent the overall structure of a program? As an example of a rather simple object-oriented approach to analysis and design, consider this advice: Write down a description of the problem. Underline all the nouns in that description. The nouns should be considered as candidates for becoming classes or objects in the program design. Similarly, underline all the verbs. These